Logic

Aria Ruler(Forouzandeh)

October 31, 2026

This page intentionally left blank.

Contents

0 Atomes **1**
0.1 Basic tools 1

1 propositional logic (zeroth-order logic) **5**
1.1 What is proposition? 5
1.2 Rules of propositions. 6

2 first-order logic **13**
2.1 what is formula? 13
2.2 Rules of first order logic. 15

3 The beginning of set theory **21**
3.1 Axioms of set theory 21
3.2 Another set theory 31
3.3 combination of two theories and height of the Level 31

4 Relations and Functions **33**
4.1 Ordered pair and Cartesian product 33
4.2 Relations 34
4.3 Functions 37
4.4 Family of sets 41

5 Equivalences and Orderings **47**
5.1 Structures and morphisms 47
5.2 Equivalences and Partitions 48
5.3 Orderings 52

6 The beginning of Abstract Algebra **61**
6.1 Group theory 61
6.2 Ring theory 64

7 Natural numbers **71**
7.1 Introduction to Natural numbers 71
7.2 Properties of Natural numbers 72
7.3 The Recursion theorem 74
7.4 Arithmetic of Natural numbers 80

7.5 Peano arithmetic 81

8 From Natural numbers to Complex numbers **85**
8.1 Integers“$\mathbb{Z}$” 85
8.2 Rational numbers “$\mathbb{Q}$” 86
8.3 Real numbers “$\mathbb{R}$” 88
8.4 Complex numbers“$\mathbb{C}$” 91

9 Finite and Infinite sets **95**
9.1 Equipotent and Order in sets 95
9.2 Finite sets 101
9.3 Countable sets 105
9.4 Uncountable sets 114

10 Ordinals **117**
10.1 Well ordered sets 117
10.2 Ordinal numbers 119
10.3 Transfinite Induction 126
10.4 Transfinite Recurtion 127
10.5 Arithmetic of Ordinals 130
10.6 Transfinite Peano arithmetic 130

11 Cardinals **133**

12 Axiom of choice **135**

13 fantacy numbers($\mathbb{Z}(Ord), \mathbb{Q}(Ord), \mathbb{R}(Ord), \mathbb{C}(Ord)$) **137**
13.1 fantacy Integers“$\mathbb{Z}(Ord)$” 137
13.2 fantacy Rational numbers “$\mathbb{Q}(Ord)$” 138
13.3 fantacy Real numbers “$\mathbb{R}(Ord)$” 140
13.4 fantacy Complex numbers“$\mathbb{C}(Ord)$” 142

14 hyperreal numbers($^*\mathbb{N}, {}^*\mathbb{Z}, {}^*\mathbb{Q}, {}^*\mathbb{R}$) **145**
14.1 Filters 145
14.2 Ultrafilters 145
14.3 Nonprincipal Ultrafilters 147
14.4 Definition of $^*\mathbb{N}$, $^*\mathbb{Z}$, $^*\mathbb{Q}$, $^*\mathbb{R}$, and $^*\mathbb{R}_{Ord}$ 148
14.5 The Ultrapower Construction of the Hyperreals 150
14.6 (Un)limited, Infinitesimal, and Appreciable Numbers 153
14.7 Arithmetic of Hyperreals 154
14.8 Halos, Galaxies, and Real Comparisons 155
14.9 Standard Parts 156

15 New beginning for calculus **159**
15.1 Limits 159
15.2 Continuity 165
15.3 Derivatives 167
15.4 Partial Derivatives 167

15.5 Riemann Integral 168

16 Well-Ordered Integral(more powerful than Lebesgue Integral) 173

17 New beginning for Probability and Measeure theory 175
17.1 Finite Approach 175
17.1.1 Expected Value in the Finite Approach 177
17.2 Continuous Approach 178
17.2.1 Expected Value in the Continuous Approach 181
17.3 Nonstandard Approach 189
17.4 Countable Probability 191
17.5 Decision Theory 193
17.5.1 The State Space 193
17.5.2 The Action Space 194
17.5.3 The Utility Function 194
17.5.4 Probability on the State Space 195
17.5.5 Expected Utility 195
17.5.6 The Maximizing Action 196
17.5.7 A Decision Problem with Infinitesimal Probabilities 196

18 Measure 199
18.1 Construction of Lebesgue Measure 199

This page intentionally left blank.

Chapter 0

Atomes

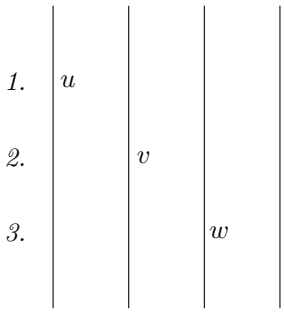
0.1 Basic tools

we start the book by building logic step by step.block by block in first it might seem a little annoying but I advice you to be patient.book will show its power soon. like euclidian geometry this book has undefined concepts. I tried to explain them for you to understand.but they don't have any definition.

Undefined 0.1.1 (Urelements)**.** *are the most fundamental entities in a logical. They cannot be constructed, defined, or described in terms of other things. Their existence is primitive—accepted without internal structure or relational properties. Urelements simply are.*

In this book, we will denote individual urelements using lowercase letters u, v, w,...

Undefined 0.1.2. ***Type*** *parallel lines used to show that an object exists (you will undrestnd what is object soon.) (we will show you how to use them).*



Rule 0.1.1. ***fusion*** *it's the Rule that explain how sets made. (this rule is schema.it means it works everywhere for every object.)*

$m.$	u_0	
	$\vdots$	
$m+n.$	u_n	
$o.$		$\{u_0, u_1 ... u_n\}$

Undefined 0.1.3. ***Set*** *we can't say what exactly sets are but we know sets can be made by fusion (if "a" is a set we write "set(a)" from now we denote sets with uppercase letters like A, B, C, ...).*

definition 0.1.1. ***Object*** *urelements and sets.(we write obj instead of object). (from now we denote objects with lowercase letters like a, b, c, ...).*

Rule 0.1.2. ***fission*** *it show us how to break an obj. (this rule is scheme it means it works everywhere for every obj.)*

$o.$		$\{a_0, a_1 ... a_n\}$
$m.$	a_0	
	$\vdots$	
$m+n.$	a_n	

Example 0.1.1. $\{a, b, c\} / \therefore \{a, b\}$ *(prove if* $\{a, b, c\}$ *exists then* $\{a, c\}$ *exists)*

1.		$\{a, b, c\}$	
2.	a		*fi1*
3.	b		*fi1*
4.	c		*fi1*
5.		$\{a, c\}$	*fu2,3,4*

Example 0.1.2. $\{a, b\} / \therefore \{b, a\}$

1.		$\{a, b\}$	
2.	a		*fi1*
3.	b		*fi1*
4.		$\{b, a\}$	*fu2,3*

Rule 0.1.3. ***promotion*** *it's the Rule that let us promote an obj to higher level. (this rule is schema.it means it works everywhere for every object.)*

1.	a	
2.		a

This page intentionally left blank.

Chapter 1

propositional logic (zeroth-order logic)

1.1 What is proposition?

definition 1.1.1. ***predicate*** *"set() " and identity "=" and membership "∈" symbols are predicates. "set() " is unary and identity and membership are binary.*

capital letter "(P, Q, R)" showes **predicates**.
definition 1.1.2. ***Atomic proposition*** *we define atomic propositions like this.*

1. *if "a" is an obj then "set(a) " is an atomic proposition.*
2. *if "a, b" are obj then "a = b" is an atomic proposition.*
3. *if "a, b" are obj then "a ∈ b" is an atomic proposition.*

we denote atomic propositions with "p_i".
Undefined 1.1.1. ***logical operators*** *($\neg, \rightarrow, \wedge, \vee$) tool that used between atoms to make more complex propositions (all of the opratores are binary except "¬" that is unary).*

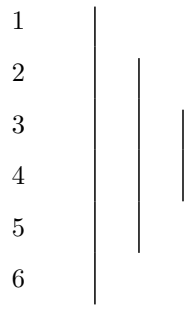
you can easily see that formula are made up by atomic propositions.in general we say formula is an atomic formula or combination of them.
definition 1.1.3. ***Propositions*** *these are propositions.*

1. *atomic propositions "(p_i)".*
2. *if "p" is a formula then "¬p" is a formula too.*
3. *if "p, q" are two propositions then "$p \bigcirc q$" is a formula too.($\bigcirc$ can be "$\rightarrow, \wedge, \vee$").*

for example "$(p_0 \rightarrow p_1, \neg p_3)$"
we denote propositions with "$p, q, r, ...$".

1.2 Rules of propositions.

definition 1.2.1. ***proposition schema(variable proposition)*** *a proposition that is unknown for us and it works as variable.(we show them with "$\varphi, \psi, \theta, \ldots$")*
Undefined 1.2.1. ***Assumption for conditional proof(acp)*** *is a tool for assuming a proposition. (any proposition can't get out but it can get in. right line must be shoter than left one.(it can't fall out). and every hypothesis that opens must be closed.)*



Rule 1.2.1. ***rules of logical operations*** $(\to I, \to E, \wedge I, \wedge E, \vee I, \vee E, \neg I, \neg E, \bot I, \bot E)$ *this rule is scheme it means it works everywhere for every propositions. (thats why we use φ, ψ instead of p, q)*

$\begin{array}{\|l} \varphi \\ \vdots \\ \psi \end{array}$ $\therefore \varphi \to \psi \qquad \to I$	$\varphi \to \psi$ φ $\therefore \psi \qquad \to E$

Example 1.2.1. *for "p, q, r" we can prove these.*

1. $p \to q, q \to r / \therefore p \to r$

2. $q / \therefore p \to q$

1	$p \to q$	ass
2	$q \to r$	ass
3	$\|\ p$	acp
4	$\|\ q$	$\to E1, 3$
5	$\|\ r$	$\to E2, 4$
6	$p \to r$	$\to I$

1	q	ass
2	$\|\ p$	acp
3	$\|\ q$	ass
4	$p \to q$	$\to I$

φ ψ $\therefore \varphi \wedge \psi \quad \wedge I$	$\dfrac{\varphi \wedge \psi}{\therefore \varphi} \wedge E \qquad \dfrac{\varphi \wedge \psi}{\therefore \psi} \wedge E$

Example 1.2.2. *for "p, q, r" we can prove these.*

1. $p \wedge q / \therefore q \wedge p$ *(Commutativity of Conjunction)*

2. $(p \wedge q) \wedge r / \therefore p \wedge (q \wedge r)$ *(Associativity of Conjunction)*

3. $p \to q / \therefore (p \wedge r) \to q$

4. $(p \wedge q) \to r / \therefore p \to (q \to r)$

5. $p \to (q \to r) / \therefore (p \wedge r) \to q$

1	$p \wedge q$	ass
2	p	$\wedge E, 1$
3	q	$\wedge E, 1$
4	$q \wedge p$	$\wedge I, 2, 3$

1	$(p \wedge q) \wedge r$	ass
2	$p \wedge q$	$\wedge E, 1$
3	r	$\wedge E, 1$
4	p	$\wedge E, 2$
5	q	$\wedge E, 2$
6	$q \wedge r$	$\wedge I, 3, 5$
7	$p \wedge (q \wedge r)$	$\wedge I, 4, 6$

1	$p \to q$	ass
2	$\quad p \wedge r$	acp
3	$\quad p$	$\wedge E, 2$
4	$\quad q$	$\to E, 1, 3$
5	$(p \wedge r) \to q$	$\to I$

1	$(p \wedge q) \to r$	ass
2	$\quad p$	acp
3	$\quad\quad q$	acp
4	$\quad\quad p \wedge q$	$\wedge I 2, 3$
5	$\quad\quad r$	$\to E 1, 4$
6	$\quad q \to r$	$\to I$
7	$p \to (q \to r)$	$\to I$

1	$p \to (q \to r)$	ass
2	$\quad p \wedge q$	acp
3	$\quad p$	$\wedge E, 2$
4	$\quad q \to r$	$\to E, 1, 3$
5	$\quad q$	$\wedge E, 2$
6	$\quad r$	$\to E, 4, 5$
7	$(p \wedge q) \to r$	$\to I$

$$\frac{\varphi}{\therefore \varphi \vee \psi}\ \vee I \qquad \frac{\varphi}{\therefore \psi \vee \varphi}\ \vee I$$

$$\begin{array}{l} \varphi \vee \psi \\ \quad \left|\begin{array}{l} \varphi \\ \vdots \\ \theta \end{array}\right. \\ \quad \left|\begin{array}{l} \psi \\ \vdots \\ \theta \end{array}\right. \\ \therefore \theta \qquad \vee E \end{array}$$

Example 1.2.3. *for "p, q, r" we can prove these.*

1. $p \vee q / \therefore q \vee p$ *(Commutativity of Disjunction)*

2. $(p \vee q) \vee r / \therefore p \vee (q \vee r)$ *(Associativity of Disjunction)*

3. $p \wedge (q \vee r) / \therefore (p \wedge q) \vee (p \wedge r)$ *(Distribution and Factorization)*

4. $(p \to q) \wedge (r \to s) / \therefore (p \vee r) \to (q \vee s)$

1	$p \vee q$	*ass*
2	$\quad p$	*acp*
3	$\quad p \vee q$	$\vee I, 2$
4	$\quad q$	*acp*
5	$\quad q \vee p$	$\vee I, 4$
6	$q \vee p$	$\vee E, 1$

1	$(p \vee q) \vee r$	*ass*
2	$\quad p \vee q$	*acp*
3	$\quad\quad p$	*acp*
4	$\quad\quad p \vee (q \vee r)$	$\vee I, 3$
5	$\quad\quad q$	*acp*
6	$\quad\quad q \vee r$	$\vee I, 5$
7	$\quad\quad p \vee (q \vee r)$	$\vee I, 6$
8	$\quad p \vee (q \vee r)$	$\vee E, 2$
9	$\quad r$	*acp*
10	$\quad q \vee r$	$\vee I, 9$
11	$\quad p \vee (q \vee r)$	$\vee I, 10$
12	$p \vee (q \vee r)$	$\vee E, 1$

1	$p \wedge (q \vee r)$	ass
2	$\quad p$	$\wedge E1$
3	$\quad q \vee r$	$\wedge E1$
4	$\quad\quad q$	acp
5	$\quad\quad p \wedge q$	$\wedge E2, 4$
6	$\quad\quad (p \wedge q) \vee (p \wedge r)$	$\vee I5$
7	$\quad\quad r$	acp
8	$\quad\quad p \wedge r$	$I\wedge\ 2,7$
9	$\quad\quad (p \wedge q) \vee (p \wedge r)$	$\vee I8$
10	$\quad (p \wedge q) \vee (p \wedge r)$	$\vee I8$
11	$(p \wedge q) \vee (p \wedge r)$	$\vee E3$

1	$(p \to q) \wedge (r \to s)$	ass
2	$(p \to q)$	$\wedge E1$
3	$(r \to s)$	$\wedge E1$
4	$\quad p \vee r$	acp
5	$\quad\quad p$	acp
6	$\quad\quad q$	$\to E2, 5$
7	$\quad\quad q \vee s$	$\vee I, 6$
8	$\quad\quad r$	acp
9	$\quad\quad s$	$\to E3, 8$
10	$\quad\quad q \vee s$	$\vee I9$
11	$\quad q \vee s$	$\vee E, 4$
12	$(p \vee r) \to (q \vee s)$	$\to I$

definition 1.2.2. $\bot$ *:* $\varphi \wedge \neg\varphi$
definition 1.2.3. $\top$ *:* $\varphi \vee \neg\varphi$

$\quad\vdots$ $\quad\bot$ $\therefore \neg\varphi \qquad \neg I$	$\quad\vdots$ $\quad\bot$ $\therefore \varphi \qquad \neg E$

definition 1.2.4. ***schema of prove*** *you've got knowed by now what is schema of a rule. if you want to prove a theorem that it's schema form.you can't prove it directly. here we use a technique calls* ***"schema of prove"****. it let us prove for all situations at the same time.*
Theorem 1.2.1. $(schema)$ $(\bot I, \bot E)$

φ $\neg\varphi$ $\therefore \bot \qquad \bot I$	$\bot$ $\therefore \varphi \qquad \bot E$

1	φ	ass
2	$\neg\varphi$	ass
3	$\varphi \wedge \neg\varphi$	$\wedge I$
4	$\bot$	$def.3$

1	$\bot$	ass
2	$\quad\neg\varphi$	acp
3	$\quad\vdots$	
4	$\quad\bot$	$rp1$
5	φ	$\neg E$

Attention 1.2.1. *Intuitionists can skip the "¬E" rule that is the "RAA" rule in old notation*

and use "$\bot E$" as a rule instead.

Example 1.2.4. *for "p, q, r" we can prove these.*

1. $p \to q, \neg q / \therefore \neg p$ *(Modus Tollens)*

2. $p \to q / \therefore \neg p \vee q$ *(Implication–Disjunction Equivalence)*

1	$p \to q$	ass
2	$\neg q$	ass
3	$\quad p$	acp
4	$\quad q$	$\to E, 1, 3$
5	$\quad \bot$	$\bot I, 2, 4$
6	$\neg p$	$\neg I$

1	$p \to q$	ass
2	$\quad \neg(\neg p \vee q)$	acp
3	$\quad\quad p$	acp
4	$\quad\quad q$	$\to E, 1, 3$
5	$\quad\quad \neg p \vee q$	$\vee I, 4$
6	$\quad\quad \bot$	$\bot I, 2, 5$
7	$\quad \neg p$	$\neg I$
8	$\quad \neg p \vee q$	$\vee I, 7$
9	$\quad \bot$	$\bot I, 2, 8$
10	$\neg p \vee q$	$\neg I$

definition 1.2.5. $\varphi \leftrightarrow \psi$ *:* $(\varphi \to \psi) \wedge (\psi \to \varphi)$

Theorem 1.2.2. $(schema)$ $(\leftrightarrow I, \leftrightarrow E)$

$\quad\vert\ \varphi$ $\quad\vert\ \vdots$ $\quad\vert\ \psi$ $\quad\vert\ \psi$ $\quad\vert\ \vdots$ $\quad\vert\ \varphi$ $\therefore \varphi \leftrightarrow \psi \qquad \leftrightarrow I$	$\dfrac{\varphi \leftrightarrow \psi \quad \varphi}{\therefore \psi} \leftrightarrow E$ $\dfrac{\varphi \leftrightarrow \psi \quad \psi}{\therefore \varphi} \leftrightarrow E$

1	φ	acp
	$\vdots$	
n	ψ	
$n+1$	$\varphi \to \psi$	$\to, I$
$n+2$	ψ	acp
	$\vdots$	
m	φ	
$m+1$	$\psi \to \varphi$	$\to, I$
$m+2$	$(\varphi \to \psi) \wedge (\psi \to \varphi)$	$\wedge I$
$m+3$	$\varphi \leftrightarrow \psi$	$def., m+3$

1	$\varphi \leftrightarrow \psi$	ass
2	φ	ass
3	$(\varphi \to \psi) \wedge (\psi \to \varphi)$	$def.1$
4	$\varphi \to \psi$	$\wedge E3$
5	ψ	$\to E1, 4$

Exercise 1.2.1. *for "p, q, r" prove.*

1. $/ \therefore p \vee \neg p$ *(Law of Excluded Middle)*
2. $\neg p, p \vee q / \therefore q$ *(Disjunctive Syllogism)*
3. $\neg p / \therefore p \to q$ *(Vacuous Implication)*
4. $p \to q \therefore / \therefore \neg q \to \neg p$ *(Contraposition)*
5. $(p \vee q) \wedge (p \vee r) \therefore / \therefore (p \vee (q \wedge r))$ *(Distribution and Factorization)*
6. $\neg p \wedge \neg q \therefore / \therefore \neg(p \vee q)$ *(De morgan)*
7. $\neg p \vee \neg q \therefore / \therefore \neg(p \wedge q)$ *(De morgan)*
8. $(\neg p \to q) / \therefore ((p \to q) \to q)$
9. $/ \therefore (p \wedge q \to r) \wedge (p \wedge s \to r) \to (p \wedge (q \vee s) \to r)$
10. $/ \therefore (p \wedge q \to r) \vee (p \wedge s \to r) \to (p \wedge (q \wedge s) \to r)$
11. $/ \therefore (p \to (q \to r)) \to ((p \to q) \to (p \to r))$
12. $/ \therefore (p \to q) \to ((q \to r) \to (p \to r))$

Attention 1.2.2. *we can rewrite **Example** and **Exercise** propositions and proofs with proposition schemas and schema of proves.so they are all rules.*

This page intentionally left blank.

Chapter 2

first-order logic

2.1 what is formula?

obj are two types.we know them and we know what exactly are we talking about and unknowns are mystery.all the obj we have used until now were known ones we call them **"constants"**.
Undefined 2.1.1. ***variables*** *A variable is a symbol (usually a letter like $x, y, or z$) that represents an unknown obj.*
definition 2.1.1. ***Atomic formula*** *we define atomic formulas like this.*

1. *if "x" is an obj(constant or variable) then "$Set(x)$" is an atomic formula.*
2. *if "x, y" are obj(constant or variable) then "$x = y$" is an atomic formula.*
3. *if "x, y" are obj(constant or variable) then "$x \in y$" is an atomic formula.*

we denote atomic formulas with "A_i".
definition 2.1.2. ***well formed formula(WWF)*** *these are formulas.*

1. *atomic formulas (A_i).*
2. *if A is a formula then $\neg A$ is a formula too.*
3. *if A, B are two formulas then $A \bigcirc B$ is a formula too.($\bigcirc$ can be $\rightarrow, \wedge, \vee$).*

for example $(A_0 \rightarrow A_1, \neg A_3)$
we easily call WWF "formula" and also we denote WWF with $A, B, C, ...$
definition 2.1.3. ***open formula*** *formulas that variables used in one of its atomic formulas.*

the formulas that are not open call **"closed formulas"** or **"sentence"**. Now question is "are very closed formulas proposition?" (propositions in classic logic are sentences that are true or false?) the answer is unfortunately **No**.
definition 2.1.4. ***Substitution for obj*** *we define Substitution an obj with variable like this.*

1. *for variables if $x = y$ then $y[a/x] := a$ and if $x \neq y$ then $y[a/x] := y$.*
2. *for constants $c[a/x] := c$.*

definition 2.1.5. ***Substitution for formulas*** *we define Substitution for a formula like this.*

1. *for atomic formulas we define it like this. if it's a single object we do it as we explain in "**Substitution for obj**". if it's like $x = y$ or $x \in y$ (x, y can be both constants and variables). we define it like this.*

$$Set(y)[a/x] := Set(y[a/x])$$
$$(x = y)[a/z] := (x[a/z] = y[a/z])$$
$$(x \in y)[a/z] := (x[a/z] \in y[a/z])$$

2. *if formula A is complex we define it like this.*

$$(B \bigcirc C)[a/x] := (B[a/x] \bigcirc C[a/x])$$

($\bigcirc$ can be $\rightarrow, \wedge, \vee$).

$$(\neg B)[a/x] := (\neg B[a/x])$$

simply means in a formula put an obj like a instead of a variable like x.we denote it by $A[a/x]$ or $A[\frac{a}{x}]$

definition 2.1.6. ***Open proposition*** *assume formula "A" with variables "$x_0, x_1, .., x_n$" in it and a set "U". we say formula "A"is a "open proposition in U" when for every obj "$a_0, a_1, .., a_n$" in "U"*

$$A([a_0/x_0][a_1/x_1]...[a_n/x_n])$$

is a proposition we call it open proposition and denote it with "$(P(x), Q(x, y), ..)$"

definition 2.1.7. ***Big "AND" and "OR"*** *assume a set like "$\{a_0, a_1, a_2, ..., a_n\}$" we define **big "AND"** like this.*

$$\bigwedge_{i=0}^{n} P(a_i) : P(a_0) \wedge P(a_1) \wedge P(a_2) \wedge ... \wedge P(a_n)$$

*and **big "OR"***

$$\bigvee_{i=0}^{n} P(a_i) : P(a_0) \vee P(a_1) \vee P(a_2) \vee ... \vee P(a_n)$$

definition 2.1.8. ***Universe of discourse*** *we specify a set like "$\{a_0, a_1, a_2, ..., a_n\}$" as universe of a Logic world.*

Rule 2.1.1. ***use of open propositions*** *assume universe of discourse is "$\{a_0, a_1, a_2, ..., a_n\}$"*

$c \quad P(x)$

means :

$c_0 \quad P[a_0/x]$

$c_1 \quad P[a_1/x]$

$c_2 \quad P[a_2/x]$

$\vdots$

$c_n \quad P[a_n/x]$

if Px was been assemed then like this

$$\begin{array}{l|l} c & P(x) \\ & \vdots \end{array}$$

means :

$$\begin{array}{l|l} c_0 & P[a_0/x] \\ & \vdots \\ c_1 & P[a_1/x] \\ & \vdots \\ c_2 & P[a_2/x] \\ & \vdots \\ c_n & P[a_n/x] \\ & \vdots \end{array}$$

definition 2.1.9. ***predicate schema(variable predicate)*** *a predicate that is unknown for us and it works as variable.(we show them with $\phi, \psi, \theta, ...$)*
definition 2.1.10. ***Universal quantification*** *assume universe of discourse is "$\{a_0, a_1, a_2, ..., a_n\}$" we define "$\forall x P(x)$"*

$$\forall x \phi(x) : \bigwedge_{i=0}^{n} \phi[a_i/x]$$

definition 2.1.11. ***Existential quantification*** *assume universe of discourse is "$\{a_0, a_1, a_2, ..., a_n\}$" we define "$\exists x P x$"*

$$\exists x \phi(x) : \bigvee_{i=0}^{n} \phi[a_i/x]$$

as you can see "$\forall x \phi(x)$" and "$\exists x \phi(x)$" are sentences.
definition 2.1.12. ***Free variables*** *variables that are not bound by a quantifier. (you can easily see that every formula with no free variables are sentences) (previously we sayed that every sentence is not necessary a proposition)*

2.2 Rules of first order logic.

Theorem 2.2.1. ******(schema)******** $(\forall I, \forall E, \exists I, \exists E)$

Assume that the universe of discourse is

$$D = \{a_0, a_1, a_2, \ldots, a_n\}.$$

Rules	*Terms and Conditions*
$\dfrac{\phi(x)}{\therefore \forall x\, \phi(x) \qquad \forall I}$	*1. x is a variable.* *2. x is arbitrary.* *3. x does not occur free in any assumption on which $\phi(x)$ depends.*
$\dfrac{\forall x\, \phi(x)}{\therefore \phi[a/x] \qquad \forall E}$	*1. a is an object in the universe of discourse.* *2. a is free for x in $\phi(x)$.*
$\dfrac{\phi[a/x]}{\therefore \exists x\, \phi(x) \qquad \exists I}$	*1. a is an object in the universe of discourse.* *2. a is free for x in $\phi(x)$.*
$\exists x\, \phi(x)$ $\quad \vert\ \phi(x)$ $\quad \vert\ \vdots$ $\quad \vert\ \theta$ $\therefore \theta \qquad \exists E$	*1. x does not occur free in any assumption on which θ depends.* *2. x does not occur free in θ.* *3. $\phi(x)$ is assumed only inside the subproof.*

Proof.

1	$\phi(x)$	ass
2	$\phi[a_0/x]$	$def.1$
	$\vdots$	
n	$\phi[a_n/x]$	$def.1$
$n+1$	$\bigwedge_{i=0}^{n} \phi[a_i/x]$	$\wedge I, 2, 3, ..., n+2$
$n+2$	$\forall x \phi(x)$	$def.n+3$

1	$\forall x \phi(x)$	ass
2	$\bigwedge_{i=0}^{n} \phi[a_i/x]$	$def.1$
3	$\phi[a/x]$	$\wedge E2$

1	$\phi[a/x]$	
2	$\bigvee_{i=0}^{n} \phi[a_i/x]$	$\vee I1$
3	$\exists x\phi(x)$	$def2$

1	$\exists x\phi(x)$		
2	$\bigvee_{i=0}^{n} \phi[a_i/x]$		
3	|	$\phi[a_0/x]$	acp
	|	$\vdots$	
n	|	θ	
$n+1$	|	$\phi[a_1/x]$	acp
	|	$\vdots$	
m	|	θ	
	$\vdots$		
l	|	$\phi[a_2/x]$	acp
	|	$\vdots$	
r	|	θ	
$r+1$	θ		$\exists E, 3$

□

Rule 2.2.1. $(schema)$ $(= I, = E)$

$\therefore a = a \qquad = I$	$\dfrac{a = b \quad \phi(a)}{\therefore \phi(b)} = E$ $\qquad$ $\dfrac{a = b \quad \phi(a)}{\therefore \phi(a)} = E$

definition 2.2.1. ***Uniqueness*** *we define "*$\exists! x\phi(x)$*"*

$$\exists! x\phi(x) : \exists x\phi(x) \wedge \forall x_0, x_1(\phi(x_0) \wedge \phi(x_1) \to x_0 = x_1)$$

Example 2.2.1. *for P,Q,R we can prove these.Proof of last ones Is Left As An Exercise To The Reader.*😀

1. $\forall x(P(x) \wedge Q(x)) \therefore / \therefore \forall x P(x) \wedge \forall x Q(x)$ *(Distribution and Factorization of* $\forall\wedge$*)*
2. $\exists x(P(x) \vee Q(x)) \therefore / \therefore \exists x P(x) \vee \exists x Q(x)$ *(Distribution and Factorization of* $\exists\vee$*)*
3. $\forall x P(x) \vee \forall x Q(x) / \therefore \forall x(P(x) \vee Q(x))$ *(Factorization of* $\forall\vee$*)*
4. $\exists x(P(x) \wedge Q(x)) / \therefore \exists x P(x) \wedge \exists x Q(x)$ *(Distribution of* $\exists\wedge$*)*
5. $\forall x(P(x) \to Q(x)) / \therefore \forall x P(x) \to \forall x Q(x)$ *(Distribution of* $\forall \to$*)*
6. $\exists x P(x) \to \exists x Q(x) / \therefore \exists x(P(x) \to Q(x))$ *(Factorization of* $\exists \to$*)*

1	$\forall x(P(x) \wedge Q(x))$	ass
2	$P(x) \wedge Q(x)$	acp
3	$P(x)$	$\wedge E$
4	$Q(x)$	$\wedge E$
5	$\forall x P(x)$	$\forall I$
6	$\forall x Q(x)$	$\forall I$
7	$\forall x P(x) \wedge \forall x Q(x)$	$\wedge I$

1	$\forall x P(x) \wedge \forall x Q(x)$	ass
2	$\forall x P(x)$	$\wedge E$
3	$\forall x Q(x)$	$\wedge E$
4	$P(x)$	$\forall E$
5	$Q(x)$	$\forall E$
6	$P(x) \wedge Q(x)$	$\wedge I$
7	$\forall x(P(x) \wedge Q(x))$	$\forall I$

1	$\exists x(P(x) \vee Q(x))$	ass
2	$\quad P(x) \vee Q(x)$	acp
3	$\quad\quad P(x)$	acp
4	$\quad\quad \exists x P(x)$	$\exists I$
5	$\quad\quad \exists x P(x) \vee \exists x Q(x)$	$\vee I$
6	$\quad\quad Q(x)$	acp
7	$\quad\quad \exists x Q(x)$	$\exists I$
8	$\quad\quad \exists x P(x) \vee \exists x Q(x)$	$\vee I$
9	$\quad \exists x P(x) \vee \exists x Q(x)$	$\vee E$
10	$\exists x P(x) \vee \exists x Q(x)$	$\exists E$

1	$\exists x P(x) \vee \exists x Q(x)$	ass
2	$\quad \exists x P(x)$	acp
3	$\quad\quad P(x)$	acp
4	$\quad\quad P(x) \vee Q(x)$	$\vee I$
5	$\quad\quad \exists x(P(x) \vee Q(x))$	$\exists I$
6	$\quad \exists x(P(x) \vee Q(x))$	$\exists E$
7	$\quad \exists x Q(x)$	acp
8	$\quad\quad Q(x)$	acp
9	$\quad\quad P(x) \vee Q(x)$	$\vee I$
10	$\quad\quad \exists x(P(x) \vee Q(x))$	$\exists I$
11	$\quad \exists x(P(x) \vee Q(x))$	$\exists E$
12	$\exists x(P(x) \vee Q(x))$	$\vee E$

1	$\forall x P(x) \vee \forall x Q(x)$	ass
2	$\quad \forall x P(x)$	acp
3	$\quad\quad P(x)$	$\forall E$
4	$\quad\quad P(x) \vee Q(x)$	$\vee I$
5	$\quad\quad \forall x(P(x) \vee Q(x))$	$\forall I$
6	$\quad \forall x Q(x)$	acp
7	$\quad\quad Q(x)$	$\forall E$
8	$\quad\quad P(x) \vee Q(x)$	$\vee I$
9	$\quad\quad \forall x(P(x) \vee Q(x))$	$\forall I$
10	$\forall x(P(x) \vee Q(x))$	$\vee E$

1	$\exists x(P(x) \wedge Q(x))$	ass
2	$\quad P(x) \wedge Q(x)$	acp
3	$\quad\quad P(x)$	$\wedge E$
4	$\quad\quad Q(x)$	$\wedge E$
5	$\quad\quad \exists x P(x)$	$\exists I$
6	$\quad\quad \exists x Q(x)$	$\exists I$
7	$\exists x P(x)$	$\exists E$
8	$\exists x Q(x)$	$\exists E$
9	$\exists x P(x) \wedge \exists x Q(x)$	$\wedge I$

1	$\forall x(P(x) \to Q(x))$	ass
2	$\quad \forall x P(x)$	acp
3	$\quad P(x)$	$\forall E$
4	$\quad P(x) \to Q(x)$	$\forall E$
5	$\quad Q(x)$	$\to E$
6	$\quad \forall x Q(x)$	$\forall I$
7	$\forall x P(x) \to \forall x Q(x)$	$\to I$

1	$\exists x P(x) \to \exists x Q(x)$	$\to I$
2	$\exists x P(x) \vee \neg(\exists x P(x))$	$ex.$
3	$\quad \exists x P(x)$	acp
4	$\quad \exists Q(x)$	$\to I$
5	$\quad\quad Q(x)$	acp
6	$\quad\quad P(x) \to Q(x)$	$ex.$
7	$\quad\quad \exists x(P(x) \to Q(x))$	$\exists I$
8	$\quad \exists x(P(x) \to Q(x))$	$\exists E$
9	$\quad \neg(\exists x P(x))$	acp
10	$\quad \forall x \neg P(x)$	$De\ morgan$
11	$\quad \neg P(x)$	$\forall E$
12	$\quad P(x) \to Q(x)$	$ex.$
13	$\quad \exists x(P(x) \to Q(x))$	$\exists I$
14	$\exists x(P(x) \to Q(x))$	$\vee E$

Exercise 2.2.1. *for P,Q,R prove.*

1. $\forall x P(x) \wedge Q \therefore / \therefore \forall x(P(x) \wedge Q)$ *(Distribution and Factorization of $\forall\wedge$)*
2. $\forall x P(x) \vee Q \therefore / \therefore \forall x(P(x) \vee Q)$ *(Distribution and Factorization of $\forall\vee$)*
3. $\exists x P(x) \wedge Q \therefore / \therefore \exists x(P(x) \wedge Q)$ *(Distribution and Factorization of $\exists\wedge$)*
4. $\exists x P(x) \vee Q \therefore / \therefore \exists x(P(x) \vee Q)$ *(Distribution and Factorization of $\exists\vee$)*
5. $Q \to \forall x P(x) \therefore / \therefore \forall x(Q \to P(x))$ *(Distribution and Factorization of $\forall \to$)*
6. $Q \to \exists x P(x) \therefore / \therefore \exists x(Q \to P(x))$ *(Distribution and Factorization of $\exists \to$)*
7. $\forall x P(x) \to Q \therefore / \therefore \exists x(P(x) \to Q)$ *(Factorization of $\forall \to$)*
8. $\exists x P(x) \to Q \therefore / \therefore \forall x(P(x) \to Q)$ *(Distribution of $\exists \to$)*
9. $\exists x \neg P(x) \therefore / \therefore \neg \forall x P(x)$ *(De morgan)*
10. $\forall x \neg P(x) \therefore / \therefore \neg \exists x P(x)$ *(De morgan)*
11. $\exists x \forall y P(x,y) / \therefore \forall x \exists y P(x,y)$ *(Quantifier Exchange Fallacy)*
12. $\exists x\Big(P(x) \wedge \exists y Q(y) \to \forall z R(z)\Big) / \therefore \forall y \forall z\Big(\forall x P(x) \wedge Q(y) \to R(z)\Big)$
13. $\forall x \exists y\Big(P(x) \wedge Q(y) \to \forall z R(z)\Big) / \therefore \forall z\Big(\exists x P(x) \wedge \forall y Q(y) \to R(z)\Big)$
14. $\forall x \forall z \exists y \exists w\Big(P(y,z) \vee Q(z) \to R(x,w)\Big) / \therefore \Big(\exists z \forall y P(y,z) \vee \exists z Q(z)\Big) \to \forall x \exists w R(x,w)$
15. $\exists x P(x) \vee \exists x\Big(Q(x) \wedge R(x)\Big) / \therefore \exists x\Big(P(x) \vee Q(x)\Big) \wedge \exists x\Big(P(x) \vee R(x)\Big)$

16. $\exists x \exists z \forall y \Big(P(x,y) \to Q(z) \vee R(x,z) \Big) / \therefore \forall x \exists y P(x,y) \to \Big(\exists z Q(z) \vee \exists x \exists z R(x,z) \Big)$

17. $\forall x \exists y \Big(\exists z P(x,y,z) \wedge Q(x,y) \Big) \vee \forall x \exists y \exists z \Big(P(x,y,z) \wedge R(x,y) \Big) / \therefore \forall x \exists y \exists z \Big(P(x,y,z) \wedge (Q(x,y) \vee R(x,y)) \Big)$

Attention 2.2.1. *we can rewrite* ***Example*** *and* ***Exercise*** *propositions and proofs with proposition schemas and schema of proves.so they are all rules.*

Chapter 3

The beginning of set theory

In language of set theory if we are talking about "$\forall X(P(X))$" it means "$\forall x(Set(x) \to P(x))$" and when we are talking about "$\exists X(P(X))$" it means "$\exists x(Set(x) \wedge P(x))$". similarly if we are talking about "$\forall x \in A \quad P(x)$" it means "$\forall x(x \in A \to P(x))$" and when we are talking about "$\exists x \in A \quad P(x)$" it means "$\exists x(x \in A \wedge P(x))$".

and we have a predicate "$Type_0(x)$".

$$Type_0(x) : \exists Y \quad x \in Y$$

3.1 Axioms of set theory

we emphasis before that set has no definition.but we can describe what set is with rules and axioms.

Axiom 3.1.1. ***Existence of universe of set theory*** *There exists a set V_1 (we denote it as V_1.it simply means the set of all sets), known as the Von Neumann Universe, which contains every set. (this axiom can't be formolized in first order logic). (there must be another axiom let us use logic for set theory but we ignore that). (from now on when we are talking about proposition "ϕ" in "set theory" we mean proposition "ϕ" in "V_1").*

definition 3.1.1. ***Subset*** *a subset is a set where every element is also an element of another, larger set called the superset*

$$A \subseteq B : \forall x(x \in A \to x \in B)$$

or i.e.

Theorem 3.1.1. (*schema*) $(\subseteq I, \subseteq E)$

$\begin{array}{l} \vert\ x \in A \\ \vert\ \vdots \\ \vert\ x \in B \\ \therefore A \subseteq B \qquad \subseteq I \end{array}$	$\begin{array}{l} A \subseteq B \\ a \in A \\ \hline \therefore a \in B \qquad \subseteq E \end{array}$

1	$x \in A$	acp
	$\vdots$	
n	$x \in B$	
$n+1$	$x \in A \to x \in B$	$I \to, 1, n$
$n+2$	$\forall x(x \in A \to x \in B)$	$\forall I, n+1$
$n+3$	$A \subseteq B$	$def.n+2$

1	$A \subseteq B$	ass
2	$a \in A$	ass
3	$\forall x(x \in A \to x \in A)$	$def.1$
4	$a \in A \to a \in B$	$\forall E, 4$
5	$a \in B$	$E \to 3, 5$

Example 3.1.1. $A \subseteq B, B \subseteq C / \therefore A \subseteq C$

1	$A \subseteq B$	ass
2	$B \subseteq C$	ass
3	$x \in A$	acp
4	$x \in B$	$\subseteq E, 1, 3$
5	$x \in C$	$\subseteq E, 2, 4$
6	$A \subseteq C$	$\subseteq I, 3, 5$

Axiom 3.1.2. ***Extensionality****Two sets are equal (are the same set) if they have the same elements.*

$$\forall X, Y(\forall x(x \in X \leftrightarrow x \in Y) \to X = Y)$$

so we can rewrite axiom of Extensionality like this.

$$\forall X, Y((X \subseteq Y \wedge Y \subseteq X) \to X = Y)$$

Theorem 3.1.2. $(schema)$ $(= I, = E)$

$x \in A$ $\vdots$ $x \in B$	
$x \in B$ $\vdots$ $x \in A$	$A = B \quad a \in A \;/\; \therefore a \in B \quad = E$ $A = B \quad a \in B \;/\; \therefore a \in A \quad = E$
$\therefore A = B \quad = I$	

1	$x \in A$	acp
	$\vdots$	
n	$x \in B$	
$n+1$	$A \subseteq B$	$\subseteq I, 1, n$
$n+2$	$x \in B$	acp
	$\vdots$	
m	$x \in A$	
$m+1$	$B \subseteq A$	$\subseteq I, n+2, m$
$m+2$	$\forall X, Y((X \subseteq Y \wedge Y \subseteq X) \to X = Y)$	Extensionality
$m+3$	$(A \subseteq B \wedge B \subseteq A) \to A = B$	$\forall E, m+2$
$m+4$	$A \subseteq B \wedge B \subseteq A$	$\wedge I, n+1, m+1$
$m+5$	$A = B$	$E \to, m+3, m+4$

Axiom 3.1.3. ***Weak comprehension*** *(schema) this axioms says for every open proposition like "$\phi(x, x_0, x_1, ..., x_n)$" there is a set like "$X$" such that every "$x$" with "$Type_0$" is a member of that set if and only if it holds "$\phi(x, x_0, x_1, ..., x_n)$".*

$$\forall x_0, x_1, ..., x_n \exists X \forall x (Type_0(x) \wedge \phi(x, x_0, x_1, ..., x_n) \leftrightarrow x \in X)$$

by axiom of extentionality we can show that X is unique. so we denote it as "$\{x \mid \phi(x, x_0, x_1, ..., x_n)\}$". sometimes instead of "$\{x \mid x \in A \wedge \phi(x, x_0, x_1, ..., x_n)\}$" we write "$\{x \in A \mid \phi(x, x_0, x_1, ..., x_n)\}$".

Lemma 3.1.1 (Russell's paradox)**.** *assume the comprehension axiom $(\forall x_0, x_1, ..., x_n \exists X \forall x (\phi(x, x_0, x_1, ..., x_n) \leftrightarrow x \in X))$ and prove inconsistency i.e.*

$$\forall x_0, x_1, ..., x_n \exists X \forall x (\phi(x, x_0, x_1, ..., x_n) \leftrightarrow x \in X) / \therefore \bot$$

1	$\exists X \forall x (x \notin x \leftrightarrow x \in X)$	*comprehension axiom*
2	$\forall x (x \notin x \leftrightarrow x \in R)$	*acp*
3	$R \notin R \leftrightarrow R \in R$	*acp*
4	$R \in R$	*acp*
5	$R \notin R$	$\leftrightarrow E$
6	$\bot$	$\bot I$
7	$R \notin R$	*acp*
8	$R \in R$	$\leftrightarrow E$
9	$\bot$	$\bot I$
10	$\bot$	$\exists E$

Lemma 3.1.2. $(\forall X \forall x(x \in X \wedge Type_0(x) \leftrightarrow x \in X))$

Let's talk about some axioms in set theoty.These daxioms will come in handy later.
Axiom 3.1.4. ***Existence*** *there is a "Set" with no member i.e..*

$$\exists x \quad Type_0(x)$$

definition 3.1.2. ***Empty set*** *also known as the null set, is the unique mathematical set containing no elements at all. (uniqueness can be proven.)*

$$\varnothing := \{x \mid \bot\}$$

therefore.

$$\forall x(x \in \varnothing \leftrightarrow \bot \wedge Type_0(x))$$

therefore.

$$\forall x(x \notin \varnothing)$$

Example 3.1.2. $/ \therefore \forall X(\varnothing \subseteq X)$

1	$x \in \varnothing$	acp
2	$\forall x(x \notin \varnothing)$	def
3	$x \notin \varnothing$	$\forall E, 2$
4	$\bot$	$\bot I, 1, 3$
5	$x \in X$	$\bot E, 4$
6	$\varnothing \subseteq X$	$\subseteq I, 1, 5$
7	$\forall X(\varnothing \subseteq X)$	$\forall I, 6$

Lemma 3.1.3. $A = \varnothing \therefore / \therefore \forall y(y \notin A)$

1	$\forall y(y \notin A)$	ass
2	$y \in A$	acp
3	$y \notin A$	$\forall E, 1$
4	$\bot$	$\bot I, 2, 3$
5	$y \in \varnothing$	$\bot E, 4$
6	$y \in \varnothing$	acp
7	$\forall y(y \notin \varnothing)$	$def.$
8	$y \notin \varnothing$	$\forall E, 7$
9	$\bot$	$\bot I, 6, 8$
10	$y \in A$	$\bot E, 9$
11	$A = \varnothing$	$I =$

1	$A = \varnothing$	*ass*
2	$\forall y(y \notin \varnothing)$	*def.*
3	$\forall y(y \notin A)$	$= E, 1, 2$

Theorem 3.1.3. (*schema*) $(\varnothing \neq I, \varnothing \neq E, \varnothing = I, \varnothing = E)$

$\dfrac{a \in A}{\therefore A \neq \varnothing} \qquad \varnothing \neq I$	$A \neq \varnothing$ $\left\vert \begin{array}{l} x \in A \\ \vdots \\ \theta \end{array} \right.$ $\therefore \theta \qquad \varnothing \neq E$
$\left\vert \begin{array}{l} x \in A \\ \vdots \\ \bot \end{array} \right.$ $\therefore A = \varnothing \qquad \varnothing = I$	$\dfrac{A = \varnothing}{\therefore a \notin A} \qquad \varnothing = E$

1	$a \in A$	*ass*
2	$\exists x(x \in A)$	$\exists I, 1$
3	$\neg(\forall x(x \notin A))$	*De Morgan* 2
4	$A \neq \varnothing$	*lemma* 4.4.1, 3

1	$x \in A$	*acp*
$\vdots$	$\vdots$	
n	$\bot$	
$n+1$	$x \notin A$	$\neg I$
$n+2$	$\forall x(x \notin A)$	$\forall I, n+1$
$n+3$	$A = \varnothing$	*lemma* 4.4.1, 1

1	$A = \varnothing$	*ass*
2	$\forall x(x \notin A)$	*lemma* 4.4.1, 1
3	$a \notin A$	$\forall E2$

definition 3.1.3. ***Pairing set***

$$\{a_0, a_1, a_2, ..., a_n\} := \{x \mid x = a_0 \vee x = a_1 \vee x = a_2 \vee ... \vee x = a_n\}$$

therefore.

$$\forall x(x \in \{a_0, a_1, a_2, ..., a_n\} \leftrightarrow (Type_0(x) \wedge x = a_0 \vee x = a_1 \vee x = a_2 \vee ... \vee x = a_n))$$

Theorem 3.1.4. (*schema*) $(\{\}I, \{\}E)$

$\dfrac{a = a_i}{\therefore a \in \{a_0, a_1, a_2, ..., a_n\}} \quad \{\}I$	$a \in \{a_0, a_1, a_2, ..., a_n\}$ $\vert\ a = a_0$ $\vert\ \vdots$ $\vert\ \theta$ $\vdots$ $\vert\ a = a_n$ $\vert\ \vdots$ $\vert\ \theta$ $\therefore \theta \quad \{\}E$

Proof Is Left As An Exercise To The Reader.😀

Axiom 3.1.5. ***Pairing*** (*schema*)

$$\forall x_0, ..., x_n \quad Type_0(x_0) \wedge ... \wedge Type(X_n) \rightarrow Type_0(\{x_0, x_1, x_2, ..., x_n\})$$

definition 3.1.4. ***Union and Intersection***
Intersection:

$$A \cap B := \{x \mid x \in A \wedge x \in B\}$$

therefore.

$$\forall x(x \in A \cap B \leftrightarrow (x \in A \wedge x \in B))$$

Union:

$$A \cup B := \{x \mid x \in A \vee x \in B\}$$

therefore.0

$$\forall x(x \in A \cup B \leftrightarrow ((x \in A \vee x \in B)))$$

Theorem 3.1.5. (*schema*) $(\cap I, \cap E, \cup I, \cup E)$

$\dfrac{a \in A \quad a \in B}{\therefore a \in A \cap B} \cap I$	$\dfrac{a \in A \cap B}{\therefore a \in A} \cap E \qquad \dfrac{a \in A \cap B}{\therefore a \in B} \cap E$
$\dfrac{a \in A}{\therefore a \in A \cup B} \cup I \qquad \dfrac{a \in A}{\therefore a \in B \cup A} \cup I$	$a \in A \cup B$ $\quad \vert\ a \in A$ $\quad \vert\ \vdots$ $\quad \vert\ \theta$ $\quad \vert\ a \in B$ $\quad \vert\ \vdots$ $\quad \vert\ \theta$ $\therefore \theta \quad \cup E$

Proof Is Left As An Exercise To The Reader.😀

sometimes all members of a set is also set we call it "set of sets".

definition 3.1.5. ***Big Union and Intersection*** *if "A" is "set of sets" and "$A \neq \varnothing$" we define Big Intersection:*

$$\bigcap A := \{x \mid \forall X(X \in A \to x \in X)\}$$

i.e.

$$\bigcap A := \{x \mid \forall X \in A \quad x \in X\}$$

therefore.

$$\forall x\Big(x \in \bigcap A \leftrightarrow (Type_0(x) \wedge \forall X(X \in A \to x \in X))\Big)$$

Big Union:

$$\bigcup A := \{x \mid \exists X(X \in A \wedge x \in X)\}$$

i.e.

$$\bigcup A := \{x \mid \exists X \in A \quad x \in X\}$$

therefore.

$$\forall x\Big(x \in \bigcup A \leftrightarrow (Type_0(x) \wedge \exists X(X \in A \wedge x \in X))\Big)$$

Theorem 3.1.6. *(schema)* $(\bigcap I, \bigcap E, \bigcup I, \bigcup E)$

$\begin{array}{l} \vert\ X \in A \\ \vert\ \vdots \\ \vert\ a \in X \\ \therefore a \in \bigcap A \qquad \bigcap I \end{array}$	$\begin{array}{l} a \in \bigcap A \\ \underline{B \in A} \\ \therefore a \in B \qquad \bigcap E \end{array}$
$\begin{array}{l} a \in B \\ \underline{B \in A} \\ \therefore a \in \bigcup A \qquad \bigcup I \end{array}$	$\begin{array}{l} a \in \bigcup A \\ \quad \vert\ a \in X \\ \quad \vert\ X \in A \\ \quad \vert\ \vdots \\ \quad \vert\ \theta \\ \therefore \theta \qquad \bigcup E \end{array}$

Proof Is Left As An Exercise To The Reader.
definition 3.1.6. ***Complement***

$$A^c := \{x \mid x \notin A\}$$

therefore.

$$\forall x(x \in A^c \leftrightarrow Type_0(x) \wedge x \notin A)$$

Theorem 3.1.7. *(schema)* $(()^cI, ()^cE)$

$\begin{array}{l} \vert\ a \in A \\ \vert\ \vdots \\ \vert\ \bot \\ \therefore a \in A^c \qquad ()^cI \end{array}$	$\begin{array}{l} \vert\ a \in A^c \\ \vert\ \vdots \\ \vert\ \bot \\ \therefore a \in A \qquad ()^cE \end{array}$

Proof Is Left As An Exercise To The Reader.
Observation 3.1.1. *we can show Universal set with...*

$$V = \{x \mid \top\}$$

therefore.

$$\forall x(x \in V \leftrightarrow Type_0(x) \wedge \top)$$

therefore.

$$\forall x(x \in V)$$

definition 3.1.7. ***Subtraction***

$$A - B := \{x \mid x \in A \wedge x \notin B\}$$

therefore.

$$\forall x\Big(x \in A - B \leftrightarrow (x \in A \wedge x \notin B)\Big)$$

Theorem 3.1.8. (*schema*) $(-I, -E)$

$\dfrac{\begin{array}{l} a \in A \\ a \notin B \end{array}}{\therefore a \in A - B} \quad -I$	$\dfrac{a \in A - B}{\therefore a \in A} \quad -E \qquad \dfrac{a \in A - B}{\therefore a \notin B} \quad -E$

Proof Is Left As An Exercise To The Reader.
Lemma 3.1.4. $/\therefore A - B = A \cap B^c$

Proof Is Left As An Exercise To The Reader.
Axiom 3.1.6. ***Separation***

$$\forall X, Y \quad Type_0(X) \to Type_0(X \cap Y)$$

therefore.

$$Type_0(A) \to Type_0(A \cap B)$$

therefore.

$$Type_0(\bigcap A)$$

therefore.

$$Type_0(A) \to Type_0(A - B)$$

Lemma 3.1.5.

$$Type_0(\varnothing)$$

Axiom 3.1.7. ***union***

$$\forall X \quad Type_0(X) \to Type_0(\bigcup X)$$

therefore.

$$Type_0(A) \wedge Type_0(B) \to Type_0(A \cup B)$$

definition 3.1.8. ***Powerset***

$$\mathcal{P}(A) := \{X \mid X \subseteq A\}$$

therefore.

$$\forall X \Big(X \in \mathcal{P}(A) \leftrightarrow (Type_0(X) \wedge X \subseteq A) \Big)$$

Theorem 3.1.9. (*schema*) $(\mathcal{P}I, \mathcal{P}E)$

$\dfrac{A \subseteq B}{\therefore A \in \mathcal{P}(B)} \quad \mathcal{P}I$	$\dfrac{A \in \mathcal{P}(B)}{\therefore A \subseteq B} \quad \mathcal{P}E$

Proof Is Left As An Exercise To The Reader.
Lemma 3.1.6. *prove.*

1. $A \in \mathcal{P}(A)$.
2. $\varnothing \in \mathcal{P}(A)$.

3. $\mathcal{P}(\varnothing) = \{\varnothing\}$.

Axiom 3.1.8. ***Powerset***

$$\forall X \quad Type_0(X) \to Type_0(P(X))$$

Exercise 3.1.1. *for A,B,C prove.*

1. $A \cap B = B \cap A$ *(Commutativity of Intersection)*
2. $A \cup B = B \cup A$ *(Commutativity of Union)*
3. $(A \cap B) \cap C = A \cap (B \cap C)$ *(Associativity of Intersection)*
4. $(A \cup B) \cup C = A \cup (B \cup C)$ *(Associativity of Union)*
5. $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ *(Distributivity of Intersection over Union)*
6. $(A \cup B) \cap (A \cup C) = A \cup (B \cap C)$ *(Distributivity of Union over Intersection)*
7. $B \cap \bigcup A = \bigcup\{B \cap X \mid X \in A\}$ *(Distributivity of Intersection over Union)*
8. $B \cup \bigcap A = \bigcap\{B \cup X \mid X \in A\}$. *(Distributivity of Union over Intersection)*
9. $A \cup A^c = V$ *(Law of Excluded Middle for sets)*
10. $A \cap A^c = \varnothing$ *(The Law of Contradiction for sets)*
11. $A^c \cap B^c = (A \cup B)^c$ *(De Morgan)*
12. $A^c \cup B^c = (A \cap B)^c$ *(De Morgan)*
13. $(\bigcup A)^c = \bigcap\{X^c \mid X \in A\}$ *(De Morgan)*
14. $(\bigcap A)^c = \bigcup\{X^c \mid X \in A\}$. *(De Morgan)*
15. $\bigcap A \subseteq \bigcup A$.
16. $A - (B \cup C) = (A - B) \cap (A - C)$ *(De Morgan)*
17. $A - (B \cap C) = (A - B) \cup (A - C)$ *(De Morgan)*
18. $(A \cup B) - C = (A - C) \cup (B - C)$.
19. $(A \cap B) - C = (A - C) \cap (B - C)$.
20. $A - (B - A) = A$.
21. $A - (A - B) = A \cap B$.
22. $A - B = \varnothing \therefore / \therefore A \subseteq B$.
23. $A - B = A \therefore / \therefore A \cap B = \varnothing$.
24. $A - B = B - A \therefore / \therefore A = B$.
25. $\mathcal{P}(A) \cap \mathcal{P}(B) = \mathcal{P}(A \cap B)$.
26. $\mathcal{P}(A) \cup \mathcal{P}(B) \subseteq \mathcal{P}(A \cup B)$.
27. $\mathcal{P}(A) \subseteq \mathcal{P}(B) \therefore / \therefore A \subseteq B$.
28. $\mathcal{P}(A - B) = \{X \in \mathcal{P}(A) \mid X \cap B = \varnothing\}$.

29. $\mathcal{P}(A) - \mathcal{P}(B) = \{X \mid X \subseteq A \wedge X \nsubseteq B\}$.

from now on we don't get in to all details of proofs because it will be too long and hard to read. so we will take it easy from now and don't write all details like reasons of lines and all lines we left it to reader to understand what happened there.

3.2 Another set theory

In this section we want to construct a universe of sets beginning with some objects that are not sets. These objects are called **urelements**. We then construct sets from these urelements and continue this process through the ordinals.

Axiom 3.2.1. ***Extensionality****Two sets are equal (are the same set) if they have the same elements.*

$$\forall X, Y(\forall x(x \in X \leftrightarrow x \in Y) \rightarrow X = Y)$$

Axiom 3.2.2. ***Existence of universe of set theory*** *There exists a set* V_1 *(we denote it as* V_1*.it simply means the set of all sets), known as the Von Neumann Universe, which contains every set. (this axiom can't be formolized in first order logic). (there must be another axiom let us use logic for set theory but we ignore that). (from now on when we are talking about proposition "*ϕ*" in "set theory" we mean proposition "*ϕ*" in "*V_1*").*

Axiom 3.2.3 (Existence of set of obj)**.** *There exists a nonempty set* U *whose elements are obj.*

definition 3.2.1. ***The first universe***

We define

$$V_0 = U.$$

Thus, V_0 *contains the urelements from which we will construct sets.*

Axiom 3.2.4 (Extending set of obj to set theory universe)**.** ***Bounded comprehension*** (schema) *assume you have set of obj* U *you can extend it to a set theory universe. we call it* V*. For every formula* $\phi(x, x_0, \ldots, x_n)$ *and every existing set of obj* U*, there exists a set* X *such that*

$$\forall x_0, ..., x_n \quad \exists X \quad \forall x \quad \big(x \in X \leftrightarrow (x \in U \wedge \phi(x, x_0, \ldots, x_n))\big).$$

By this axiom, there is a universe we call V_1. we can form sets whose elements are selected from an already existing set(V_0).

definition 3.2.2. ***The next universe(Level)***

We define V_2 *the extension universe of* V_1 *and so on.*

3.3 combination of two theories and height of the Level

We now combine the two theories. We begin with a set V_0 whose elements satisfies $Type_0(x)$. We then extend V_0 to V_1 by adding all sets that can be formed from V_0 using the axioms of comprehension. We then extend V_1 to V_2 in the same way, and so on.

definition 3.3.1. ***Type***

$$Type_0(x) : x \in V_0$$

$$Type_1(x) : x \in V_1$$

Axiom 3.3.1 (Union)**.** *For every $X \in V_0$, the union $\bigcup X$ is a set, and we add it to V_1:*

$$\forall X \in V_1 \quad Type_1\Big(\bigcup X\Big).$$

Axiom 3.3.2 (Powerset)**.** *For every $X \in V_0$, the powerset $\mathcal{P}(X)$ is a set, and we add it to V_1:*

$$\forall X \in V_1 \quad Type_1\big(\mathcal{P}(X)\big).$$

Axiom 3.3.3 (Pairing and separation)**.** *For every $a, b \in V_1$, the pair $\{a, b\}$ is a set in V_1. More generally, for every formula $\phi(x, x_0, \ldots, x_n)$ and every $x_0, \ldots, x_n \in V_1$, the set*

$$\{x \in V_1 \mid \phi(x, x_0, \ldots, x_n)\}$$

is a set in V_1.

definition 3.3.2. ***The next level*** *We define V_1 to be the smallest set containing V_0 and closed under the operations of union, powerset, pairing, and comprehension over V_0. That is,*

$$V_1 := V_0 \cup \Big\{\bigcup X \mid X \in V_0\Big\} \cup \Big\{\mathcal{P}(X) \mid X \in V_0\Big\} \cup \Big\{\{a, b\} \mid a, b \in V_0\Big\} \cup \cdots$$

We then define V_2 from V_1 in exactly the same way, and in general V_{n+1} from V_n.

definition 3.3.3. ***Height of the Level*** *The* height *of V_1 is the least ordinal κ such that*

$$V_1 = V_\kappa,$$

where the cumulative hierarchy V_α is defined by

$$V_0 = V_0, \qquad V_{\alpha+1} = V_\alpha \cup \mathcal{P}(V_\alpha), \qquad V_\lambda = \bigcup_{\beta<\lambda} V_\beta \quad (\lambda \text{ a limit ordinal}).$$

We denote this height by κ_1. Thus V_1 is itself a set, and its height κ_1 is an ordinal.

Observation 3.3.1. *With this definition, V_1 is a set of height κ_1. The universe of the first theory is recovered by taking the union of all levels V_n, which is V_κ for $\kappa = \sup_n \kappa_n$.*

Example 3.3.1. *If V_0 contains only the empty set, then V_1 is the set of hereditarily finite sets, and its height is ω.*

Theorem 3.3.1. (*schema*) *For every $n \in \omega$,*

$$V_n \subseteq V_{n+1} \subseteq V_1.$$

Proof Is Left As An Exercise To The Reader.😀

Chapter 4

Relations and Functions

4.1 Ordered pair and Cartesian product

definition 4.1.1. ***Ordered pair***

$$(x, y) := \{\{x\}, \{x, y\}\}.$$

so $\forall x, y \quad Type_0(x) \wedge Type_0(y) \rightarrow Type_0((x, y))$

definition 4.1.2. ***Tuples****(n is a arbitrary number.)(this is a temporary definition, we will give a better one in the next chapter.)*

$$(x_1, x_2, ..., x_n) := ((x_1, x_2, ..., x_{n-1}), x_n)$$

Lemma 4.1.1. $/\therefore \forall x_0, x_1 \quad (x_0, y_0) = (x_1, y_1) \leftrightarrow ((x_0 = x_1) \wedge (y_0 = y_1))$

Proof Is Left As An Exercise To The Reader.😀

definition 4.1.3. ***Cartesian product***

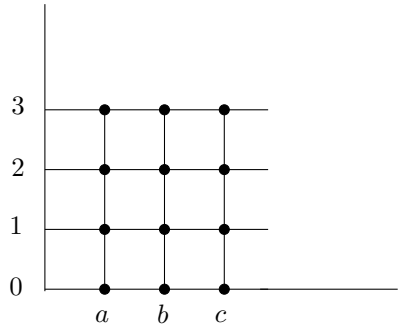
$$A \times B := \{u \mid \exists x, y\Big((x, y) = u \wedge x \in A \wedge y \in B\Big)\}.$$

or i.e.

$$A \times B := \{(x, y) \mid x \in A \wedge y \in B\}.$$

Example 4.1.1. *if* $A = \{a, b, c\}$ *and* $B = \{0, 1, 2, 3\}$ *then*

$$A \times B = \{(a, 0), (a, 1), (a, 2), (a, 3), (b, 0), (b, 1), (b, 2), (b, 3), (c, 0), (c, 1), (c, 2), (c, 3)\}$$



so $\forall X, Y \quad Type_0(X) \wedge Type_0(Y) \to Type_0(X \times Y)$
Theorem 4.1.1. $(schema)$ $(\times I, \times E)$

$\dfrac{a \in A \qquad b \in B}{\therefore (a,b) \in A \times B} \times I$	$a \in A \times B$ $\quad \vert \; a = (x,y)$ $\quad \vert \; x \in A$ $\quad \vert \; y \in B$ $\quad \vert \; \vdots$ $\quad \vert \; \theta$ $\therefore \theta \qquad \times E$

Proof Is Left As An Exercise To The Reader.
Theorem 4.1.2. $(schema)$

$\dfrac{(a,b) \in A \times B}{\therefore a \in A}$	$\dfrac{(a,b) \in A \times B}{\therefore b \in B}$

Proof Is Left As An Exercise To The Reader.

4.2 Relations

definition 4.2.1. ***Relation*** *we call a set like* $\boldsymbol{R}$ *a* ***Relation*** *when there are two sets like* X *and* Y *s.t.* $R \subseteq X \times Y$*.*

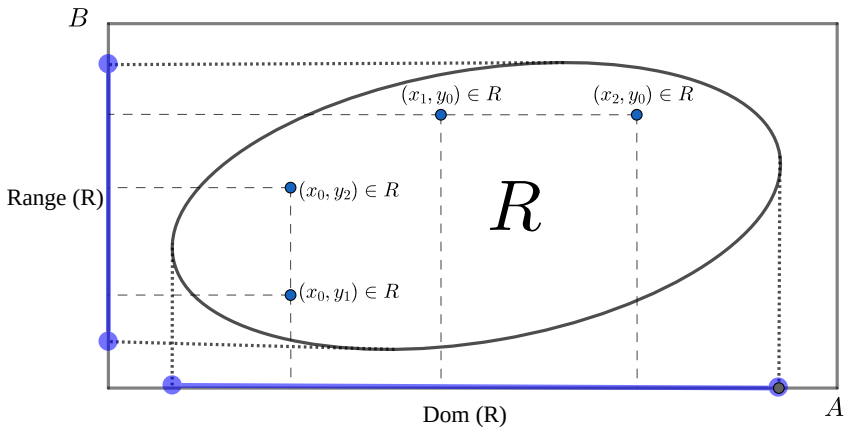
$$Rel(R) : \exists X, Y (R \subseteq X \times Y).$$

sometimes we use different notations in this book. you have to be smart and recognize it by yourself. sometimes instead of "$x \in R$" we denote it by "$R(x)$". and instead of "$(x_0, x_1, ..., x_n) \in R$" we denote it by "$R(x_0, x_1, ..., x_n)$". if tuple is binary, instead of "$(x, y) \in R$" we denote it by "xRy". or "$R(x, y)$" instead of "$R\big((x, y)\big)$"
so if we have a definition like "$R(x) : \phi(x)$" s.t. "$\phi(x)$" is a first order proposition. "weak comprehension" show us there is a straight relatioon between "propositions" and "sets" so you can see it as "$R := \{x \mid \phi(x)\}$". and sometimes we might show a formula in natural language for example below instead of "$Rel(X)$" we can say "X is relation". if it is confusing so forget it.
definition 4.2.2. ***Domain and Range(also called Image)***
Domain of a set "R" is all "x" that there exists a "y" s.t. "xRy"

$$Dom(R) := \{x \mid \exists y \quad xRy\}$$

Range(Image) of a set "R" is all "y" that there exists a "x" s.t. "xRy"

$$Range(R) := \{y \mid \exists x \quad xRy\}.$$



Theorem 4.2.1. $(schema)$ $(DomI, DomE)$

$\dfrac{aRb}{\therefore a \in Dom(R)} \quad DomI$	$a \in Dom(R)$ $\vert \; aRy$ $\vert \; \vdots$ $\vert \; \theta$ $\therefore \theta \qquad DomE$

Proof Is Left As An Exercise To The Reader.
Theorem 4.2.2. $(schema)$ $(RangeI, RangeE)$

$\dfrac{aRb}{\therefore b \in Range(R)} \quad RangeI$	$b \in Range(R)$ $\vert \; xRb$ $\vert \; \vdots$ $\vert \; \theta$ $\therefore \theta \qquad RangeE$

Proof Is Left As An Exercise To The Reader.
definition 4.2.3. ***Inverse of a Relation***

$$R^{-1} = \{u \mid \exists x, y\Big(u = (y, x) \wedge xRy\Big)\}$$

or i.e.

$$R^{-1} = \{(y, x) \mid xRy\}$$

therefore.

$$\forall x, y\Big(xR^{-1}y \leftrightarrow yRx\Big)$$

Theorem 4.2.3. (*schema*) $(()^{-1}I, ()^{-1}E)$

$\dfrac{aRb}{\therefore bR^{-1}a} \quad ()^{-1}I$	$a \in R^{-1}$ $\quad \vert \; a = (y, x)$ $\quad \vert \; xRy$ $\quad \vert \; \vdots$ $\quad \vert \; \theta$ $\therefore \theta \quad ()^{-1}E$

Proof Is Left As An Exercise To The Reader.
definition 4.2.4. ***Combination of a Relations***

$$S \circ R = \{u \mid \exists x, y\Big(u = (x, y) \wedge \exists z\big(xRz \wedge zSy\big)\Big)\}$$

or i.e.

$$S \circ R = \{(x, y) \mid \exists z\Big(xRz \wedge zSy\Big)\}$$

therefore.

$$\forall x, y\Big((x, y) \in S \circ R \leftrightarrow \exists z\big(xRz \wedge zSy\big)\Big)$$

Theorem 4.2.4. (*schema*) $(\circ I, \circ E)$

aSb bRc $\overline{\therefore aR \circ Sc} \quad \circ I$	$a \in R \circ S$ $\quad \vert \; a = (x, y)$ $\quad \vert \; xSz$ $\quad \vert \; zRy$ $\quad \vert \; \vdots$ $\quad \vert \; \theta$ $\therefore \theta \quad \circ E$

Proof Is Left As An Exercise To The Reader.

definition 4.2.5. ***Image***

$$R[A] := \{y \mid \exists x \in A \quad xRy\}$$

you can olso see by definition of "inverse of a Relation" and "image" this equation holds.

$$R^{-1}[B] = \{x \mid \exists y \in B \quad xRy\}$$

*we cal this "**Pre-image**"*
Theorem 4.2.5. $(schema)$ $(R[]I, R[]E, R^{-1}[]I, R^{-1}[]E)$

$a \in A$ aRb $\therefore b \in R[A]$ $\quad R[]I$	$b \in R[A]$ $\quad \vert \; x \in A$ $\quad \vert \; xRb$ $\quad \vert \; \vdots$ $\quad \vert \; \theta$ $\therefore \theta \quad R[]E$
$b \in B$ aRb $\therefore a \in R^{-1}[B]$ $\quad R^{-1}[]I$	$a \in R^{-1}[B]$ $\quad \vert \; y \in B$ $\quad \vert \; aRy$ $\quad \vert \; \vdots$ $\quad \vert \; \theta$ $\therefore \theta \quad R^{-1}[]E$

Proof Is Left As An Exercise To The Reader.😀

4.3 Functions

definition 4.3.1. ***Function*** *we call a relation "**f**" a "**function**" when every input has exactly one output i.e.*

$$Func(f) : Rel(f) \wedge \forall x, y_1, y_2 \Big((x, y_1) \in f \wedge (x, y_2) \in f \rightarrow y_1 = y_2 \Big)$$

Theorem 4.3.1. $(schema)$ $(FuncI, FuncE)$

$\begin{array}{l} \vert\ xfy \\ \vert\ xfz \\ \vert\ \vdots \\ \vert\ y = z \\ \therefore Func(f) \qquad FuncI \end{array}$	$\begin{array}{l} Func(f) \\ afb \\ afc \\ \hline \therefore b = c \qquad FuncE \end{array}$

Proof Is Left As An Exercise To The Reader.

definition 4.3.2. ***Restriction of a function*** *to a set "X" (usually a subset of "$dom(f)$") is the function*

$$f \restriction X := \{(x, y) \in f \mid x \in X\}$$

Axiom 4.3.1 (output of a function)**.** *assume a function* ***f*** *and $(x, y) \in f$, we can use "$f(x)$" as "$f(x) = y$" and we call "$f(x)$" a "**output of a function**". and the reverse is also true, if we use "$f(x)$", we have a function "**f**" and we can say "$x \in Dom(f)$". "output of a function" is object too.*

Theorem 4.3.2. (*schema*)

$\begin{array}{l} Func(f) \\ (x, y) \in f \\ \hline \therefore f(x) = y \qquad f()E \end{array}$	$\begin{array}{l} f(x) = y \\ \hline \therefore Func(f) \qquad f()E \\ \therefore (x, y) \in f \qquad f()E \end{array}$

Attention 4.3.1. *input of a function can be a ordered pair itself, so we can have $f((x, y))$ and it is a "**output of a function**". we define "$f(x, y) := f((x, y))$" for simplicity. this is true for tuples as well, so we can have "$f((x_1, x_2, ..., x_n))$" and we define "$f(x_1, x_2, ..., x_n) := f((x_1, x_2, ..., x_n))$" for simplicity.*

definition 4.3.3. ***Term*** *we define term like this.*

1. *every obj is a term.*
2. *if $t_0, ...t_n$ are terms and "$Func(f)$" and "$(t_0, ...t_n) \in Dom(f)$". then "$f(t_0, ...t_n)$" is also a term.*

we denote terms with small letters like $t, s, r, ...$

definition 4.3.4. ***Substitution for term*** *we define Substitution an term with variable like this.*

1. *for variables if $x = y$ then $y[t/x] := t$ and if $x \neq y$ then $y[t/x] := y$.*
2. *for constants $c[t/x] := c$.*
3. *for terms like $f(t_0, ...t_n)$*

$$f(t_0, ...t_n)[t/x] := f(t_0[t/x], ...t_n[t/x])$$

definition 4.3.5. ***Substitution for formulas*** *we define Substitution for a formula like this.*

1. *for atomic formulas we define it like this. if it's a single term we do it as we explain in "**Substitution for term**". if it's like $t_0 = t_1$ or $t_0 \in t_1$ (t_0, t_1 are terms). we define it like*

this.

$$Set(y)[t/x] := Set(y[t/x])$$

$$(t_0 = t_1)[t/x] := (t_0[t/x] = t_1[t/x])$$

$$(t_0 \in t_1)[t/x] := (t_0[t/x] \in t_1[t/x])$$

2. *if formula A is complex we define it like this.*

$$(B \bigcirc C)[t/x] := (B[t/x] \bigcirc C[t/x])$$

($\bigcirc$ can be $\rightarrow, \wedge, \vee$).

$$(\neg B)[t/x] := (\neg B[t/x])$$

Theorem 4.3.3. ***(schema)*** $(\forall I, \forall E, \exists I, \exists E)$

Rules	*Terms and Conditions*
$\dfrac{\phi(x)}{\therefore \forall x \phi(x)} \quad \forall I$	1. *x is a variable.* 2. *"$\phi(x)$" must not be an assumption.* 3. *x must not occur free in any assumption on which $\phi(x)$ depends.*
$\dfrac{\forall x \phi(x)}{\therefore \phi[t/x]} \quad \forall E$	1. *t is a term.* 2. *t is free for x in $\phi(x)$.*
$\dfrac{\phi[t/x]}{\therefore \exists x \phi(x)} \quad \exists I$	1. *t is a term.* 2. *t is free for x in $\phi(x)$.*
$\exists x \phi(x)$ $\left\vert \begin{array}{l} \phi(x) \\ \vdots \\ \theta \end{array} \right.$ $\therefore \theta \quad \exists E$	1. *x is a variable.* 2. *"$\phi(x)$" is assumed only within the subproof.* 3. *x does not occur free in θ.* 4. *x does not occur free in any assumption on which θ depends.*

$(\forall I, \exists E)$ is the same as $(\forall I, \exists E)$ in chapter three so we have to prove $(\forall E, \exists I)$. so assume $t = f(x)$ so to it's easy to see $Func(f)$ and $x \in Dom(f)$.

1	$\forall x\phi(x)$	ass
2	$Func(f)$	1
3	$x \in Dom(f)$	1
4	$\quad (x,y) \in f$	acp
5	$\quad f(x) = y$	def
6	$\quad \phi(y)$	$\forall E$
7	$\quad \phi(f(x))$	$= E$
8	$\phi(f(x))$	$\exists E$

1	$\phi(f(x))$	ass
2	$Func(f)$	ass
3	$x \in Dom(f)$	ass
4	$\quad (x,y) \in f$	acp
5	$\quad f(x) = y$	def
6	$\quad \phi(y)$	$= E$
7	$\quad \exists x\phi(x)$	$\exists I$
8	$\exists x\phi(x)$	$\exists E$

Attention 4.3.2. *in all rules and theorems you can replace all constants like $a, b, c, ...$ with terms like $t, s, r, ...$ and rewrite them.*

definition 4.3.6. ***Compatible Function***

1. $Compatible(f,g) : \forall x \in dom(f) \cap dom(g) \quad f(x) = g(x)$.
2. $Compatible(F) : \forall f, g \in F \quad Compatible(f,g)$.

Lemma 4.3.1. .

1. $Compatible(f,g) \therefore / \therefore Func(f \cup g)$.
2. $Compatible(f,g) \therefore / \therefore f \restriction (dom(f) \cap dom(g)) = g \restriction (dom(f) \cap dom(g))$.

Proof Is Left As An Exercise To The Reader.

Theorem 4.3.4. $Compatible(F) / \therefore (1), (2), (3)$

1. $Func(\bigcup F)$.
2. $dom(\bigcup F) = \bigcup\{dom(f) \mid f \in F\}$.
3. $\forall f \in F \quad Compatible(f, \bigcup F)$.

Proof Is Left As An Exercise To The Reader.

definition 4.3.7. ***Function from set to set*** *we call a function* ***f*** *from* ***A*** *to* ***B*** *when it's domain is* ***A*** *and it's range is subset of* ***B*** *i.e.*

$$(f : A \to B) : Func(f) \wedge Dom(f) = A \wedge Range(f) \subseteq B$$

definition 4.3.8. ***Set of functions from set to set*** *we denote set of all "$f : A \to B$" with "B^A" i.e.*

$$B^A : \{f \mid f : A \to B\}$$

definition 4.3.9. ***Injective function(Relation)*** *we call a function(Relation)* ***f*** *injective when every output has at most one input i.e.*

$$Injective(f) : Rel(f) \wedge \forall y, x_1, x_2 \Big((x_1, y) \in f \wedge (x_2, y) \in f \to x_1 = x_2 \Big).$$

we call a function ***f*** *injective from* ***A*** *to* ***B*** *when it's function from* ***A*** *to* ***B*** *and it's injective i.e.*

$$(f : A \xrightarrow{inj} B) : (f : A \to B) \wedge Injective(f)$$

Theorem 4.3.5. $(schema)$ $(InjectiveI, InjectiveE)$

$\begin{array}{l} \vert\ xfz \\ \vert\ yfz \\ \vert\ \vdots \\ \vert\ x = y \\ \therefore Injective(f) \qquad InjectiveI \end{array}$	$\begin{array}{l} Injective(f) \\ tfr \\ sfr \\ \hline \therefore s = r \qquad\qquad InjectiveE \end{array}$

Proof Is Left As An Exercise To The Reader.
definition 4.3.10. ***Surjective function*** *we call a function* ***f*** *from* ***A*** *to* ***B*** *when it's domain is* ***A*** *and it's range is equal to* ***B*** *i.e.*

$$(f : A \xrightarrow[surj]{} B) : (f : A \to B) \wedge Range(f) = B$$

or i.e.

$$(f : A \xrightarrow[surj]{} B) : (f : A \to B) \wedge (\forall y \in B \quad \exists x \in A \quad (x, y) \in f)$$

definition 4.3.11. ***Bijection function*** *we call a function* ***f*** *from* ***A*** *to* ***B*** *when it's domain is* ***A*** *and it's range is equal to* ***B*** *i.e.*

$$(f : A \xrightarrow[surj]{inj} B) : (f : A \xrightarrow{inj} B) \wedge (f : A \xrightarrow[surj]{} B)$$

4.4 Family of sets

Attention 4.4.1. ***Indexes*** *are objects (like numbers) allows us to give a unique "tag" to every object. we use $i, j, k, ...$ for variable indexes and we use $m, n, l, ...$ for constant indexes.*
definition 4.4.1. ***Family of sets*** *assume $A : I \to \mathcal{F}$ is a function from a nonempty set of indices I to a family of sets $\mathcal{F}$, then we call $(Range[A])$ an indexed family of sets. we can write A_i instead of $A(i)$ and we can write $(A_i)_{i\in I}$ instead of "A" and $\{A_i\}_{i\in I}$ instead of "Range[A]" or "A[I]".*

$$(A_i)_{i\in I} := A$$

and

$$\{A_i\}_{i\in I} := A[I]$$

definition 4.4.2. ***Intersrction and union of an family of sets***

$$\bigcap_{i\in I} A_i := \bigcap \{A_i\}_{i\in I}$$

$$\bigcup_{i\in I} A_i := \bigcup \{A_i\}_{i\in I}$$

Attention 4.4.2. *we can write $\forall i \in I \quad P(i)$ instead of $\forall i(i \in I \to P(i))$ and we can write $\exists i \in I \quad P(i)$ instead of $\exists i(i \in I \wedge P(i))$.*

Lemma 4.4.1.

$$/\therefore \bigcap_{i\in I} A_i = \{x \mid \forall i \in I \quad x \in A_i\}$$

1	$x \in \bigcap_{i\in I} A_i$	acp
2	$x \in \bigcap\{A_i\}_{i\in I}$	$def.1$
3	$\quad Y \in A[I]$	acp
4	$\quad x \in Y$	acp
5	$\quad\quad i \in I$	acp
6	$\quad\quad A_i = Y$	acp
7	$\quad\quad x \in A_i$	$= E$
8	$\quad i \in I \to x \in A_i$	$\to I$
9	$\quad \forall i \in I \quad x \in A_i$	$\forall I$
10	$\forall i \in I \quad x \in A_i$	$\bigcap E$
11	$x \in \{x \mid \forall i \in I \quad x \in A_i\}$	def
12	$x \in \{x \mid \forall i \in I \quad x \in A_i\}$	acp
13	$\forall i \in I \quad x \in A_i$	def
14	$\quad Y \in A[I]$	acp
15	$\quad\quad i \in I$	acp
16	$\quad\quad A_i = Y$	acp
17	$\quad\quad i \in I \to x \in A_i$	$= E$
18	$\quad\quad x \in A_i$	$\to E$
19	$\quad\quad x \in Y$	$= E$
20	$\quad x \in Y$	$R[]E$
21	$x \in \bigcap\{A_i\}_{i\in I}$	$\bigcap I$
22	$x \in \bigcap_{i\in I} A_i$	def
23	$\bigcap_{i\in I} A_i = \{x \mid \forall i \in I \quad x \in A_i\}$	$= I$

$$/\therefore \bigcup_{i\in I} A_i = \{x \mid \exists i \in I \quad x \in A_i\}$$

1	$x \in \bigcup_{i \in I} A_i$	acp
2	$x \in \bigcup \{A_i\}_{i \in I}$	$def.1$
3	$\quad Y \in A[I]$	acp
4	$\quad x \in Y$	acp
5	$\quad\quad i \in I$	acp
6	$\quad\quad A_i = Y$	acp
7	$\quad\quad x \in A_i$	$= E$
8	$\quad\quad i \in I \wedge x \in A_i$	$\wedge I$
9	$\quad\quad \exists i \in I \quad x \in A_i$	$\exists I$
10	$\quad \exists i \in I \quad x \in A_i$	$R[]E$
11	$\exists i \in I \quad x \in A_i$	$\bigcup E$
12	$x \in \{x \mid \exists i \in I \quad x \in A_i\}$	def
13	$x \in \{x \mid \exists i \in I \quad x \in A_i\}$	acp
14	$\exists i \in I \quad x \in A_i$	def
15	$\quad i \in I$	acp
16	$\quad x \in A_i$	acp
17	$\quad A_i \in A[I]$	$R[]I$
18	$\quad x \in \bigcup \{A_i\}_{i \in I}$	$\bigcup I$
19	$x \in \bigcup \{A_i\}_{i \in I}$	$\exists E$
20	$x \in \bigcup_{i \in I} A_i$	def
21	$\bigcup_{i \in I} A_i = \{x \mid \exists i \in I \quad x \in A_i\}$	$= I$

Theorem 4.4.1. $(schema)$ $(\bigcap_I I, \bigcap_I E, \bigcup_I I, \bigcup_I E)$

$\begin{array}{\|l} i \in I \\ \vdots \\ t \in A_i \end{array}$ $\therefore t \in \bigcap_{i\in I} A_i \qquad \bigcap_I I$	$t \in \bigcap_{i\in I} A_i$ $n \in I$ $\therefore t \in A_n \qquad \bigcap_I E$
$t \in A_n$ $n \in I$ $\therefore t \in \bigcup_{i\in I} A_i \qquad \bigcup_I I$	$t \in \bigcup_{i\in I} A_i$ $\begin{array}{\|l} i \in I \\ t \in A_i \\ \vdots \\ \theta \end{array}$ $\therefore \theta \qquad \bigcup_I E$

Proof Is Left As An Exercise To The Reader.😀

Exercise 4.4.1. *Prove the following for sets A, B, C, Relations R, S, T and functions f, g whenever the expressions are defined.*

1. $A \times (B \cup C) = (A \times B) \cup (A \times C)$.
2. $A \times (B \cap C) = (A \times B) \cap (A \times C)$.
3. $(A \cup B) \times C = (A \times C) \cup (B \times C)$.
4. $(A \cap B) \times C = (A \times C) \cap (B \times C)$.
5. $A \times B = \varnothing \therefore / \therefore (A = \varnothing \vee B = \varnothing)$.
6. $A \times B = B \times A \therefore / \therefore (A = B \vee A = \varnothing \vee B = \varnothing)$.
7. $Dom(R^{-1}) = Range(R)$.
8. $Range(R^{-1}) = Dom(R)$.
9. $(R^{-1})^{-1} = R$.
10. $Dom(R \cup S) = Dom(R) \cup Dom(S)$.
11. $Range(R \cup S) = Range(R) \cup Range(S)$.
12. $Dom(R \cap S) \subseteq Dom(R) \cap Dom(S)$.
13. $Range(R \cap S) \subseteq Range(R) \cap Range(S)$.
14. $(R \cup S)^{-1} = R^{-1} \cup S^{-1}$.
15. $(R \cap S)^{-1} = R^{-1} \cap S^{-1}$.
16. $(R \circ S)^{-1} = S^{-1} \circ R^{-1}$.
17. $R \circ (S \circ T) = (R \circ S) \circ T$.

18. $R \circ (S \cup T) = (R \circ S) \cup (R \circ T)$.
19. $(R \cup S) \circ T = (R \circ T) \cup (S \circ T)$.
20. $R[A \cup B] = R[A] \cup R[B]$.
21. $R[A \cap B] \subseteq R[A] \cap R[B]$.
22. $A \subseteq Dom(R) / \therefore A \subseteq R^{-1}[R[A]]$.
23. $f : A \to B, g : B \to C / \therefore g \circ f : A \to C$.
24. $f : A \xrightarrow{inj} B, g : B \xrightarrow{inj} C / \therefore g \circ f : A \xrightarrow{inj} C$.
25. $f : A \xrightarrow[surj]{} B, g : B \xrightarrow[surj]{} C / \therefore g \circ f : A \xrightarrow[surj]{} C$.
26. $f : A \xrightarrow{inj} B / \therefore f^{-1} : Range(f) \to A$.
27. $f : A \xrightarrow[surj]{inj} B / \therefore f^{-1} : B \xrightarrow[surj]{inj} A$.
28. $Func(f) / \therefore f[f^{-1}[B]] \subseteq B$.
29. $Func(f) / \therefore f^{-1}[D \cap C] = f^{-1}[D] \cap f^{-1}[C]$.
30. $Func(f) / \therefore f^{-1}[B - C] = f^{-1}[B] - f^{-1}[C]$.
31. $Func(f) / \therefore f[A] - f[B] \subseteq f[A - B]$.
32. $f : A \xrightarrow{inj} B, C \subseteq A / \therefore f[A - B] = f[A] - f[B]$.
33. $f : A \to B, g : B \to C \therefore / \therefore (g \circ f)[A] = g[f[A]]$.
34. $\bigcup_{i \in I}(A_i \cap B_i) \subseteq \left(\bigcup_{i \in I} A_i\right) \cap \left(\bigcup_{i \in I} B_i\right)$.
35. $\left(\bigcap_{i \in I} A_i\right) \cup \left(\bigcap_{i \in I} B_i\right) \subseteq \bigcap_{i \in I}(A_i \cup B_i)$.
36. $\bigcup_{i \in I}(A_i \cup B_i) = \left(\bigcup_{i \in I} A_i\right) \cup \left(\bigcup_{i \in I} B_i\right)$.
37. $\bigcap_{i \in I}(A_i \cap B_i) = \left(\bigcap_{i \in I} A_i\right) \cap \left(\bigcap_{i \in I} B_i\right)$.
38. $\bigcup_{i \in I}(A_i - B_i) \subseteq \left(\bigcup_{i \in I} A_i\right) - \left(\bigcap_{i \in I} B_i\right)$.
39. $\bigcup_{i \in I}(A_i \cap B) = \left(\bigcup_{i \in I} A_i\right) \cap B$.
40. $\bigcap_{i \in I}(A_i \cup B) = \left(\bigcap_{i \in I} A_i\right) \cup B$.
41. $\bigcup_{i \in I}(A_i - B) = \left(\bigcup_{i \in I} A_i\right) - B$.
42. $\bigcap_{i \in I}(A_i - B) = \left(\bigcap_{i \in I} A_i\right) - B$.
43. $\bigcap_{i \in I} A_i \subseteq \bigcup_{i \in I} A_i$.
44. $\bigcap_{i \in I} A_i = \bigcup_{i \in I} A_i \therefore / \therefore \forall i, j \in I \ (A_i = A_j)$.
45. $\bigcap_{i \in I} A_i = \varnothing \therefore / \therefore \exists i \in I \ (A_i = \varnothing)$.
46. $\bigcup_{i \in I} A_i = \varnothing \therefore / \therefore \forall i \in I \ (A_i = \varnothing)$.
47. $\bigcup_{i \in I} A_i \subseteq B \therefore / \therefore \forall i \in I \ (A_i \subseteq B)$.
48. $B \subseteq \bigcap_{i \in I} A_i \therefore / \therefore \forall i \in I \ (B \subseteq A_i)$.

49. $\bigcup_{i \in I} A_i \subseteq \bigcap_{i \in I} A_i \therefore / \therefore \forall i, j \in I \ (A_i \subseteq A_j)$.

50. $Dom(f \restriction A) = Dom(f) \cap A$.

51. $Range(f \restriction A) = f[A \cap Dom(f)]$.

52. $f \restriction Dom(f) = f$.

53. $f \restriction A = f \restriction B \therefore / \therefore f \restriction (A \cap Dom(f)) = f \restriction (B \cap Dom(f))$.

54. $f \restriction A \subseteq f$.

55. $f \restriction A = f \therefore / \therefore Dom(f) \subseteq A$.

Chapter 5

Equivalences and Orderings

5.1 Structures and morphisms

definition 5.1.1. ***Structure*** *"$\mathfrak{M}$" is a tuple defined like this.*

$$\mathfrak{M} := (M, f_0, f_1, ..., f_n, R_0, R_1, ..., R_m, c_0, c_1, ..., c_k)$$

"M" is a ***nonempty*** *set shows our universe "$\forall i \in \{0, ..., n\} \quad f_i : M^{n'}$ → M" are functions and "$\forall i \in \{0, ..., m\} \quad R_i \subseteq M^{m'}$" are relations and "$\forall i \in \{0, ..., k\} \quad c_i \in M$" are constants. ($M^n$ means $\overbrace{M \times M \times ... \times M}^{n \text{ times}}$ we will define it later) and for the case we have arbitrary family of functions and relations and constants.*

$$\mathfrak{M} := (M, \{f_i\}_{i \in I_f}, \{R_i\}_{i \in I_R}, \{c_i\}_{i \in I})$$

definition 5.1.2. ***Homorphism*** *if "$\mathfrak{M} := (M, \{f_{i,m}\}_{i \in I_f}, \{R_{i,m}\}_{i \in I_R}, \{c_{i,m}\}_{i \in I})$" and "$\mathfrak{N} := (N, \{f_{i,n}\}_{i \in I_f}, \{R_{i,n}\}_{i \in I_R}, \{c_{i,n}\}_{i \in I})$" are two structures. function "$h : M \to N$"is called "(weak) Homorphism" when this conditions holds*

1. *for eveery relation "$R_{i,m}$"*

$$\forall x_0, x_1, x_2, ..., x_k \in M \quad R_{i,m}(x_0, x_1, x_2, ..., x_k) \to R_{i,n}(h(x_0), h(x_1), h(x_2), ..., h(x_k))$$

.

2. *for eveery function "$f_{i,m}$"*

$$\forall x_0, x_1, x_2, ..., x_k \in M \quad h(f_{i,m}(x_0, x_1, x_2, ..., x_k)) = f_{i,n}(h(x_0), h(x_1), h(x_2), ..., h(x_k))$$

.

3. *for eveery constant "$c_{i,m}$"*

$$h(c_{i,m}) = c_{i,n}$$

.

in condition (1) if "$\leftrightarrow$" be instead of "$\rightarrow$" then it's called "(strong) Homorphism"

few notices: when we are talking about "Homorphism" we usually talk about "(strong) Homorphism" .
in the case (in definition "(strong) Homorphism") we denote it by "$Hom(h, \mathfrak{M}, \mathfrak{N})$". if exists function "$h$" "$Hom(h, \mathfrak{M}, \mathfrak{N})$" we say "$\mathfrak{M}$" and "$\mathfrak{N}$" are homomorphic. we denote it by "$Hom(\mathfrak{M}, \mathfrak{N})$".
if "$h : M \xrightarrow{inj} N$" we call it "Monomorphism" or "Embedding".we say "$\mathfrak{M}$" and "$\mathfrak{N}$" are momomorphic.
if "$h : M \xrightarrow[surj]{} N$" we call it "Epimorphism".we say "$\mathfrak{M}$" and "$\mathfrak{N}$" are Eomomorphic.
if "$h : M \xrightarrow[surj]{inj} N$" we call it "Isomorphism".we say "$\mathfrak{M}$" and "$\mathfrak{N}$" are Isomomorphic. and denote it by "$\mathfrak{M} \cong \mathfrak{N}$".
if "$k : M \to M$" is a function and "$Hom(k, \mathfrak{M}, \mathfrak{M})$". we call "$k$" "Indomorphism".
if "$k : M \xrightarrow[surj]{inj} M$" is a bijection and "$Hom(k, \mathfrak{M}, \mathfrak{M})$". we call "$k$" "Automorphism".

5.2 Equivalences and Partitions

definition 5.2.1. ***Equivalence relation*** *assume a structure "$\mathfrak{A} := (A, \sim)$"("$\sim$" is a relation). we say "$\sim$" is a "Equivalence relation" in "$\mathfrak{A}$" if it's reflexive,symmetric, and transitive.*

$$(reflexive) : \forall x \in A \quad x \sim x$$

$$(symmetric) : \forall x, y \in A \quad x \sim y \to y \sim x$$

$$(transitive) : \forall x, y, z \in A \quad x \sim y \wedge y \sim z \to x \sim z$$

Lemma 5.2.1. *"$\cong$" is a equivalence relation in "$(Structure, \cong)$"*
Proof Is Left As An Exercise To The Reader.
definition 5.2.2. ***Equivalence class of "x"*** *Let "$\sim$" be an equivalence relation on "$\mathfrak{A}$". For every "$x \in A$"*

$$[x]_\sim := \{y \in A \mid y \sim x\}$$

you can also see it as a function $[_]_\sim : A \to V$.
Theorem 5.2.1. $(schema)$ $([]_\sim I, []_\sim E)$

$\dfrac{t \sim s}{\therefore t \in [s]_\sim} \quad []_\sim I$	$\dfrac{t \in [s]_\sim}{\therefore t \sim s} \quad []_\sim E$

Proof Is Left As An Exercise To The Reader.
Lemma 5.2.2. *For every "$a, b \in A$"*

1. $a \sim b \therefore / \therefore [a]_\sim = [b]_\sim$
2. $a \nsim b \therefore / \therefore [a]_\sim \cap [b]_\sim = \varnothing$

prove.1

$/\therefore$

1	$a \sim b$	ass
2	$\quad x \in [a]_\sim$	acp
3	$\quad x \sim a$	$[]_\sim E$
4	$\quad x \sim b$	$trans$
5	$\quad x \in [b]_\sim$	$[]_\sim I$
6	$\quad x \in [b]_\sim$	acp
7	$\quad x \sim b$	$[]_\sim E$
8	$\quad x \sim a$	$trans$
9	$\quad x \in [a]_\sim$	$[]_\sim I$
10	$[a]_\sim = [b]_\sim$	

$\therefore /$

1	$[a]_\sim = [b]_\sim$	ass
2	$a \sim a$	$reflex$
3	$a \in [a]_\sim$	$[]_\sim I$
4	$a \in [b]_\sim$	$= E$
5	$a \sim b$	$[]_\sim E$

prove.2

$/\therefore$

1	$a \nsim b$	ass
2	$\quad x \in [a]_\sim \cap [b]_\sim$	acp
3	$\quad x \in [a]_\sim$	$\cap E$
4	$\quad x \in [b]_\sim$	$\cap E$
5	$\quad x \sim a$	$[]_\sim E$
6	$\quad x \sim b$	$[]_\sim E$
7	$\quad a \sim b$	$trans$
8	$\quad \bot$	$\bot I$
9	$[a]_\sim \cap [b]_\sim = \varnothing$	$= \varnothing I$

$\therefore /$

1	$[a]_\sim \cap [b]_\sim = \varnothing$	
2	$\quad a \sim b$	acp
3	$\quad a \sim a$	$reflex$
4	$\quad a \in [a]_\sim$	$[]_\sim I$
5	$\quad a \in [b]_\sim$	$[]_\sim I$
6	$\quad a \in [a]_\sim \cap [b]_\sim$	$\cap I$
7	$\quad \bot$	$\bot I$
8	$a \nsim b$	$\neg I$

definition 5.2.3. ***Partition*** *is a "set of sets" like "P" of a set like "A" with this conditions.*

1. $\forall X \in P \quad X \neq \varnothing$
2. $\forall X, Y \in P \quad X \neq Y \to X \cap Y = \varnothing$
3. $\bigcup P = A$

   if these condition holds then we say "$Part(P, A)$".

definition 5.2.4. ***Set of Equivalence classes***

$$A/\sim := \{[x]_\sim \mid x \in A\}$$

Theorem 5.2.2. *if "$\sim$" is a equivalence class in "$(A, \sim)$" then "$A/\sim$" is a partition for "A"*

i.e. "$Part(A/\sim, A)$".

1		$[x]_\sim \in A/\sim$	acp
2		$x \sim x$	$reflex$
3		$x \in [x]_\sim$	$[]_\sim I$
4		$[x]_\sim \neq \varnothing$	$\neq \varnothing I$
5	$\forall [x]_\sim \in A/\sim$	$[x]_\sim \neq \varnothing$	

1		$[x]_\sim \neq [y]_\sim$	acp
2		$x \nsim y$	$Lemma$
3		$[x]_\sim \cap [y]_\sim = \varnothing$	$Lemma$
4	$\forall [x]_\sim, [y]_\sim \in A/\sim$	$[x]_\sim \neq [y]_\sim \rightarrow [x]_\sim \cap [y]_\sim = \varnothing$	

1		$x \in \bigcup(A/\sim)$	acp
2		$[y]_\sim \in A/\sim$	acp
3		$x \in [y]_\sim$	acp
4		$[y]_\sim \subseteq A$	
5		$x \in A$	$\subseteq E$
6		$x \in A$	$\bigcup E$
7		$x \in A$	acp
8		$x \in [x]_\sim$	
9		$[x]_\sim \in (A/\sim)$	
10		$x \in \bigcup(A/\sim)$	$\bigcup I$
11	$A = \bigcup(A/\sim)$		$= I$

Theorem 5.2.3. *if "P" is a partition for "A" we define*

$$\sim_P := \{(x, y) \mid \exists Y \in P \quad (x \in Y \wedge y \in Y)\}$$

then "$\sim_P$" is a equivalence class in "$(A, \sim_P)$".

1	|	$x \in A$	acp
2	|	$x \in \bigcup P$	Def
3	| |	$x \in Y$	acp
4	| |	$Y \in P$	acp
5	| |	$\exists Y \in P \quad (x \in Y \wedge x \in Y)$	$\wedge I, \exists I, 3, 4$
6	| |	$x \sim_P x$	$Def.5$
7	|	$x \sim_P x$	$\bigcup E, 2$
8		$\forall x \in A \quad x \sim_P x$	$\rightarrow I, \forall I$

1	|	$x, y \in A$	acp
2	| |	$x \sim_P y$	acp
3	| |	$\exists Y \in P \quad (x \in Y \wedge y \in Y)$	$Def2$
4	| | |	$Y \in P \wedge x \in Y \wedge y \in Y$	acp
5	| | |	$Y \in P \wedge y \in Y \wedge x \in Y$	$commutative\wedge, 4$
6	| | |	$\exists Y \in P \quad (y \in Y \wedge x \in Y)$	$\exists I, 5$
7	| | |	$y \sim_P x$	$Def6$
8	| |	$y \sim_P x$	$\exists E, 3$
9	|	$x \sim_P y \rightarrow y \sim_P x$	$\rightarrow I$
10		$\forall x, y \in A \quad x \sim_P y \rightarrow y \sim_P x$	$\rightarrow I, \forall I$

1	$x, y, z \in A$	acp
2	$\quad x \sim_P y$	acp
3	$\quad y \sim_P z$	acp
4	$\quad \exists Y \in P \quad (x \in Y \wedge y \in Y)$	$Def2$
5	$\quad \exists Y \in P \quad (y \in Y \wedge z \in Y)$	$Def3$
6	$\quad\quad Y \in P \wedge x \in Y \wedge y \in Y$	acp
7	$\quad\quad\quad Y' \in P \wedge y \in Y' \wedge z \in Y'$	acp
8	$\quad\quad\quad y \in Y \cap Y'$	$\wedge E, \cap I$ 6,7
9	$\quad\quad\quad Y \cap Y' \neq \varnothing$	$\varnothing \neq I$
10	$\quad\quad\quad Y = Y'$	$partition - condition, contraposition, 8$
11	$\quad\quad\quad Y \in P \wedge x \in Y \wedge z \in Y$	$= E, \wedge I$
12	$\quad\quad\quad \exists Y \in P \quad (x \in Y \wedge z \in Y)$	$\exists I$
13	$\quad\quad\quad x \sim_P z$	$Def52$
14	$\quad\quad x \sim_P z$	$\exists E$
15	$\quad x \sim_P z$	$\exists E$
16	$x \sim_P y \wedge y \sim_p z \to x \sim_P z$	$\to I$
17	$\forall x, y, z \in A \quad x \sim_P y \wedge y \sim_p z \to x \sim_P z$	$\to I, \forall I$

5.3 Orderings

definition 5.3.1. ***(partial) Order relation*** *assume a structure "$\mathfrak{A} := (A, \preceq)$"("$\preceq$" is a relation). we say "$\preceq$" is a "(partial) Order relation" in "$\mathfrak{A}$" if it's reflexive,antisymmetric, and transitive.*

$$(reflexive) : \forall x \in A \quad x \preceq x$$

$$(antisymmetric) : \forall x, y \in A \quad x \preceq y \wedge y \preceq x \to x = y$$

$$(transitive) : \forall x, y, z \in A \quad x \preceq y \wedge y \preceq z \to x \preceq z$$

definition 5.3.2. ***(partial) Strict order relation*** *assume a structure "$\mathfrak{A} := (A, \prec)$"("$\prec$" is a relation). we say "$\prec$" is a "(partial) Strict order relation" in "$\mathfrak{A}$" if it's irreflexive, and transitive.*

$$(irreflexive) : \forall x \in A \quad x \not\prec x$$

$$(transitive) : \forall x, y, z \in A \quad x \prec y \wedge y \prec z \to x \prec z$$

by two conditions bellow we can proof asymmetric too.

$$(asymmetric) : \forall x, y \in A \quad x \prec y \to y \not\prec x$$

now we will show you relation between "(partial) Order relation" and "(partial) Strict order relation".

Theorem 5.3.1. .

if "$\preceq$" is a "(partial) Order relation" in "$\mathfrak{A}$" define "$a \prec b : a \preceq b \wedge a \neq b$" prove that "$\prec$" is "(partial) Strict order relation".

1		$x \in A$	acp
2		$x \prec x$	acp
3		$x \neq x$	$\wedge E, 2$
4		$x = x$	$= I$
5		$\bot$	$\bot I, 3, 4$
6		$x \not\prec x$	$\neg I$
7	$\forall x \in A$	$x \not\prec x$	$\rightarrow I, \forall I$

1		$x, y, z \in A$	acp
2		$x \prec y$	acp
3		$y \prec z$	acp
4		$x \preceq y$	$\wedge E, 2$
5		$y \preceq z$	$\wedge E, 3$
6		$x \preceq z$	$trans, 4, 5$
7		$x \neq y$	$\wedge E, 2$
8		$y \neq z$	$\wedge E, 3$
9		$x = z$	acp
10		$y \preceq x$	$= E, 5, 9$
11		$x = y$	$antisym, 5, 10$
12		$\bot$	$\bot I, 7, 11$
13		$x \neq z$	$\neg I$
14		$x \prec z$	$\wedge I, 6, 13$
15		$x \prec y \wedge y \prec z \rightarrow x \prec z$	$\rightarrow I$
16	$\forall x, y, z \in A$	$x \prec y \wedge y \prec z \rightarrow x \prec z$	$\rightarrow I, \forall I$

Theorem 5.3.2. .

if "$\prec$" is a "(partial) Strict Order relation" in "$\mathfrak{A}$" define "$a \preceq b : a \prec b \vee a = b$" prove that "$\preceq$" is "(partial) order relation".

1	$x \in A$	acp
2	$x = x$	$= I$
3	$x \preceq x$	$\vee I, 2$
4	$\forall x \in A \quad x \preceq x$	$\rightarrow I, \forall I$

1	$x, y \in A$	acp
2	$x \preceq y \wedge y \preceq x$	acp
3	$x \prec y \wedge y \prec x$	acp
4	$\bot$	$asym, 3$
5	$x = y$	$\bot I, 4$
6	$x \prec y \wedge y = x$	acp
7	$x = y$	$\wedge E, 6$
8	$x = y \wedge y \prec x$	acp
9	$x = y$	$\wedge E, 8$
10	$x = y \wedge y = x$	acp
11	$x = y$	$\wedge E, 10$
12	$x = y$	$\vee E, 2$
13	$x \preceq y \wedge y \preceq x \rightarrow x = y$	$\rightarrow I$
14	$\forall x, y \in A \quad x \preceq y \wedge y \preceq x \rightarrow x = y$	$\rightarrow I, \forall I$

1	$x, y, z \in A$	acp
2	$\quad x \preceq y \wedge y \preceq z$	acp
3	$\quad\quad x \prec y \wedge y \prec z$	acp
4	$\quad\quad x \prec z$	$trans, 3$
5	$\quad\quad x \preceq z$	$\vee I, 4$
6	$\quad\quad x \prec y \wedge y = z$	acp
7	$\quad\quad x \prec z$	$= E, 6$
8	$\quad\quad x \preceq z$	$\vee I, 7$
9	$\quad\quad x = y \wedge y \prec z$	acp
10	$\quad\quad x \prec z$	$= E, 9$
11	$\quad\quad x \preceq z$	$\vee I, 10$
12	$\quad\quad x = y \wedge y = z$	acp
13	$\quad\quad x = z$	$= E, 12$
14	$\quad\quad x \preceq z$	$\vee I, 13$
15	$\quad x \preceq z$	$\vee E, 2$
16	$x \preceq y \wedge y \prec z \to x \preceq z$	$\to I$
17	$\forall x, y, z \in A \quad x \preceq y \wedge y \prec z \to x \preceq z$	$\to I, \forall I$

definition 5.3.3. ***(Total-Linear) Order relation*** *assume a structure "$\mathfrak{A} := (A, \preceq)$"("$\preceq$" is a Order relation). we say "$\preceq$" is a "(Total-Linear) Order relation" in "$\mathfrak{A}$" if every two elements are comparable.*

$$(comparablity) : \forall x, y \in A \quad x \preceq y \vee y \preceq x$$

definition 5.3.4. ***(Total-Linear) Strict order relation*** *assume a structure "$\mathfrak{A} := (A, \prec)$"("$\prec$" is a Strict order relation). we say "$\prec$" is a "(Total-Linear) Strict order relation" in "$\mathfrak{A}$" if every two elements are comparable.*

$$(trichotomy) : \forall x, y \in A \quad x \prec y \vee x = y \vee y \prec x$$

Theorem 5.3.3. .

1. *if "$\preceq$" is a "(Total-Linear) Order relation" in "$\mathfrak{A}$" define "$a \prec b : a \preceq b \wedge a \neq b$" prove that "$\prec$" is "(Total-Linear) Strict order relation".*

2. *if "$\prec$" is a "(Total-Linear) Strict Order relation" in "$\mathfrak{A}$" define "$a \preceq b : a \prec b \vee a = b$" prove that "$\preceq$" is "(Total-Linear) order relation".*

1	$x, y \in A$	acp
2	$x = y \vee x \neq y$	LEM
3	$\quad x = y$	acp
4	$\quad x \prec y \vee x = y \vee y \prec x$	$\vee I$ 2
5	$\quad x \neq y$	acp
6	$\quad x \preceq y \vee y \preceq x$	comparablity of $\preceq$ 1
7	$\quad\quad x \preceq y$	acp
8	$\quad\quad x \preceq y \wedge x \neq y$	$\wedge I$ 7, 6
9	$\quad\quad x \prec y$	def $\prec$ 8
10	$\quad\quad x \prec y \vee x = y \vee y \prec x$	$\vee I$ 9
11	$\quad\quad y \preceq x$	acp
12	$\quad\quad y \preceq x \wedge y \neq x$	$\wedge I$ 12, 13
13	$\quad\quad y \prec x$	def $\prec$ 12
14	$\quad\quad x \prec y \vee x = y \vee y \prec x$	$\vee I$ 15
15	$\quad x \prec y \vee x = y \vee y \prec x$	$\vee E$ 6
16	$x \prec y \vee x = y \vee y \prec x$	$\vee E$ 2
17	$\forall x, y \in A \quad x \prec y \vee x = y \vee y \prec x$	$\to I, \forall I$

1	$x, y \in A$	acp
2	$x \prec y \vee x = y \vee y \prec x$	comparablity of $\prec$
3	$\quad x \prec y \vee x = y$	acp
4	$\quad x \preceq y$	def $\preceq$ 3
5	$\quad x \preceq y \vee y \preceq x$	$\vee I$ 4
6	$\quad y \prec x$	acp
7	$\quad y \prec x \vee y = x$	$\vee I$ 6
8	$\quad y \preceq x$	def $\preceq$ 7
9	$\quad x \preceq y \vee y \preceq x$	$\vee I$ 8
10	$x \preceq y \vee y \preceq x$	$\vee E$ 2
11	$\forall x, y \in A \quad x \preceq y \vee y \preceq x$	$\to I, \forall I$

Lemma 5.3.1. *assume a structure "$\mathfrak{A} := (A, \preceq)$"("$\preceq$" is a Order relation). and $B, C \subseteq A$ prove*

1. *$\exists b \in B \quad \forall x \in B \quad b \preceq x / \therefore \exists! b \in B \quad \forall x \in B \quad b \preceq x$ in this case we say "B" has least element or Minimum or "$HaveLeast(B)$"*

2. $\exists c \in C \quad \forall x \in C \quad x \preceq c / \therefore \exists! c \in C \quad \forall x \in C \quad x \preceq c$ *in this case we say "C" has greatest element or has Maximum or "HaveGreatest(C)"*

1	$\exists b \in B \; \forall x \in B (b \preceq x)$	*ass*
2	$\quad b_1, b_2 \in B$	acp
3	$\quad\quad \forall x \in B \quad b_1 \preceq x$	acp
4	$\quad\quad \forall x \in B \quad b_2 \preceq x$	acp
5	$\quad\quad b_1 \preceq b_2$	$\forall E$ 3
6	$\quad\quad b_2 \preceq b_1$	$\forall E$ 4
7	$\quad\quad b_1 = b_2$	antisym of $\preceq$ 5 ,6
8	$\quad ((\forall x \in B \quad b_1 \preceq x) \wedge (\forall x \in B \; b_2 \preceq x)) \rightarrow b_1 = b_2$	$\rightarrow I$
9	$\exists! b \in B \quad \forall x \in B (b \preceq x)$	$\exists!$ def 1, 10

1	$\exists c \in C \; \forall x \in C (x \preceq c)$	*ass*
2	$\quad c_1, c_2 \in C$	acp (assumption for uniqueness)
3	$\quad\quad \forall x \in C \; x \preceq c_1$	$\wedge E$ 3
4	$\quad\quad \forall x \in C \; x \preceq c_2$	$\wedge E$ 3
5	$\quad\quad c_2 \preceq c_1$	$\forall E$ 3
6	$\quad\quad c_1 \preceq c_2$	$\forall E$ 4
7	$\quad\quad c_1 = c_2$	antisym of $\preceq$ 5 , 6
8	$\quad ((\forall x \in C \; x \preceq c_1) \wedge (\forall x \in C \; x \preceq c_2)) \rightarrow c_1 = c_2$	$\rightarrow I$
9	$\exists! c \in C \; \forall x \in C (x \preceq c)$	$\exists!$ def 1, 10

Theorem 5.3.4. ***Minimum and Maximum*** *assume a structure "$\mathfrak{A} := (A, \preceq)$"("$\preceq$" is a Order relation). and $\mathcal{B} := \{B \subseteq A \mid HaveLeast(B)\}$ and $\mathcal{C} := \{C \subseteq A \mid HaveGreatest(C)\}$ we define like this*

$$min_{\mathfrak{A}} := \{(B, b) \in \mathcal{B} \times A \mid \forall x \in B \quad b \preceq x\}$$
$$max_{\mathfrak{A}} := \{(C, c) \in \mathcal{C} \times A \mid \forall x \in C \quad x \preceq c\}$$

we must prove these two relations are two functions "$min_{\mathfrak{A}} : \mathcal{B} \rightarrow A$" and "$max_{\mathfrak{A}} : \mathcal{C} \rightarrow A$".

1	$\quad (B, b_1) \in min_{\mathfrak{A}}$	*acp*
2	$\quad (B, b_2) \in min_{\mathfrak{A}}$	*acp*
3	$\quad (\forall x \in B \quad b_1 \preceq x)$	*def $min_{\mathfrak{A}}$ 1*
4	$\quad (\forall x \in B \quad b_2 \preceq x)$	*def $min_{\mathfrak{A}}$ 2*
5	$\quad b_1 = b_2$	*Lemma 5.7.1 (uniqueness)*
6	$Func(min_{\mathfrak{A}})$	*FuncI*

1	$(C, c_1) \in max_{\mathfrak{A}}$	*acp*
2	$(C, c_2) \in max_{\mathfrak{A}}$	*acp*
3	$(\forall x \in C \quad x \preceq c_1)$	*def* $max_{\mathfrak{A}}$ *1*
4	$(\forall x \in C \quad x \preceq c_2)$	*def* $max_{\mathfrak{A}}$ *2*
5	$c_1 = c_2$	*Lemma 5.7.1 (uniqueness)*
6	$Func(max_{\mathfrak{A}})$	*FuncI*

definition 5.3.5. ***Minimal and Maximal*** *assume a structure "*$\mathfrak{A} := (A, \preceq)$*"("*$\preceq$*" is a Order relation). we define "Minimal and Maximal" like this*

$$Minimal_{\mathfrak{A}}(b) : \forall x \in A \quad x \not\prec b$$

i.e. "b" is the minimal in "$\mathfrak{A}$*".*

$$Maximal_{\mathfrak{A}}(c) : \forall x \in A \quad c \not\prec x$$

i.e. "c" is the maximal in "$\mathfrak{A}$*".*

definition 5.3.6. ***Well ordering*** *assume a structure "*$\mathfrak{A} := (A, \preceq)$*"("*$\preceq$*" is a Linear Order relation). we say "*$\mathfrak{A}$ *" is "Well ordered" when every nonempty subset of "A " has a "Minimum " i.e.*

$$WellOrdered(\mathfrak{A}) : \forall X \subseteq A \quad X \neq \varnothing \to HaveLeast(X)$$

definition 5.3.7. ***Lower and Upper Bound*** *assume a structure "*$\mathfrak{A} := (A, \preceq)$*"("*$\preceq$*" is a Order relation).* $B, C \subseteq A$*. we define "Lower and Upper Bound" like this*

$$Lowerb_{\mathfrak{A}}(b, B) : \forall x \in B \quad b \preceq x$$

i.e. "b" is the lowerbound of B *in "*$\mathfrak{A}$*".*

$$Upperb_{\mathfrak{A}}(c, C) : \forall x \in C \quad x \preceq c$$

i.e. "C" is the upperbound of B *in "*$\mathfrak{A}$*".*

if the structure clear to us we don't write structure index "$min(_), Minimal(_), Lowerb(_, _)$". and we don't mention it.

Theorem 5.3.5. ***Infimum and Supremum*** *assume a structure "*$\mathfrak{A} := (A, \preceq)$*"("*$\preceq$*" is a Order relation). and* $\mathcal{B} := \{B \subseteq A \mid HaveGreatest\{b \in A \mid Lowerb_{\mathfrak{A}}(b, B)\}\}$ *and* $\mathcal{C} := \{C \subseteq A \mid HaveLeast\{c \in A \mid Upperb_{\mathfrak{A}}(c, C)\}\}$ *we define "*$inf_{\mathfrak{A}} : \mathcal{B} \to V$*" and "*$sup_{\mathfrak{A}} : \mathcal{C} \to V$*" like this*

$$inf_{\mathfrak{A}}(B) := max_{\mathfrak{A}}\{b \in A \mid Lowerb_{\mathfrak{A}}(b, B)\}$$

$$sup_{\mathfrak{A}}(B) := min_{\mathfrak{A}}\{c \in A \mid Upperb_{\mathfrak{A}}(c, C)\}$$

Theorem 5.3.6. *assume a structure "*$\mathfrak{A} := (A, \preceq)$*"("*$\preceq$*" is a Order relation). and* $B \subseteq A$ *prove*

1. $min_{\mathfrak{A}}(B) = b / \therefore inf_{\mathfrak{A}}(B) = b$.

1	$min_{\mathfrak{A}}(B) = b$	*ass*
2	$b \in B$	*def* $min_{\mathfrak{A}}$ *1*
3	$\forall y \in B \quad b \preceq y$	*def* $min_{\mathfrak{A}}$ *1*
4	$Lowerb_{\mathfrak{A}}(b, B)$	*def* $Lowerb_{\mathfrak{A}}$ *4*
5	$b \in \{x \in A \mid Lowerb_{\mathfrak{A}}(x, B)\}$	*Def 5*
6	$\quad z \in \{x \in A \mid Lowerb_{\mathfrak{A}}(x, B)\}$	*acp*
7	$\quad Lowerb_{\mathfrak{A}}(z, B)$	*def 7*
8	$\quad \forall y \in B \quad z \preceq y$	*def* $Lowerb_{\mathfrak{A}}$ *8*
9	$\quad z \preceq b$	$\forall E$ *9*
10	$\forall z \in \{x \in A \mid Lowerb_{\mathfrak{A}}(x, B)\} \quad z \preceq b$	$\forall I$
11	$max_{\mathfrak{A}}(\{x \in A \mid Lowerb_{\mathfrak{A}}(x, B)\}) = b$	*def* $max_{\mathfrak{A}}$ *5 , 10*
12	$inf_{\mathfrak{A}}(B) = b$	*def* $inf_{\mathfrak{A}}$ *(13)*

2. $inf_{\mathfrak{A}}(B) = b, b \in B / \therefore min_{\mathfrak{A}}(B) = b.$

1	$inf_{\mathfrak{A}}(B) = b$	*ass*
2	$b \in B$	*ass*
3	$max_{\mathfrak{A}}(\{x \in A \mid Lowerb_{\mathfrak{A}}(x, B)\}) = b$	*def* $inf_{\mathfrak{A}}$ *2*
4	$b \in \{x \in A \mid Lowerb_{\mathfrak{A}}(x, B)\}$	*def* $max_{\mathfrak{A}}$ *4*
5	$Lowerb_{\mathfrak{A}}(b, B)$	*Def 6*
6	$\forall y \in B \quad b \preceq y$	*def* $Lowerb_{\mathfrak{A}}$ *7*
7	$min_{\mathfrak{A}}(B) = b$	*def* $min_{\mathfrak{A}}$ *2 , 6*

this theorem holds for "max, sup*" too.*

This page intentionally left blank.

Chapter 6

The beginning of Abstract Algebra

we want to introduce some functions "$\bigcirc : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$" to you. we use "$m \bigcirc n$" instead of "$\bigcirc(m, n)$".

6.1 Group theory

definition 6.1.1. *A structure* $\mathfrak{G} := (G, \cdot)$

1. *If* $\forall x, y, z \in G \quad (xy)z = x(yz)$ *(associative* $\times$*),* $\mathfrak{G}$ *is a* ***semigroup****.*
2. *If* $\mathfrak{G}$ *is a* ***semigroup*** *and* $\forall x \in G \quad \exists y \in G \quad \forall x \in G \quad xy = yx = x$ *(has identity* $\times$*),* $\mathfrak{G}$ *is a* ***monoid****. you can prove for every* ***monoid*** $\mathfrak{G}$ *identity element is unique we denote it by "e"(in* $\times$ *with "*1*" and in* $+$ *with* 0*).*
3. *If* $\mathfrak{G}$ *is a* ***monoid*** *and* $\forall x \in G \quad \exists y \in G \quad xy = yx = e$ *(has inverse* $\times$*),* $\mathfrak{G}$ *is a group.*
4. *If* $\mathfrak{G}$ *is a* ***semigroup/monoid*** *and* $\forall x, y, z \in G \quad xz = yz \to x = y$ *(cancellative* $\times$*),* $\mathfrak{G}$ *is a* ***cancellative semigroup/monoid****. (you can prove every group is cancellative as an exercise.*😀*)*
5. *If* $\mathfrak{G}$ *is a* ***semigroup/monoid/group*** *and* $\forall x, y \in G \quad xy = yx$ *(commutative(abelian)* $\times$*),* $\mathfrak{G}$ *is a* ***commutative(abelian) semigroup/monoid/group****.*

Theorem 6.1.1 (Grothendieck Group Construction)**.** *Let* $(S, +, 0)$ *be a cancellative commutative monoid.*

Then there exists an abelian group $G(S)$ *together with a monoid mbedding*

$$\eta : S \longrightarrow G(S)$$

such that for every abelian group A *and every monoid mbedding*

$$f : S \to A,$$

there exists a unique group mbedding

$$\tilde{f} : G(S) \to A$$

satisfying

$$\tilde{f} \circ \eta = f.$$

The canonical map is

$$\eta(a) = [(a, 0)].$$

Proof. **Step 0: Define relation.**

Define a relation $\sim$ on $S \times S$ by

$$(a, b) \sim (c, d) \leftrightarrow a + d = b + c.$$

Claim 0: $\sim$ is an equivalence relation.

Reflexivity

1	$(a, b) \in S \times S$	
2	$a + b = b + a$	
3	$(a, b) \sim (a, b)$	

Symmetry

1	$(a, b) \sim (c, d)$	
2	$a + d = b + c$	
3	$c + b = d + a$	
4	$(c, d) \sim (a, b)$	

Transitivity

1	$(a, b) \sim (c, d)$	
2	$(c, d) \sim (e, f)$	
3	$a + d = b + c$	
4	$c + f = d + e$	
5	$a + d + c + f = b + c + d + e$	add and combine, 3, 4
6	$a + f + (c + d) = b + e + (c + d)$	rearrange, 5
7	$a + f = b + e$	cancel $+(c + d)$, 6
8	$(a, b) \sim (e, f)$	

Step 1: Define Grothendieck group

$$G(S) := (S \times S)/\sim, \quad [(a,b)] \in G(S).$$

Step 2: Define operation

$$[(a,b)] + [(c,d)] := [(a+c, b+d)]$$

$$-[(a,b)] := [(b,a)]$$

—

Step 3: Well-definedness

(Addition and inverse are well-defined by compatibility of $\sim$ with $+$.)

—

Step 4: Group structure

(a,b)] + [(b,a)] = [(a+b, a+b)][(b,a)] = [(a+b, a+b)] (a+b,a+b)] = 01	+
0 2	=
3	every element has inverse
4	$G(S)$ is an abelian group

—

Step 5: Define extension

Let $f : S \to A$ be a monoid mbedding into an abelian group A.

$$\tilde{f}([(a,b)]) := f(a) - f(b).$$

—

Step 6: Well-definedness

1	$(a,b) \sim (c,d)$
2	$a + d = b + c$
3	$f(a) + f(d) = f(b) + f(c)$
4	$f(a) - f(b) = f(c) - f(d)$

—

Step 7: Homomorphism property

(Straightforward from group operations in A.)

—

Step 8: Factorization

$$\tilde{f}([(a,0)]) = f(a)$$

—

Step 9: Uniqueness

Any mbedding is forced by:

$$[(a,b)] = [(a,0)] - [(b,0)].$$

□

6.2 Ring theory

definition 6.2.1. *A structure* $\mathfrak{R} := (R, +, \cdot)$

1. *If* $(R,+)$ *is a abelian* ***monoid/group*** *and* $\forall x,y,z \in R \quad (xy)z = x(yz)$ *(associative* $\times$*),* $\forall x,y,z \in R \quad \Big(x(y+z) = xy + xz\Big) \wedge \Big((x+y)z = xz + yz\Big)$ *(distributive* $\times$*),* $\mathfrak{R}$ *is a* ***Semiring/Ring****.*
2. *If* $\mathfrak{R}$ *is a* ***Semiring/Ring*** *and* $\exists y \in R \quad \forall x \in R \quad xy = yx = x$ *(has identity* $\times$*),* $\mathfrak{R}$ *is a* ***Semiring/Ring with identity****. you can prove for every* ***Semiring/Ring with identity*** $\mathfrak{R}$ *identity element is unique we denote it by "*1*".*
3. *If* $\mathfrak{R}$ *is a* ***Semiring/Ring*** *and* $\forall x,y \in R \quad xy = 0 \to x = 0 \vee y = 0$ *(zero divisor free* $\times$*),* $\mathfrak{R}$ *is a* ***zero divisor free Semiring/Ring****. (you can prove every* ***division Semiring/Ring*** *is zero divisor free as an exercise.☺)*
4. *If* $\mathfrak{R}$ *is a* ***Semiring/Ring with identity*** *and* $\forall x \in R, \neq 0 \quad \exists y \in R \quad xy = yx = 1$ *(has inverse* $\times$*),* $\mathfrak{R}$ *is a* ***division Semiring/Ring*** *with.*
5. *If* $\mathfrak{R}$ *is a* ***Semiring/Ring*** *and* $\forall x,y \in R \quad xy = yx$ *(commutative(abelian)* $\times$*),* $\mathfrak{R}$ *is a* ***commutative(abelian) Semiring/Ring****.*
6. *If* $\mathfrak{R}$ *is a* ***zero divisor free Semiring/Ring*** *and* ***commutative(abelian) Semiring/Ring****.* $\mathfrak{R}$ *is a* ***Semidomain/Domain*** *.*
7. *If* $\mathfrak{R}$ *is a* ***division Semiring/Ring*** *and* ***commutative(abelian) Semiring/Ring****.* $\mathfrak{R}$ *is a* ***SemiField/Field*** *.*

Theorem 6.2.1 (Grothendieck Construction for Semirings)**.** *Let* $(S,+,\cdot)$ *be a semiring whose additive structure* $(S,+,0)$ *is a cancellative commutative monoid.*

Then there exists a ring $G(S)$ *together with a semiring mbedding*

$$\eta : S \longrightarrow G(S)$$

such that for every ring T *and every semiring mbedding*

$$f : S \to T,$$

there exists a unique ring mbedding

$$\tilde{f} : G(S) \to T$$

satisfying

$$\tilde{f} \circ \eta = f.$$

The canonical map is

$$\eta(a) = [(a, 0)].$$

Proof. **Step 0: Define the underlying relation.**

Define a relation $\sim$ on $S \times S$ by

$$(a, b) \sim (c, d) \leftrightarrow a + d = b + c.$$

Claim 0: $\sim$ is an equivalence relation.

Reflexivity

1	$(a, b) \in S \times S$
2	$a + b = b + a$
3	$(a, b) \sim (a, b)$

Symmetry

1	$(a, b) \sim (c, d)$
2	$a + d = b + c$
3	$c + b = d + a$
4	$(c, d) \sim (a, b)$

Transitivity

1	$(a, b) \sim (c, d)$	
2	$(c, d) \sim (e, f)$	
3	$a + d = b + c$	
4	$c + f = d + e$	
5	$a + d + c + f = b + c + d + e$	add and combine, 3, 4
6	$a + f + (c + d) = b + e + (c + d)$	rearrange, 5
7	$a + f = b + e$	cancel $+(c + d)$, 6
8	$(a, b) \sim (e, f)$	

(additive commutative monoid cancellation NOT required; we instead justify via rearrangement in monoid equality)

Step 1: Define the carrier set.

Define

$$G(S) := (S \times S)/\sim, \quad [(a,b)] \in G(S).$$

Step 2: Define operations.

Define

$$[(a,b)] + [(c,d)] := [(a+c,\ b+d)],$$
$$[(a,b)] \cdot [(c,d)] := [(ac+bd,\ ad+bc)].$$

Define

$$\eta(a) := [(a,0)].$$

Step 3: Well-definedness of addition.

1	$(a,b) \sim (a',b'),\ (c,d) \sim (c',d')$
2	$a+b' = a'+b,\ c+d' = c'+d$
3	$(a+c)+(b'+d') = (a'+c')+(b+d)$
4	$[(a+c,b+d)] = [(a'+c',b'+d')]$
5	$+$ is well-defined

Step 4: Well-definedness of multiplication.

1	$(a,b) \sim (a',b'),\ (c,d) \sim (c',d')$
2	$a+b' = a'+b,\ c+d' = c'+d$
3	$ac+bd+(a'd'+b'c') = a'c'+bd+ac'+b'd$
4	$[(ac+bd,ad+bc)] = [(a'c'+b'd',a'd'+b'c')]$
5	$\cdot$ is well-defined

Step 5: Ring structure.

1	$[(a,b)] + [(c,d)] = [(a+c,b+d)]$
2	associativity inherited from S
3	$[(a,b)] \cdot [(c,d)]$ associative
4	$distributivity follows from semiring laws$
5	$(0,0)$ is additive identity
6	$[(a,a)] = 0$ in $G(S)$
7	$G(S)$ is a ring

Step 6: Define extension of a mbedding.

Let

$$f : S \to T$$

be a semiring mbedding into a ring T.

Define

$$\tilde{f}([(a,b)]) := f(a) - f(b).$$

Claim 6: $\tilde{f}$ is well-defined.

1 $(a,b) \sim (c,d)$

2 $a + d = b + c$

3 $f(a) + f(d) = f(b) + f(c)$

4 $f(a) - f(b) = f(c) - f(d)$

5 $\tilde{f}([(a,b)]) = \tilde{f}([(c,d)])$

Step 7: $\tilde{f}$ is a ring mbedding.

1 $\tilde{f}([(a,b)] + [(c,d)]) = f(a+c) - f(b+d)$

2 $= (f(a) - f(b)) + (f(c) - f(d))$

3 $= \tilde{f}([(a,b)]) + \tilde{f}([(c,d)])$

4 $\tilde{f}([(a,b)] \cdot [(c,d)]) = f(ac+bd) - f(ad+bc)$

5 $= (f(a) - f(b))(f(c) - f(d))$

6 $= \tilde{f}([(a,b)])\tilde{f}([(c,d)])$

Step 8: Factorization property.

1 $\tilde{f}(\eta(a)) = \tilde{f}([(a,0)])$

2 $= f(a)$

3 $\tilde{f} \circ \eta = f$

Step 9: Uniqueness.

1 $g : G(S) \to T$ ring mbedding

2 $g \circ \eta = f$

3 $g([(a,b)]) = g([(a,0)]) - g([(b,0)])$

4 $= f(a) - f(b)$

5 $= \tilde{f}([(a,b)])$

6 $g = \tilde{f}$

Therefore $G(S)$ satisfies the universal property of the Grothendieck construction.

□

Theorem 6.2.2 (Field of Fractions)**.** *Let R be a integral domain.*

Then there exists a field $\mathrm{Frac}(R)$ *together with a ring embedding*

$$\eta : R \longrightarrow \mathrm{Frac}(R)$$

such that for every field K and every ring embedding

$$f : R \to K,$$

there exists a unique field embedding

$$\tilde{f} : \mathrm{Frac}(R) \to K$$

satisfying

$$\tilde{f} \circ \eta = f.$$

The canonical map is

$$\eta(a) = \frac{a}{1}.$$

Proof. **Step 0: Define relation.**

Define a relation $\sim$ on $R \times (R - \{0\})$ by

$$(a, b) \sim (c, d) \leftrightarrow ad = bc.$$

Claim 0: $\sim$ is an equivalence relation.

Reflexivity

1	$(a, b) \in R \times (R - \{0\})$
2	$ab = ba$
3	$(a, b) \sim (a, b)$

Symmetry

1	$(a, b) \sim (c, d)$
2	$ad = bc$
3	$cb = da$
4	$(c, d) \sim (a, b)$

Transitivity

1	$ad = bc$	
2	$cf = de$	
3	$adf = bcf$	multiply by f, 1
4	$bcf = bde$	substitute, 2
5	$adf = bde$	
6	$(af)d = (be)d$	rearrange
7	R is an integral domain	
8	$d \neq 0$	
9	$af = be$	cancellation (no zero divisors)
10	$(a, b) \sim (e, f)$	

—

Step 1: Define field of fractions

$$\text{Frac}(R) := (R \times (R - \{0\}))/\sim$$

$$[(a, b)] := \frac{a}{b}$$

—

Step 2: Define operations

$$[(a, b)] + [(c, d)] := [(ad + bc, bd)]$$

$$[(a, b)] \cdot [(c, d)] := [(ac, bd)]$$

—

Step 3: Define embedding

$$\eta(a) := [(a, 1)]$$

—

Step 4: Well-definedness

(Follows from cross-multiplication and compatibility of $\sim$.)

—

Step 5: Field structure

1 $[(a,b)] \cdot [(b,a)] = [(ab,ab)]$

2 $[(ab,ab)] = 1$

3 every nonzero element has inverse

4 $\mathrm{Frac}(R)$ is a field

—

Step 6: Extension map

Let $f : R \to K$ be a ring embedding into a field K.

$$\tilde{f}([(a,b)]) := \frac{f(a)}{f(b)}$$

—

Step 7: Well-definedness

1 $(a,b) \sim (c,d)$

2 $ad = bc$

3 $f(a)f(d) = f(b)f(c)$

4 $\frac{f(a)}{f(b)} = \frac{f(c)}{f(d)}$

—

Step 8: Homomorphism and factorization

$$\tilde{f}(\eta(a)) = \tilde{f}([(a,1)]) = f(a)$$

—

Step 9: Uniqueness

$$[(a,b)] = \frac{[(a,1)]}{[(b,1)]} \Rightarrow \tilde{f} \text{ is forced}$$

□

Chapter 7

Natural numbers

7.1 Introduction to Natural numbers

definition 7.1.1. ***Natural numbers*** *we define "*$0, 1, 2, 3, ...$*" like this.*

$$\begin{aligned}
&0 := \varnothing \\
&1 := \{0\} = \varnothing \cup \{\varnothing\} = \{\varnothing\} \\
&2 := \{0, 1\} = \{\varnothing\} \cup \big\{\{\varnothing\}\big\} = \big\{\varnothing, \{\varnothing\}\big\} \\
&3 := \{0, 1, 2\} = \Big\{\varnothing, \{\varnothing\}, \big\{\varnothing, \{\varnothing\}\big\}\Big\} \\
&\vdots
\end{aligned}$$

definition 7.1.2. ***Successor function*** *is a function "*$s : V \to V$*" defined like this.*

$$s(x) := x \cup \{x\}$$

you can see $\forall x \quad Type_0(x) \to Type_0(s(x))$
definition 7.1.3. *we define a set* ***"Inductive"*** *like this.*

$$Ind(X) : \varnothing \in X \wedge \forall y(y \in X \to s(y) \in X)$$

notice that "$Ind(X)$" might be empty.
Axiom 7.1.1. ***The Axiom of Infinity*** *we define a set "Inductive" like this.*

$$\exists X \quad Type_0(X) \wedge Ind(X)$$

so "$Ind(X)$" is a nonempty set. so we can define
definition 7.1.4. ***Set of Natural numbers(***$\mathbb{N}$***)***

$$\mathbb{N} := \bigcap Ind$$

Lemma 7.1.1. *prove.*

1. $\mathbb{N} = \{x \mid \underbrace{\forall Y\Big(\big(\underbrace{\varnothing \in Y \wedge \forall z(z \in Y \to s(z) \in Y)}_{Ind(Y)}\big) \to x \in Y\Big)}_{\phi(x)}\}$
2. $Ind(\mathbb{N})$
   Proof Is Left As An Exercise To The Reader.

notice that now "$\phi(x)$" is a proposition. so "$\{x \mid \forall Y\Big(\big(\varnothing \in Y \wedge \forall z(z \in Y \to s(z) \in Y)\big) \to x \in Y\Big)\}$" is a set.

7.2 Properties of Natural numbers

Theorem 7.2.1. ***Induction*** $P(0), \forall n \in \mathbb{N} \quad \Big(P(n) \to P(s(n))\Big) / \therefore \forall n \in \mathbb{N} \quad P(n)$
proof.

1	$S := \{x \in \mathbb{N} \mid P(x)\}$	def
2	$P(0)$	ass
3	$\forall n \in \mathbb{N} \quad \Big(P(n) \to P(s(n))\Big)$	ass
4	$\quad n \in \mathbb{N}$	acp
5	$\quad 0 \in S$	$1, 2$
6	$\quad\quad n \in S$	acp
7	$\quad\quad P(n)$	$comprhension$
8	$\quad\quad P(n) \to P(s(n))$	$\forall E, \to E, 3, 4$
9	$\quad\quad P(s(n))$	$\to E, 7, 8$
10	$\quad\quad s(n) \in S$	$4, 9$
11	$\quad \forall n\Big(n \in S \to s(n) \in S\Big)$	$1, 3$
12	$\quad Ind(S)$	$5, 6$
13	$\quad n \in \bigcap Ind$	def
14	$\quad n \in S$	$\bigcap E$
15	$\quad P(n)$	$comprhension$
16	$\forall n \in \mathbb{N} \quad P(n)$	$\to I, \forall I$

definition 7.2.1. *A set "T" is* ***Transitive*** *if $u \in v \in T$ implies $u \in T$.*

$$Transitive(T) : \forall u, v \quad u \in v \in T \to u \in T$$

Lemma 7.2.1. .

1. $\forall n \in \mathbb{N} \quad Transitive(n)$
2. $Transitive(\mathbb{N})$ 😀

proof.

1	$\forall v \quad v \notin 0$	
2	$\forall u, v \quad u \in v \in 0 \to u \in 0$	
3	$Transitive(0)$	
4	$\quad n \in \mathbb{N}$	acp
5	$\quad Transitive(n)$	acp
6	$\quad\quad u \in v \wedge v \in s(n)$	acp
7	$\quad\quad v \in n \cup \{n\}$	$\wedge E$
8	$\quad\quad\quad v \in n$	$\wedge E$
9	$\quad\quad\quad u \in n$	$5, 6, 8$
10	$\quad\quad\quad u \in s(n)$	$\cup I$
11	$\quad\quad\quad v \in \{n\}$	$\wedge E$
12	$\quad\quad\quad v = n$	
13	$\quad\quad\quad u \in n$	$= E$
14	$\quad\quad\quad u \in s(n)$	$\cup I$
15	$\quad\quad u \in s(n)$	
16	$\quad \forall u, v \quad u \in v \in s(n) \to u \in s(n)$	
17	$\quad Transitive(s(n))$	
18	$\forall n \in \mathbb{N} \quad Transitive(n) \to Transitive(s(n))$	
19	$\forall n \in \mathbb{N} \quad Transitive(n)$	*Induction theorem*

Proof Is Left As An Exercise To The Reader.

definition 7.2.2. ***Order in Natural numbers($\mathbb{N}$)***

$$< := \{(x, y) \in \mathbb{N} \times \mathbb{N} \mid x \in y\}$$

Theorem 7.2.2. ***Order of Natural numbers($\mathbb{N}$)*** *prove*

1. $(\mathbb{N}, <)$ *is linearly ordered.* 😀
2. $\forall n \in \mathbb{N} \quad (n, <)$ *is Well ordered.*
3. $(\mathbb{N}, <)$ *is Well ordered.*

4. $\Big(\forall x \in \mathbb{N} \quad \exists y \in \mathbb{N} \quad x < y\Big)$ *(doesn't have right endpoint)*

proof.

1	$0 \not< 0$	$def\varnothing$
2	$\quad n \in \mathbb{N}$	acp
3	$\quad\quad n \not< n$	acp
4	$\quad\quad\quad s(n) < s(n)$	acp
5	$\quad\quad\quad\quad s(n) \in n$	acp
6	$\quad\quad\quad\quad n \in s(n)$	$\cup I$
7	$\quad\quad\quad\quad n \in n$	$Transitive(\mathbb{N})$
8	$\quad\quad\quad\quad \bot$	$3, 7$
9	$\quad\quad\quad\quad s(n) \in \{n\}$	acp
10	$\quad\quad\quad\quad s(n) = n$	
11	$\quad\quad\quad\quad s(n) \not< s(n)$	$= E$
12	$\quad\quad\quad\quad \bot$	$4, 11$
13	$\quad\quad\quad \bot$	$\cup E$
14	$\quad\quad s(n) \not< s(n)$	$\neg I$
15	$\forall n \in \mathbb{N} \quad n \not< n$	*Induction theorem*

1	$\forall n, m, k \in \mathbb{N} \quad n < m < k \to n < k$	*Induction theorem*

Proof Is Left As An Exercise To The Reader.

Theorem 7.2.3. ***Induction ,second version*** $\Big(\forall k < n \quad P(k)\Big) \to P(n) / \therefore \forall n \in \mathbb{N} \quad P(n)$

proof.

7.3 The Recursion theorem

definition 7.3.1. ***Tuple (new definition)-Finite sequence*** *is a function "$a : n \to V$" s.t. $n \in \mathbb{N}$ we know we can show this by "$(a_i)_{i\in n}$" we can also show it by "$(a_0, a_1, a_2, ..., a_{n-1})$"*

definition 7.3.2. ***Sequence*** *is a function "$a : \mathbb{N} \to V$" we know we can show this by "$(a_i)_{i\in\mathbb{N}}$" we can also show it by "$(a_i)_{i=0}^{\infty}$"*

now we can redefine "Cartesian product". we define finite Cartesian product "$\prod_{i\in n} A_i := \{(a_i)_{i\in n} \mid a_i \in A_i\}$" and define infinite Cartesian product "$\prod_{i\in\mathbb{N}} A_i := \{(a_i)_{i\in\mathbb{N}} \mid a_i \in A_i\}$" and define Cartesian product in general way "$\prod_{i\in I} A_i := \{(a_i)_{i\in I} \mid a_i \in A_i\}$". and you can easily see that from the definition of "B^A", "A^n" is the set of all n-length finite sequences of $a \in A$ and "$A^{\mathbb{N}}$" is the set of all infinite sequences of $a \in A$.

Theorem 7.3.1 (Recursion)**.** *For any set A, any $a \in A$, and any function $g : A \times \mathbb{N} \to A$, there is a unique function $f : \mathbb{N} \to A$ such that:*

1. $f(0) = a$.

2. $\forall n \in \mathbb{N} \quad f(s(n)) = g(f(n), n)$.

Proof. Define

$$Compute_n(t_n) : t_n : s(n) \to A \ \wedge \ t_n(0) = a \ \wedge \ \forall k < n \ \big(t(s(k)) = g(t(k), k)\big).$$

$$F := \Big\{ t_n \ \Big| \ Compute_n(t_n) \Big\}.$$

Claim 0: $\forall n \in \mathbb{N} \quad \exists t_n \in F$**.**

Claim 1: $F \neq \varnothing$**.**

1 $t_0 \in F$

2 $F \neq \varnothing$

Define

$$f := \bigcup F.$$

Claim 2: f **is a function.**

1 | $t_m, t_n \in F$
2 | | $Dom(t_m) = s(m)$
3 | | $Dom(t_n) = s(n)$
4 | | | $m \leq n$
5 | | | $t_m(0) = a$
6 | | | $t_n(0) = a$
7 | | | $t_m(0) = t_n(0)$
8 | | | | $k < m$
9 | | | | | $t_m(k) = t_n(k)$
10 | | | | | $g(t_m(k), k) = g(t_n(k), k)$
11 | | | | | $t_m(s(k)) = g(t_m(k), k)$
12 | | | | | $t_n(s(k)) = g(t_n(k), k)$
13 | | | | | $t_m(s(k)) = t_n(s(k))$
14 | | | | $\forall k \leq m \quad t_m(k) = t_n(k)$
15 | | $t_m \subseteq t_n \vee t_n \subseteq t_m$
16 $\forall t_m, t_n \in F \quad (t_m \subseteq t_n \vee t_n \subseteq t_m)$
17 $Compatible(F)$
18 $Func(f)$

Claim 3: Dom$(f) = \mathbb{N}$ and Range$(f) \subseteq A$.

1 | $n \in \mathbb{N}$
2 | $\exists t_n \in F$
3 | $n \in Dom(t_n)$
4 | $n \in Dom(f)$
5 | $n \in Dom(f)$
6 | | $t_m \in F$
7 | | $n \in Dom(t_m)$
8 | | $Dom(t_m) = s(m)$
9 | | $n \in s(m)$
10 | | $n \in \mathbb{N}$
11 | $n \in \mathbb{N}$
12 $Dom(f) = \mathbb{N}$
13 | $x \in Range(f)$
14 | $(n, x) \in f$
15 | | $t_m \in F$
16 | | $(n, x) \in t$
17 | | $t_m : s(m) \to A$
18 | | $x \in A$
19 | $x \in A$
20 $Range(f) \subseteq A$

Claim 4: f satisfies the recursion equations.

1 $f(0) = a$
2 | $n \in \mathbb{N}$
3 | $\exists t_{n+1} \in F$
4 | $t_{n+1}(s(n)) = g(t_{n+1}(n), n)$
5 | $t_{n+1} \subseteq f$
6 | $f(s(n)) = g(f(n), n)$
7 $\forall n \in \mathbb{N} \quad f(s(n)) = g(f(n), n)$

Claim 5 : Uniqueness.

1 | $h : \mathbb{N} \to A$
2 | $h(0) = a$
3 | $\forall n \in \mathbb{N} \quad h(s(n)) = g(h(n), n)$
4 | $f(0) = h(0)$
5 | | $f(n) = h(n)$
6 | | $g(f(n), n) = g(h(n), n)$
7 | | $f(s(n)) = g(f(n), n)$
8 | | $h(s(n)) = g(h(n), n)$
9 | | $f(s(n)) = h(s(n))$
10 | $\forall n \in \mathbb{N} \quad f(n) = h(n)$
11 | $f = h$
12 $\exists! f : \mathbb{N} \to A \quad \Big(f(0) = a \wedge \forall n \in \mathbb{N} \quad f(s(n)) = g(f(n), n)\Big)$

Therefore, there exists a unique function $f : \mathbb{N} \to A$ such that $f(0) = a$ and $\forall n \in \mathbb{N}\ f(s(n)) = g(f(n), n)$. □

Theorem 7.3.2 (Parametric Recursion Theorem)**.** *For any sets "A, P", any function "$a : P \to A$", and any function "$g : P \times A \times \mathbb{N} \to A$", there is a unique function "$f : P \times \mathbb{N} \to A$" such that*

1. $$\forall p \in P \quad f(p, 0) = a(p)$$

2. $$\forall p \in P\ \forall n \in \mathbb{N} \quad f\big(p, s(n)\big) = g\big(p, f(p, n), n\big)$$

Proof. Fix $p \in P$.

Define

$$G_p(x, n) := g(p, x, n).$$

Claim 1 : $G_p : A \times \mathbb{N} \to A$.

1 | $p \in P$
2 | | $(x, n) \in A \times \mathbb{N}$
3 | | $g(p, x, n) \in A$
4 | | $G(p)(x, n) \in A$
5 | $G(p) : A \times \mathbb{N} \to A$
6 $\forall p \in P \quad G(p) : A \times \mathbb{N} \to A$

For every $p \in P$, by the ordinary Recursion Theorem applied to $a(p) \in A$ and $G(p) : A \times \mathbb{N} \to A$, there exists a unique function

$$F_p : \mathbb{N} \to A$$

such that

$$F_p(0) = a(p)$$

and

$$\forall n \in \mathbb{N} \quad F_p(s(n)) = G(p)(F_p(n), n).$$

Define

$$f(p, n) := F_p(n).$$

Claim 2 : $f : P \times \mathbb{N} \to A$.

1	$(p, n) \in P \times \mathbb{N}$
2	$F_p : \mathbb{N} \to A$
3	$F_p(n) \in A$
4	$f(p, n) \in A$
5	$f : P \times \mathbb{N} \to A$

Claim 3 : f satisfies the recursion equations.

1	$p \in P$
2	$F_p(0) = a(p)$
3	$f(p, 0) = a(p)$
4	$\forall p \in P \quad f(p, 0) = a(p)$
5	$p \in P$
6	$n \in \mathbb{N}$
7	$F_p(s(n)) = G(p)(F_p(n), n)$
8	$G(p)(F_p(n), n) = g(p, F_p(n), n)$
9	$f(p, s(n)) = g(p, f(p, n), n)$
10	$\forall p \in P \; \forall n \in \mathbb{N} \quad f(p, s(n)) = g(p, f(p, n), n)$

Define

$$H_p(n) := h(p, n).$$

Claim 4 : Uniqueness.

1 | $h : P \times \mathbb{N} \to A$
2 | $\forall p \in P \quad h(p, 0) = a(p)$
3 | $\forall p \in P \; \forall n \in \mathbb{N} \quad h(p, s(n)) = g(p, h(p, n), n)$
4 | | $p \in P$
5 | | $H_p(0) = a(p)$
6 | | $\forall n \in \mathbb{N} \quad H_p(s(n)) = g(p, H_p(n), n)$
7 | | $F_p = H_p$
8 | | | $n \in \mathbb{N}$
9 | | | $f(p, n) = F_p(n)$
10 | | | $H_p(n) = h(p, n)$
11 | | | $f(p, n) = h(p, n)$
12 | | $\forall n \in \mathbb{N} \quad f(p, n) = h(p, n)$
13 | $\forall p \in P \; \forall n \in \mathbb{N} \quad f(p, n) = h(p, n)$
14 | $f = h$
15 $\exists! f : P \times \mathbb{N} \to A \quad \Big(\forall p \in P \quad f(p, 0) = a(p) \wedge \forall p \in P \forall n \in \mathbb{N} \quad f(p, s(n)) = g(p, f(p, n), n)\Big)$

Therefore, there exists a unique function $f : P \times \mathbb{N} \to A$ such that

$$\forall p \in P \quad f(p, 0) = a(p)$$

and

$$\forall p \in P \; \forall n \in \mathbb{N} \quad f(p, s(n)) = g(p, f(p, n), n).$$

□

7.4 Arithmetic of Natural numbers

Theorem 7.4.1. ***Summation*** *there is a unique function "$+ : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$" s.t.*

1. $\forall m \in \mathbb{N} \quad m + 0 = m$
2. $\forall m, n \in \mathbb{N} \quad m + s(n) = s(m + n)$

Proof Is Left As An Exercise To The Reader.
Theorem 7.4.2. ***Multiplication*** *there is a unique function "$\times : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$" s.t.*

1. $\forall m \in \mathbb{N} \quad m \times 0 = 0$
2. $\forall m, n \in \mathbb{N} \quad m \times s(n) = m \times n + m$

Proof Is Left As An Exercise To The Reader.

we denote also "Multiplication" with "$m \cdot n$" and "mn" too.

7.5 Peano arithmetic

assume a structure "$\mathfrak{N} := (N, +, \cdot, s, 0)$"(here "$+, \cdot, s, 0$" are arbitrary functions and constants and doesn't have meaning we defined before). with these properties.

1. $\forall x, y \in N \quad s(x) = s(y) \to x = y$
2. $\forall x \in N \quad s(x) \neq 0$
3. $\forall x \in N \quad x + 0 = x$
4. $\forall x, y \in N \quad x + s(y) = s(x + y)$
5. $\forall x \in N \quad x \cdot 0 = 0$
6. $\forall x, y \in N \quad x \cdot s(y) = x \cdot y + x$
7. $\phi(0) \wedge \forall y \in N \quad \Big(\phi(y) \to \phi(s(y))\Big) \to \forall y \in N \quad \phi(y)$ (this is true for arbitrary "ϕ")

we call "$\mathfrak{N}$" a model of "Peano arithmetic".
Lemma 7.5.1. *prove for "$(N, +)$"*

1. $\forall x, y, z \in N \quad (x + y) + z = x + (y + z)$ *(associative +)*
2. $\forall x \in N \quad x + 0 = x = 0 + x$ *(identity +)*
3. $\forall x, y, z \in N \quad x + z = y + z \to x = y$ *(cancellative +)*
4. $\forall x, y \in N \quad x + y = y + x$ *(commutative(abelian) +)*

1	$x, y, z \in N$	*acp*
2	$(x + y) + 0 = x + y = x + (y + 0)$	*Ax. 3*
3	$\quad (x + y) + z = x + (y + z)$	*acp*
4	$\quad (x + y) + s(z) = s\big((x + y) + z\big) = s\big(x + (y + z)\big) = x + s(y + z) = x + (y + s(z))$	*Ax. 4 , 3*
5	$(x + y) + z = x + (y + z)$	*Ax. Ind*
6	$\forall x, y, z \in N \quad (x + y) + z = x + (y + z)$	$\to I, \forall I$

1	$x \in N$	*acp*
2	$0+0=0$	*Ax. 3*
3	$\quad 0+x=x$	*acp*
4	$\quad 0+s(x)=s(0+x)=s(x)$	*Ax. 3 , 4*
5	$0+x=x$	*Ax. Ind*
6	$\forall x \in N \quad x+0=0+x=x$	$\to I, \forall I$

1	$x,y \in N$	*acp*
2	$y+s(0)=s(y+0)=s(y)=s(y)+0$	*Ax. 3*
3	$\quad y+s(x)=s(y)+x$	*acp*
4	$\quad y+s(s(x))=s(y+s(x))=s(s(y)+x)=s(y)+s(x)$	*Ax. 4 , 3*
5	$x+y=y+x$	*Ax. Ind*
6	$\forall x,y \in N \quad y+s(x)=s(y)+x$	$\to I, \forall I$

1	$x,y \in N$	*acp*
2	$x+0=0+x$	*Lemma 7.5.1.2*
3	$\quad x+y=y+x$	*acp*
4	$\quad x+s(y)=s(x+y)=s(y+x)=y+s(x)=s(y)+x$	*Ax. Ind*
5	$x+y=y+x$	*Ax. Ind*
6	$\forall x,y \in N \quad x+y=y+x$	$\to I, \forall I$

Proof Is Left As An Exercise To The Reader.☺

Lemma 7.5.2. *prove for "$(N,+,\cdot)$"*

1. $\forall x,y,z \in N \quad (xy)z=x(yz)$ *(associative $\times$)*
2. $\forall x \in N \quad x \cdot 1 = 1 \cdot x = x$ *(identity $\times$)*
3. $\forall x,y \in N \quad xy=0 \to x=0 \vee y=0$ *(zero divisor free $\times$)*
4. $\forall x,y \in N \quad xy=yx$ *(commutative(abelian) $\times$)*
5. $\forall x,y,z \in N \quad \Big(x(y+z)=xy+xz\Big) \wedge \Big((x+y)z=xz+yz\Big)$ *(distributive $\times$)*

Proof Is Left As An Exercise To The Reader.☺

1	$x \in N$	*acp*
2	$0 \cdot 0 = 0$	*Ax. 5*
3	$\quad 0 \cdot x = 0$	*acp*
4	$\quad 0 \cdot s(x) = 0 \cdot x + 0 = 0$	
5	$0 \cdot x = 0$	*Ax. Ind*
6	$\forall x, y \in N \quad 0 \cdot x = 0$	$\to I, \forall I$

1	$x, y \in N$	*acp*
2	$0 + y0 = s(y) \cdot 0$	*Ax. 5*
3	$\quad x + yx = s(y) \cdot x$	*acp*
4	$\quad s(x) + yx = x + s(yx) = s(x + yx) = s(s(y) \cdot x) = s(y) \cdot s(x)$	
5	$x + yx = s(y) \cdot x$	*Ax. Ind*
6	$\forall x, y \in N \quad x + yx = s(y) \cdot x$	$\to I, \forall I$

1	$x, y \in N$	*acp*
2	$x \cdot 0 = 0$	*Ax. 5*
3	$x \cdot 0 = 0 \cdot x$	*Lemma*
4	$\quad xy = yx$	*acp*
5	$\quad x \cdot s(y) = xy + x = yx + x = x + yx = s(y) \cdot x$	
6	$xy = yx$	*Ax. Ind*
7	$\forall x, y \in N \quad xy = yx$	$\to I, \forall I$

Now define "$x < y : \exists z \neq 0 \quad x + z = y$". You can prove for natural numbers

$$\forall m, n \in \mathbb{N} \quad m < n \leftrightarrow \exists k \neq 0 \quad m + k = n$$

Lemma 7.5.3. *prove for "$(N, +, \times, <)$"*

1. $\forall x, y, z \in N \quad x < y \to x + z < y + z$
2. $\forall x, y, z \in N \quad x < y \wedge 0 < z \to xz < yz$

Proof Is Left As An Exercise To The Reader.😀

as you can see "$(\mathbb{N}, +, \times, s, 0)$" is a model of peano arithmetic. so properties are true for them.

This page intentionally left blank.

Chapter 8

From Natural numbers to Complex numbers

8.1 Integers"$\mathbb{Z}$"

we learned about "$\mathbb{N}$" before and we constructed structure "$(\mathbb{N}, +, \cdot, <)$". if you assume "$\mathbb{N} \times \mathbb{N}$" we can define an equivalence relation "$(n_1, m_1) \sim (n_2, m_2) : n_1 + m_2 = n_2 + m_1$"(check it's an equivalence relation).

definition 8.1.1. ***Integers"$\mathbb{Z}$"*** *is a set*

$$\mathbb{Z} := \frac{\mathbb{N} \times \mathbb{N}}{\sim}$$

we denote it's members with "$z_0, z_1, \ldots$"

in this system "$0 := [(0,0)]$" and "$\forall n \in \mathbb{N} \quad n := [(n,0)]$" and "$\forall n \in \mathbb{N} \quad -n := [(0,n)]$". in general"$\forall m, n \in \mathbb{N} \quad m - n := [(m,n)]$". so we can say "$\mathbb{N} \subseteq \mathbb{Z}$".

Theorem 8.1.1. ***Summation*** *there is a unique function "$+_{\mathbb{Z}} : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$" s.t.*

$$\forall m_1, m_2, n_1, n_2 \in \mathbb{N} \quad [(m_1, n_1)] +_{\mathbb{Z}} [(m_2, n_2)] = [(m_1 + m_2, n_1 + n_2)]$$

Proof Is Left As An Exercise To The Reader.

Theorem 8.1.2. ***Multiplication*** *there is a unique function "$\times_{\mathbb{Z}} : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$" s.t.*

$$\forall m_1, m_2, n_1, n_2 \in \mathbb{N} \quad [(m_1, n_1)] \times_{\mathbb{Z}} [(m_2, n_2)] = [(m_1 m_2 + n_1 n_2, m_1 n_2 + m_2 n_1)]$$

Proof Is Left As An Exercise To The Reader.

definition 8.1.2. ***Order in*** $\mathbb{Z}$

$$[(m_1, n_1)] <_{\mathbb{Z}} [(m_2, n_2)] : m_1 + n_2 < m_2 + n_1$$

this lemma let us use "$<$" instead of "$<_{\mathbb{Z}}$", "$+$" instead of "$+_{\mathbb{Z}}$" and "$\times$" instead of "$\times_{\mathbb{Z}}$".

Lemma 8.1.1. *prove.*

1. $/\therefore \forall m,n \in \mathbb{N} \quad m <_{\mathbb{Z}} n \leftrightarrow m < n$
2. $/\therefore \forall m,n \in \mathbb{N} \quad m +_{\mathbb{Z}} n = m + n$
3. $/\therefore \forall m,n \in \mathbb{N} \quad m \times_{\mathbb{Z}} n = m \times n$

Proof Is Left As An Exercise To The Reader.

we denote "Multiplication" with "$z \cdot y$" and "zy" too.
Lemma 8.1.2. *prove for "$(\mathbb{Z},+)$"*

1. $\forall x,y,z \in \mathbb{Z} \quad (x+y)+z = x+(y+z)$ *(associative $+$)*
2. $\forall z \in \mathbb{Z} \quad z+0 = 0+z = z$ *(identity $+$)*
3. $\forall z \in \mathbb{Z} \quad \exists y \in \mathbb{Z} \quad z+y = y+z = 0$ *(inverse $+$)*
4. $\forall x,y \in \mathbb{Z} \quad x+y = y+x$ *(commutative(abelian) $+$)*

Proof Is Left As An Exercise To The Reader.
Lemma 8.1.3. *prove for "$(\mathbb{Z},+,\times)$"*

1. $\forall x,y,z \in \mathbb{Z} \quad (xy)z = x(yz)$ *(associative $\times$)*
2. $\forall z \in \mathbb{Z} \quad z \cdot 1 = 1 \cdot z = z$ *(identity $\times$)*
3. $\forall x,y \in \mathbb{Z} \quad xy = 0 \rightarrow x = 0 \vee y = 0$ *(zero divisor free $\times$)*
4. $\forall x,y \in \mathbb{Z} \quad xy = yx$ *(commutative(abelian) $\times$)*
5. $\forall x,y,z \in \mathbb{Z} \quad \Big(x(y+z) = xy + xz\Big) \wedge \Big((x+y)z = xz + yz\Big)$ *(distributive $\times$)*

Proof Is Left As An Exercise To The Reader.
Theorem 8.1.3. ***Order of Integers($\mathbb{Z}$)*** *prove for "$(\mathbb{Z},+,\times,<_{\mathbb{Z}})$"*

1. *is linearly ordered.*
2. $\Big(\forall x \in \mathbb{Z} \quad \exists y \in \mathbb{Z} \quad x < y\Big) \wedge \Big(\forall x \in \mathbb{Z} \quad \exists y \in \mathbb{Z} \quad y < x\Big)$ *(doesn't have endpoints)*
3. $\forall x,y,z \in \mathbb{Z} \quad x < y \rightarrow x+z < y+z$
4. $\forall x,y,z \in \mathbb{Z} \quad x < y \wedge 0 < z \rightarrow xz < yk$

Proof Is Left As An Exercise To The Reader.

8.2 Rational numbers "$\mathbb{Q}$"

we learned about "$\mathbb{Z}$" before and we constructed structure "$(\mathbb{Z},+,\cdot,<)$". if you assume "$\mathbb{Z} \times (\mathbb{Z} - \{0\})$" we can define an equivalence relation "$(y_1,x_1) \sim (y_2,x_2) : y_1 \cdot x_2 = y_2 \cdot x_1$"(check it's an equivalence relation).
definition 8.2.1. ***Rational numbers"$\mathbb{Q}$"*** *is a set*

$$\mathbb{Q} := \frac{\mathbb{Z} \times (\mathbb{Z} - \{0\})}{\sim}$$

we denote it's members with "$q_0, q_1, \ldots$"

in this system "$0 := [(0,1)]$" and "$1 := [(1,1)]$" and "$\forall x \in \mathbb{Z} \quad x := [(x,1)]$" and "$\forall x \in \mathbb{Z} \quad \frac{1}{x} := [(1,x)]$". in general "$\forall x,y \in \mathbb{Z} \quad \frac{x}{y} := [(x,y)]$". so we can say "$\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q}$".

Theorem 8.2.1. ***Summation*** *there is a unique function "$+_{\mathbb{Q}} : \mathbb{Q} \times \mathbb{Q} \to \mathbb{Q}$" s.t.*

$$\forall x,y,z,u \in \mathbb{Z} \quad [(x,z)] +_{\mathbb{Q}} [(y,u)] = [(xu+zy, zu)]$$

Proof Is Left As An Exercise To The Reader.

Theorem 8.2.2. ***Multiplication*** *there is a unique function "$\times_{\mathbb{Q}} : \mathbb{Q} \times \mathbb{Q} \to \mathbb{Q}$" s.t.*

$$\forall x,y,z,u \in \mathbb{Z} \quad [(x,z)] \times_{\mathbb{Q}} [(y,u)] = [(xy, zu)]$$

Proof Is Left As An Exercise To The Reader.

definition 8.2.2. ***Order in*** $\mathbb{Q}$.

if $(z > 0, u > 0)$ or $(z < 0, u < 0)$

$$[(x,z)] <_{\mathbb{Q}} [(y,u)] : x \cdot u < y \cdot z$$

if $(z > 0, u < 0)$ or $(z < 0, u > 0)$

$$[(x,z)] <_{\mathbb{Q}} [(y,u)] : x \cdot u > y \cdot z$$

this lemma let us use "$<$" instead of "$<_{\mathbb{Q}}$", "$+$" instead of "$+_{\mathbb{Q}}$" and "$\times$" instead of "$\times_{\mathbb{Q}}$".

Lemma 8.2.1. *prove.*

1. $/ \therefore \forall z,y \in \mathbb{Z} \quad z <_{\mathbb{Q}} y \leftrightarrow z < y$
2. $/ \therefore \forall z,y \in \mathbb{Z} \quad z +_{\mathbb{Q}} y = z + y$
3. $/ \therefore \forall z,y \in \mathbb{Z} \quad z \times_{\mathbb{Q}} y = z \times y$

Proof Is Left As An Exercise To The Reader.

we denote "Multiplication" with "$x \cdot y$" and "xy" too.

Lemma 8.2.2. *prove for "$(\mathbb{Q}, +)$"*

1. $\forall x,y,z \in \mathbb{Q} \quad (x+y)+z = x+(y+z)$ *(associative $+$)*
2. $\forall x \in \mathbb{Q} \quad x+0 = 0+x = x$ *(identity $+$)*
3. $\forall x \in \mathbb{Q} \quad \exists y \in \mathbb{Q} \quad x+y = y+x = 0$ *(inverse $+$)*
4. $\forall x,y \in \mathbb{Q} \quad x+y = y+x$ *(commutative(abelian) $+$)*

Proof Is Left As An Exercise To The Reader.😀

Lemma 8.2.3. *prove for "$(\mathbb{Q}, +, \times)$"*

1. $\forall x,y,z \in \mathbb{Q} \quad (xy)z = x(yz)$ *(associative $\times$)*
2. $\forall x \in \mathbb{Q} \quad x \cdot 1 = 1 \cdot x = x$ *(identity $\times$)*
3. $\forall x \in \mathbb{Q} - \{0\} \quad \exists y \in \mathbb{Q} \quad xy = yx = 1$ *(inverse $\times$)*
4. $\forall x,y \in \mathbb{Q} \quad xy = yx$ *(commutative(abelian) $\times$)*

5. $\forall x, y, z \in \mathbb{Q} \quad \Big(x(y+z) = xy + xz\Big) \wedge \Big((x+y)z = xz + yz\Big)$ *(distributive $\times$)*

Proof Is Left As An Exercise To The Reader.

Theorem 8.2.3. ***Order of Rational numbers($\mathbb{Q}$)*** *prove for "$(\mathbb{Q}, <_{\mathbb{Q}})$"*

1. *is linearly ordered.*
2. $\Big(\forall x \in \mathbb{Q} \quad \exists y \in \mathbb{Q} \quad x < y\Big) \wedge \Big(\forall x \in \mathbb{Q} \quad \exists y \in \mathbb{Q} \quad y < x\Big)$ *(doesn't have endpoints)*
3. $\forall x, y \in \mathbb{Q} \quad \exists z \in \mathbb{Q} \quad x \neq y \to x < z < y$ *(dense)*
4. $\forall x, y, z \in \mathbb{Q} \quad x < y \to x + z < y + z$
5. $\forall x, y, z \in \mathbb{Q} \quad x < y \wedge 0 < z \to xz < yk$

Proof Is Left As An Exercise To The Reader.

8.3 Real numbers "$\mathbb{R}$"

before we we get to lesson define function "$|_| : \mathbb{Q} \times \mathbb{Q} \to \mathbb{Q}$" like this.

$$|x| := \begin{cases} x & 0 \leq x \\ -x & x < 0 \end{cases}$$

if you are not familiar to this notation it means "$|_| := \{(x, y) \in \mathbb{Q} \times \mathbb{Q} \mid (0 \leq x \to y = x) \wedge (x < 0 \to y = -x)\}$". learn this notation because we want to use later.

we learned about "$\mathbb{Q}$" before and we constructed structure "$(\mathbb{Q}, +, \cdot, <)$". now assume $\mathbb{Q}^{\mathbb{N}}$(we know what does it mean by the definition. it means all the sequences of $\mathbb{Q}$). we don't want it all we just want a subset.

definition 8.3.1. ***Cauchy sequence*** *is a sequence like "$(a_i)_{i \in \mathbb{N}} \in \mathbb{Q}^{\mathbb{N}}$" s.t. elements of it get infinitely close to each other i.e.*

$$Cauchy(a_i)_{i \in \mathbb{N}} : \forall \varepsilon > 0 \quad \exists N \quad \forall n, m > N \quad |a_n - a_m| < \varepsilon$$

we usually give "N" natural numbers but it's not necessary.

we define an equivalence relation "$(a_i)_{i \in \mathbb{N}} \sim (b_i)_{i \in \mathbb{N}} : \forall \varepsilon > 0 \quad \exists N \quad \forall n > N \quad |a_n - b_n| < \varepsilon$"(check it's an equivalence relation). we denote every $[(a_i)_{i \in \mathbb{N}}]$ with $[a_i]_{i \in \mathbb{N}}$.

1	$a_n - a_n = 0$	
2	$\lvert a_n - a_n \rvert < \varepsilon$	
3	$n > N \to \lvert a_n - a_n \rvert < \varepsilon$	
4	$\forall \varepsilon > 0 \quad \exists N \quad \forall n > N \quad \lvert a_n - a_n \rvert < \varepsilon$	$\forall I, \exists I, \forall I$
5	$(a_i)_{i \in \mathbb{N}} \sim (a_i)_{i \in \mathbb{N}}$	$def. \sim$

1	$(a_i)_{i\in\mathbb{N}} \sim (b_i)_{i\in\mathbb{N}}$	ass
2	$\forall\varepsilon > 0 \quad \exists N \quad \forall n > N \quad \lvert a_n - b_n\rvert < \varepsilon$	$def.\sim$
3	$\forall\varepsilon > 0 \quad \exists N \quad \forall n > N \quad \lvert b_n - a_n\rvert < \varepsilon$	
4	$(b_i)_{i\in\mathbb{N}} \sim (a_i)_{i\in\mathbb{N}}$	$def.\sim$

1	$(a_i)_{i\in\mathbb{N}} \sim (b_i)_{i\in\mathbb{N}}$	ass
2	$(b_i)_{i\in\mathbb{N}} \sim (c_i)_{i\in\mathbb{N}}$	ass
3	$\forall\varepsilon > 0 \quad \exists N \quad \forall n > N \quad \lvert a_n - b_n\rvert < \varepsilon$	$def.\sim$
4	$\forall\varepsilon > 0 \quad \exists N \quad \forall n > N \quad \lvert b_n - c_n\rvert < \varepsilon$	$def.\sim$
5	$\exists N \quad \forall n > N \quad \lvert a_n - b_n\rvert < \frac{\varepsilon}{2}$	$def.\sim$
6	$\exists N \quad \forall n > N \quad \lvert b_n - c_n\rvert < \frac{\varepsilon}{2}$	$def.\sim$
7	$\quad \forall n > N \quad \lvert a_n - b_n\rvert < \frac{\varepsilon}{2}$	acp
8	$\quad\quad \forall n > N' \quad \lvert b_n - c_n\rvert < \frac{\varepsilon}{2}$	acp
9	$\quad\quad\quad n > max\{N, N'\}$	acp
10	$\quad\quad\quad n > N \to \lvert a_n - b_n\rvert < \frac{\varepsilon}{2}$	$\forall E$
11	$\quad\quad\quad n > N' \to \lvert b_n - c_n\rvert < \frac{\varepsilon}{2}$	$\forall E$
12	$\quad\quad\quad \lvert a_n - b_n\rvert < \frac{\varepsilon}{2}$	$\to E$
13	$\quad\quad\quad \lvert b_n - c_n\rvert < \frac{\varepsilon}{2}$	$\to E$
14	$\quad\quad\quad \lvert a_n - b_n\rvert + \lvert b_n - c_n\rvert < \varepsilon$	
15	$\quad\quad\quad \lvert a_n - c_n\rvert < \lvert a_n - b_n\rvert + \lvert b_n - c_n\rvert$	
16	$\quad\quad\quad \lvert a_n - c_n\rvert < \varepsilon$	
17	$\quad\quad n > max\{N, N'\} \to \lvert a_n - c_n\rvert < \varepsilon$	$\to I$
18	$\quad\quad \forall n > max\{N, N'\} \quad \lvert a_n - c_n\rvert < \varepsilon$	$\forall I$
19	$\quad\quad \exists N \quad \forall n > N \quad \lvert a_n - c_n\rvert < \varepsilon$	$\exists I$
20	$\exists N \quad \forall n > N \quad \lvert a_n - c_n\rvert < \varepsilon$	$\exists E$
21	$\forall\varepsilon > 0 \quad \exists N \quad \forall n > N \quad \lvert a_n - c_n\rvert < \varepsilon$	$\forall I$
22	$(a_i)_{i\in\mathbb{N}} \sim (c_i)_{i\in\mathbb{N}}$	$def.\sim$

definition 8.3.2. ***Real numbers "$\mathbb{R}$"*** *is a set*

$$\mathbb{R} := Cauchy/\sim$$

we denote it's members with "$r_0, r_1, \ldots$"

in this system "$0 := [0]_{i\in\mathbb{N}}$" and "$1 := [1]_{i\in\mathbb{N}}$" and "$\forall q \in \mathbb{Q} \quad q := [q]_{i\in\mathbb{N}}$". so we can say "$\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R}$".

Theorem 8.3.1. ***Summation*** *there is a unique function "*$+_{\mathbb{R}} : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$*" s.t.*

$$[a_i]_{i\in\mathbb{N}} +_{\mathbb{R}} [b_i]_{i\in\mathbb{N}} = [a_i + b_i]_{i\in\mathbb{N}}$$

Proof Is Left As An Exercise To The Reader.
Theorem 8.3.2. ***Multiplication*** *there is a unique function "*$\times_{\mathbb{R}} : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$*" s.t.*

$$[a_i]_{i\in\mathbb{N}} \times_{\mathbb{R}} [b_i]_{i\in\mathbb{N}} = [a_i b_i]_{i\in\mathbb{N}}$$

Proof Is Left As An Exercise To The Reader.
definition 8.3.3. ***Order in*** $\mathbb{R}$.

$$[a_i]_{i\in\mathbb{N}} <_{\mathbb{R}} [b_i]_{i\in\mathbb{N}} : \exists\varepsilon > 0 \quad \exists N \quad \forall n > N \quad b_n - a_n > \varepsilon$$

this lemma let us use "$<$" instead of "$<_{\mathbb{R}}$", "$+$" instead of "$+_{\mathbb{R}}$" and "$\times$" instead of "$\times_{\mathbb{R}}$".
Lemma 8.3.1. *prove.*

1. $/ \therefore \forall x, y \in \mathbb{Q} \quad x <_{\mathbb{R}} y \leftrightarrow x < y$
2. $/ \therefore \forall x, y \in \mathbb{Q} \quad x +_{\mathbb{R}} y = x + y$
3. $/ \therefore \forall x, y \in \mathbb{Q} \quad x \times_{\mathbb{R}} y = x \times y$

Proof Is Left As An Exercise To The Reader.

we denote "Multiplication" with "$x \cdot y$" and "xy" too.
Lemma 8.3.2. *prove.*

1. $\forall x, y, z \in \mathbb{R} \quad (x + y) + z = x + (y + z)$ *(associative +)*
2. $\forall x \in \mathbb{R} \quad x + 0 = 0 + x = x$ *(identity +)*
3. $\forall x \in \mathbb{R} \quad \exists y \in \mathbb{R} \quad x + y = y + x = 0$ *(inverse +)*
4. $\forall x, y \in \mathbb{R} \quad x + y = y + x$ *(commutative(abelian) +)*

Proof Is Left As An Exercise To The Reader.
Lemma 8.3.3. *prove.*

1. $\forall x, y, z \in \mathbb{R} \quad (xy)z = x(yz)$ *(associative ×)*
2. $\forall x \in \mathbb{R} \quad x \cdot 1 = 1 \cdot x = x$ *(identity ×)*
3. $\forall x \in \mathbb{R} - \{0\} \quad \exists y \in \mathbb{R} \quad xy = yx = 1$ *(inverse ×)*
4. $\forall x, y \in \mathbb{R} \quad xy = yx$ *(commutative(abelian) ×)*
5. $\forall x, y, z \in \mathbb{R} \quad \Big(x(y + z) = xy + xz\Big) \wedge \Big((x + y)z = xz + yz\Big)$ *(distributive ×)*

Proof Is Left As An Exercise To The Reader.
Theorem 8.3.3. ***Linear order of Real numbers(***$\mathbb{R}$***)*** *prove "*$(\mathbb{R}, <_{\mathbb{R}})$

1. *is linearly ordered.*
2. $\Big(\forall x \in \mathbb{R} \quad \exists y \in \mathbb{R} \quad x < y\Big) \wedge \Big(\forall x \in \mathbb{R} \quad \exists y \in \mathbb{R} \quad y < x\Big)$ *(doesn't have endpoints)*

3. $\forall x, y \in \mathbb{R} \quad \exists z \in \mathbb{R} \quad x \neq y \to x < z < y$ *(dense)*
4. $\forall X \subseteq \mathbb{R} \quad (X \neq \varnothing \wedge \exists y \in \mathbb{R} \quad Uppreb(X, y)) \to (\exists y \in \mathbb{R} \quad sup(X) = y)$ *(complete)*
5. $\forall x, y, z \in \mathbb{R} \quad x < y \to x + z < y + z$
6. $\forall x, y, z \in \mathbb{R} \quad x < y \wedge 0 < z \to xz < yk$

Proof Is Left As An Exercise To The Reader.

8.4 Complex numbers"ℂ"

we learned about "ℝ" before and we constructed structure "$(\mathbb{R}, +, \cdot, <)$".
definition 8.4.1. ***Complex numbers"ℂ"*** *is a set*

$$\mathbb{C} := \mathbb{R} \times \mathbb{R}$$

we denote it's members with "$z_0, x, \ldots$"

in this system "$0 := (0, 0)$" and "$\forall r \in \mathbb{R} \quad r := (r, 0)$" . in general "$\forall x, y \in \mathbb{R} \quad x + iy := (x, y)$".
so we can say "$\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}$".
Theorem 8.4.1. ***Summation*** *there is a unique function "$+_{\mathbb{C}} : \mathbb{C} \times \mathbb{C} \to \mathbb{C}$" s.t.*

$$\forall x, y, z, u \in \mathbb{R} \quad (x, z) +_{\mathbb{C}} (y, u) = (x + y, z + u)$$

Proof Is Left As An Exercise To The Reader.
Theorem 8.4.2. ***Multiplication*** *there is a unique function "$\times_{\mathbb{C}} : \mathbb{C} \times \mathbb{C} \to \mathbb{C}$" s.t.*

$$\forall x, y, z, u \in \mathbb{R} \quad (x, z) \times_{\mathbb{C}} (y, u) = (xy - zu, xu + yz)$$

Proof Is Left As An Exercise To The Reader.

this lemma let us use "$+$" instead of "$+_{\mathbb{C}}$" and "$\times$" instead of "$\times_{\mathbb{C}}$".
Lemma 8.4.1. *prove.*

1. $/ \therefore \forall x, y \in \mathbb{R} \quad x +_{\mathbb{C}} y = x + y$
2. $/ \therefore \forall x, y \in \mathbb{R} \quad x \times_{\mathbb{C}} y = x \times y$

Proof Is Left As An Exercise To The Reader.

we denote "Multiplication" with "$z \cdot y$" and "zy" too.
Lemma 8.4.2. *prove for "$(\mathbb{C}, +)$"*

1. $\forall x, y, z \in \mathbb{C} \quad (x + y) + z = x + (y + z)$ *(associative $+$)*
2. $\forall z \in \mathbb{C} \quad z + 0 = 0 + z = z$ *(identity $+$)*
3. $\forall z \in \mathbb{C} \quad \exists y \in \mathbb{C} \quad z + y = y + z = 0$ *(inverse $+$)*
4. $\forall x, y \in \mathbb{C} \quad x + y = y + x$ *(commutative(abelian) $+$)*

Proof Is Left As An Exercise To The Reader.
Lemma 8.4.3. *prove for "$(\mathbb{C}, +, \times)$"*

1. $\forall x, y, z \in \mathbb{C} \quad (xy)z = x(yz)$ *(associative $\times$)*
2. $\forall z \in \mathbb{C} \quad z \cdot 1 = 1 \cdot z = z$ *(identity $\times$)*
3. $\forall x, y \in \mathbb{C} \quad xy = yx$ *(commutative(abelian) $\times$)*
4. $\forall x, y, z \in \mathbb{C} \quad \Big(x(y+z) = xy + xz\Big) \wedge \Big((x+y)z = xz + yz\Big)$ *(distributive $\times$)*

Proof Is Left As An Exercise To The Reader.

Theorem 8.4.3. ***Exponents*** *there is a unique function "$(_)^{(_)} : \mathbb{C} \times \mathbb{N} \to \mathbb{C}$" s.t.*

1. $\forall x \in \mathbb{C} \quad x^0 = 1$
2. $\forall x \in \mathbb{C} \quad \forall n \in \mathbb{N} \quad x^{s(n)} = x^n \cdot x$

Proof Is Left As An Exercise To The Reader.

Lemma 8.4.4. *prove for "$(\mathbb{C}, +, \cdot, (_)^{(_)})$"*

1. $\forall x \in \mathbb{Z} \quad \forall m, n \in \mathbb{N} \quad x^m \cdot x^n = x^{m+n}$
2. $\forall x \in \mathbb{Z} \quad \forall m, n \in \mathbb{N} \quad (x^m)^n = x^{mn}$

Proof Is Left As An Exercise To The Reader.

Theorem 8.4.4. ***Exponents*** *there is a unique function "$(_)^{(_)} : \mathbb{C} \times \mathbb{Z} \to \mathbb{C}$" s.t.*

1. $\forall x \in \mathbb{C} \quad x^0 = 1$
2. $\forall x \in \mathbb{C} \quad \forall m, n \in \mathbb{N} \quad x^{m-n} = \frac{x^m}{x^n}$

Proof Is Left As An Exercise To The Reader.

Lemma 8.4.5. *prove for "$(\mathbb{C}, +, \cdot)$"*

1. $\forall x \in \mathbb{C} \quad \forall y, z \in \mathbb{Z} \quad x^y \cdot x^z = x^{y+z}$
2. $\forall x \in \mathbb{C} \quad \forall y, z \in \mathbb{Z} \quad (x^y)^z = x^{yz}$

Proof Is Left As An Exercise To The Reader.

Theorem 8.4.5. ***Big Sum*** *if "$a : \mathbb{N} \to \mathbb{C}$" there is a unique function "$\sum_{i=0}^{(_)}(_) : \mathbb{C} \times \mathbb{N} \to \mathbb{C}$" s.t.*

1. $\forall a \quad \sum_{i=0}^{0}(a_0) = a_0$
2. $\forall a \quad \forall n \in \mathbb{N} \quad \sum_{i=0}^{s(n)} a_{s(n)} = \sum_{i=0}^{n} a_n + a_{s(n)}$

Proof Is Left As An Exercise To The Reader.

Lemma 8.4.6. *prove for "$(\mathbb{C}, +, \cdot)$"*

1. $\forall n \in \mathbb{N} \quad \forall a, b \quad \sum_{i=0}^{n} a_n + \sum_{i=0}^{n} b_n = \sum_{i=0}^{n}(a_n + b_n)$
2. $\forall n \in \mathbb{N} \quad \forall a \quad k \sum_{i=0}^{n} a_n = \sum_{i=0}^{n} k a_n$

Proof Is Left As An Exercise To The Reader.

Theorem 8.4.6. ***Big Product*** *if "$a : \mathbb{N} \to \mathbb{C}$" there is a unique function "$\prod_{i=0}^{(_)}(_) : \mathbb{C} \times \mathbb{N} \to \mathbb{C}$" s.t.*

1. $\forall a \quad \prod_{i=0}^{0}(a_0) = a_0$
2. $\forall a \quad \forall n \in \mathbb{N} \quad \prod_{i=0}^{s(n)} a_{s(n)} = \prod_{i=0}^{n} a_n \cdot a_{s(n)}$

Proof Is Left As An Exercise To The Reader.

Lemma 8.4.7. *prove for "*$(\mathbb{C}, +, \cdot)$*"*

1. $\forall n \in \mathbb{N} \quad \forall a, b \quad \prod_{i=0}^{n} a_n \prod_{i=0}^{n} b_n = \prod_{i=0}^{n} (a_n b_n)$
2. $\forall n \in \mathbb{N} \quad \forall a \quad (\prod_{i=0}^{n} a_n)^k = \prod_{i=0}^{n} (a_n)^k$

Proof Is Left As An Exercise To The Reader.

This page intentionally left blank.

Chapter 9

Finite and Infinite sets

9.1 Equipotent and Order in sets

we talked about "Isomorphism" before now assume two Structures $\mathfrak{A} := (A)$ and $\mathfrak{B} := (B)$. because these Structures have only a set "Isomorphism" between them only means there is a bijection function $f : A \xrightarrow[surj]{inj} B$.

definition 9.1.1. ***Equipotent*** *in situation above we say "A" and "B" are "Equipotent". we denote it with "$A \cong B$" i.e.*

$$A \cong B : \exists f(f : A \xrightarrow[surj]{inj} B)$$

Lemma 9.1.1. *The relation "$\cong$" on V is an equivalence relation.*

Proof. We prove reflexivity, symmetry, and transitivity.

Reflexivity. For every $X \in V$, the identity function $\mathrm{id}_X : X \to X$ is both injective and surjective. Hence

$$\mathrm{id}_X : X \xrightarrow[surj]{inj} X.$$

Therefore, by the definition of $\cong$, we have $X \cong X$.

1	$\mathrm{id}_X : X \xrightarrow[surj]{inj} X$	
2	$\exists f; f : X \xrightarrow[surj]{inj} X$	$\exists I$
3	$X \cong X$	def.

Symmetry. Suppose $X \cong Y$. Then, by definition, there exists a bijection $f : X \to Y$. Let f be such a function. Since the inverse of a bijection is again a bijection,

$$f^{-1} : Y \xrightarrow[surj]{inj} X.$$

Hence there exists a bijection from Y to X, and therefore $Y \cong X$.

1	$X \cong Y$	*ass*
2	$\exists f; f : X \xrightarrow[surj]{inj} Y$	def.
3	$\quad f : X \xrightarrow[surj]{inj} Y$	*acp*
4	$\quad f^{-1} : Y \xrightarrow[surj]{inj} X$	lemma
5	$\quad \exists g\ g : Y \xrightarrow[surj]{inj} X$	$\exists I$
6	$\quad Y \cong X$	def.
7	$Y \cong X$	$\exists E$

Transitivity. Suppose $X \cong Y$ and $Y \cong Z$. Then there exist bijections $f : X \to Y$ and $g : Y \to Z$. Since the composition of two bijections is a bijection,

$$g \circ f : X \xrightarrow[surj]{inj} Z.$$

Thus there exists a bijection from X to Z, and consequently $X \cong Z$.

1	$X \cong Y$	*ass*
2	$Y \cong Z$	*ass*
3	$\exists f\ f : X \xrightarrow[surj]{inj} Y$	def.
4	$\exists g\ g : Y \xrightarrow[surj]{inj} Z$	def.
5	$\quad f : X \xrightarrow[surj]{inj} Y$	*acp*
6	$\quad\quad g : Y \xrightarrow[surj]{inj} Z$	*acp*
7	$\quad\quad g \circ f : X \xrightarrow[surj]{inj} Z$	lemma
8	$\quad\quad \exists h\ h : X \xrightarrow[surj]{inj} Z$	$\exists I$
9	$\quad\quad X \cong Z$	def.
10	$X \cong Z$	$\exists E$

□

so "$V/\cong$" is a partition for "V" and every partition equals to a $[A]_\cong$ we denote it by $[A]$.

definition 9.1.2. ***Embedding*** *if there is a function $f : A \xrightarrow{inj} B$. in situation above we say "A" is less than or equipotent "B". we denote it with "$A \precsim B$" i.e.*

$$A \precsim B : \exists f(f : A \xrightarrow{inj} B)$$

similarly we define "$[A] \preceq [B] : \forall X \in [A], Y \in [B] \quad X \precsim Y$"

Lemma 9.1.2. $A \precsim B \therefore / \therefore [A] \preceq [B]$

1	$A \precsim B$	
2	$\quad X \in [A]$	
3	$\quad Y \in [B]$	
4	$\quad X \cong A$	
5	$\quad Y \cong B$	
6	$\quad \exists f \quad f : X \xrightarrow[surj]{inj} A$	$def.4$
7	$\quad \exists f \quad f : A \xrightarrow{inj} B$	$def.1$
8	$\quad \exists f \quad f : B \xrightarrow[surj]{inj} Y$	$def.5$
9	$\quad\quad f : X \xrightarrow[surj]{inj} A$	acp
10	$\quad\quad\quad h : A \xrightarrow{inj} B$	acp
11	$\quad\quad\quad\quad g : B \xrightarrow[surj]{inj} Y$	acp
12	$\quad\quad\quad\quad g \circ h \circ f : X \xrightarrow{inj} Y$	$lemma$
13	$\quad\quad\quad\quad \exists f \quad f : X \xrightarrow{inj} Y$	$\exists I$
14	$\quad\quad\quad\quad X \precsim Y$	$def.13$
15	$\quad X \precsim Y$	$\exists E$
16	$[A] \preceq [B]$	

Theorem 9.1.1. ***Schroeder-Bernstein*** $A \precsim B, B \precsim A / \therefore A \cong B$
Proof Is Left As An Exercise To The Reader.

$$h(y) := \begin{cases} f \circ g(y)) & \bigcup_{i \in \mathbb{N}} A_i \\ y & \bigcup_{i \in \mathbb{N}} B_i \end{cases}$$

$$C_0 := B$$

$$C_{s(n)} := f \circ g[C_n]$$

$$D_0 := f[A]$$

$$D_{s(n)} := f \circ g[D_n]$$

$$A_n := C_n - D_n$$

$$B_n := D_n - C_{s(n)}$$

Claim 0: $\exists n \in \mathbb{N} \quad C_{s(n)} = C_n \rightarrow A \cong B$**.**

1	$\exists n \in \mathbb{N} \quad C_{s(n)} = C_n$	ass
2	$C_{s(n)} = C_n$	acp
3	$C_n \subseteq D_n$	$lemma$
4	$D_n \subseteq C_n$	$lemma$
5	$C_n = D_n$	$\subseteq E$
6	$A_n = C_n - D_n$	$def.$
7	$A_n = \varnothing$	$=$
8	$B_n = D_n - C_{s(n)}$	$def.$
9	$B_n = \varnothing$	$=$
10	$C_n = D_n = C_{s(n)}$	$\wedge I$
11	$C_{s(n+1)} = C_{s(n)}$	$def.$
12	$C_{s(n+1)} = C_n$	$=$
13	$\cdots = C_n$	$induction$
14	$C_i = C_n \quad (i \geq n)$	$induction$
15	$\bigcup_{i \in \mathbb{N}} C_i = C_0$	$def.$
16	$B = C_0$	$def.$
17	$B \cong f[A]$	$lemma$
18	$A \cong f[A]$	$lemma$
19	$A \cong B$	$trans$

Claim 1: $\bigcup_{i \in \mathbb{N}} A_i \cap \bigcup_{i \in \mathbb{N}} B_i = \varnothing$

1	$\exists x \left(x \in \bigcup_{i\in\mathbb{N}} A_i \cap \bigcup_{j\in\mathbb{N}} B_j\right)$	
2	$\quad x \in \bigcup_{i\in\mathbb{N}} A_i$	
3	$\quad x \in \bigcup_{j\in\mathbb{N}} B_j$	
4	$\quad \exists i \in \mathbb{N} \quad x \in A_i$	$\cup E$
5	$\quad \exists j \in \mathbb{N} \quad x \in B_j$	$\cup E$
6	$\quad\quad x \in A_i$	acp
7	$\quad\quad x \in C_i$	$def.$
8	$\quad\quad x \notin D_i$	$def.$
9	$\quad\quad\quad x \in B_j$	acp
10	$\quad\quad\quad x \in D_j$	$def.$
11	$\quad\quad\quad x \notin C_{s(j)}$	$def.$
12	$\quad\quad\quad i \leq j \ \vee \ j < i$	LEM
13	$\quad\quad\quad\quad i \leq j$	acp
14	$\quad\quad\quad\quad D_j \subseteq D_i$	$nesting$
15	$\quad\quad\quad\quad x \in D_i$	$\subseteq E$
16	$\quad\quad\quad\quad \bot$	$\neg E$
17	$\quad\quad\quad\quad j < i$	acp
18	$\quad\quad\quad\quad C_i \subseteq C_{s(j)}$	$nesting$
19	$\quad\quad\quad\quad x \in C_{s(j)}$	$\subseteq E$
20	$\quad\quad\quad\quad \bot$	$\neg E$
21	$\quad\quad\quad \bot$	$\vee E$
22	$\quad \bot$	$\vee E$
23	$\neg\exists x \left(x \in \bigcup_{i\in\mathbb{N}} A_i \cap \bigcup_{j\in\mathbb{N}} B_j\right)$	$\neg I$

Claim 2: $\bigcup_{i\in\mathbb{N}} A_i \cup \bigcup_{i\in\mathbb{N}} B_i = B$

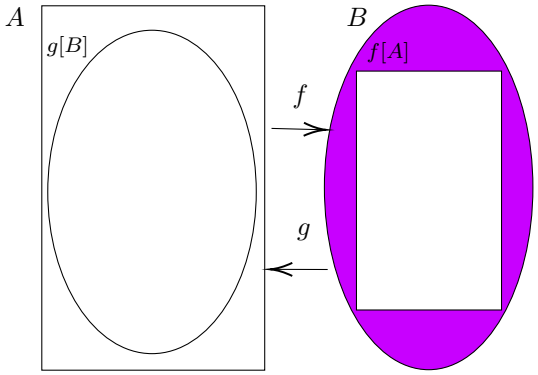
1	$\bigcap_{i\in\mathbb{N}} C_i = \varnothing$	ass
2	$C_{s(n)} \subseteq D_n \subseteq C_n$	$nesting$
3	$C_n - D_n \ \cup \ D_n - C_{s(n)} = C_n - C_{s(n)}$	$lemma$
4	$A_n \cup B_n = C_n - C_{s(n)}$	$def.$
5	$\bigcup_{n\in\mathbb{N}}(A_n \cup B_n) = \bigcup_{n\in\mathbb{N}}(C_n - C_{s(n)})$	$\cup E$
6	$\bigcup_{n\in\mathbb{N}}(C_n - C_{s(n)}) = C_0 - \bigcap_{n\in\mathbb{N}} C_n$	$lemma$
7	$C_0 - \bigcap_{n\in\mathbb{N}} C_n = C_0$	ass
8	$C_0 = B$	$def.$
9	$\bigcup_{n\in\mathbb{N}}(A_n \cup B_n) = B$	$\exists E$
10	$\bigcup_{n\in\mathbb{N}} A_n \cup \bigcup_{n\in\mathbb{N}} B_n = B$	$\cup$

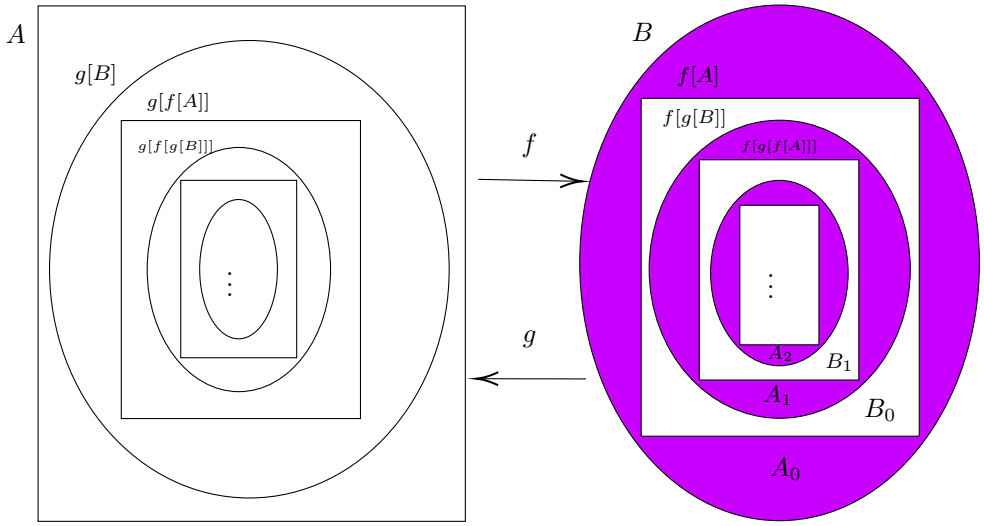
Claim 3: $Func(h)$

Claim 4: $injective(h)$

Claim 5: $h : B \xrightarrow[surj]{inj} f[A]$

1	$h : B \xrightarrow[surj]{inj} f[A]$
2	$B \cong f[A]$
3	$A \cong f[A]$
4	$A \cong B$





Lemma 9.1.3. *"$\preceq$" is a order relation in "$(V/\cong, \preceq)$"*
Proof Is Left As An Exercise To The Reader.☺

9.2 Finite sets

definition 9.2.1. *if there is a "$n \in \mathbb{N}$" s.t. $A \cong n$ we call it* ***"Finite"****$(Finite(A))$. if a set is not finite we call it* ***"Infinite"****$(Infinite(A))$.*
Lemma 9.2.1. $/ \therefore \forall n \in \mathbb{N} \quad X \subset n \to X \ncong n$

1 $\forall X \quad X \subset 0 \to X \ncong 0$ *vacuous truth*

2 | $\forall X \quad X \subset n \to X \ncong n$

3 | | $X \subset s(n)$

4 | | | $X \cong s(n)$

5 | | | $\exists f \quad f : s(n) \xrightarrow[surj]{inj} X$

6 | | | | $f : s(n) \xrightarrow[surj]{inj} X$

7 | | | | | $n \notin X$

8 | | | | | $X \subset n \cup \{n\}$

9 | | | | | $X \subseteq n$

10 | | | | | $X - \{f(n)\} \subset n$

11 | | | | | $X - \{f(n)\} \subset n \to X - \{f(n)\} \ncong n$

12 | | | | | $X - \{f(n)\} \ncong n$

13 | | | | | $f \restriction n : n \xrightarrow[surj]{inj} X - \{f(n)\}$

14 | | | | | $X - \{f(n)\} \cong n$

15 | | | | | $\bot$

16 | | | | | $n \in X$

17 | | | | | $n \in Dom(f)$

18 | | | | | $n \in Range(f)$

19 | | | | | | $(k, n) \in f$

20 | | | | | | $(n, m) \in f$

21 | | | | | | $g := f \cup \{(k, m)\} - \{(k, n), (n, m)\}$

22 | | | | | | $g : n \xrightarrow[surj]{inj} X - \{n\}$

23 | | | | | | $X \subset n \cup \{n\}$

24 | | | | | | $X - \{n\} \subset n$

25 | | | | | | $X - \{n\} \subset n \to X - \{n\} \ncong n$

26 | | | | | | $X - \{n\} \ncong n$

27 | | | | | | $X - \{n\} \cong n$

28 | | | | | | $\bot$

29 | | | | $\bot$

30 | | | $X \ncong s(n)$

31 | | $\forall X \quad X \subset s(n) \to X \ncong s(n)$

Corollary 9.2.1. .

1. $\forall m, n \in \mathbb{N} \quad m \neq n \rightarrow m \ncong n$

2. $\forall m, n \in \mathbb{N} \quad A \cong m \wedge A \cong n \rightarrow m = n$

3. $Infinite(\mathbb{N})$

4. $\forall m, n \in \mathbb{N} \quad m \prec n \leftrightarrow m < n$

Proof Is Left As An Exercise To The Reader.😀

corollary (4) let us use "$<$" instead of "$\prec$"
Lemma 9.2.2. $Finite(X), Y \subseteq X / \therefore Finite(Y)$

1 $X \cong 0 \wedge Y \subseteq X \to \exists m \in \mathbb{N} \quad Y \cong m$

2 | $X \cong n \wedge Y \subseteq X \to \exists m \in \mathbb{N} \quad Y \cong m$

3 | | $X \cong s(n)$

4 | | $\exists f \quad f : s(n) \xrightarrow[surj]{inj} X$

5 | | | $f : s(n) \xrightarrow[surj]{inj} X$

6 | | | $Y \subseteq X$

7 | | | $g := f - \{(n, f(n))\}$

8 | | | $g : n \xrightarrow[surj]{inj} X - \{f(n)\}$

9 | | | $X - \{f(n)\} \cong n$

10 | | | $Y - \{f(n)\} \subseteq X - \{f(n)\}$

11 | | | $X - \{f(n)\} \cong n \wedge Y - \{f(n)\} \subseteq X - \{f(n)\} \to \exists m \in \mathbb{N} \quad Y - \{f(n)\} \cong m$

12 | | | $\exists m \in \mathbb{N} \quad Y - \{f(n)\} \cong m$

13 | | | | $Y - \{f(n)\} \cong m$

14 | | | | | $h : m \xrightarrow[surj]{inj} Y - \{f(n)\}$

15 | | | | | | $f(n) \in Y$

16 | | | | | | $Y = (Y - \{f(n)\}) \cup \{f(n)\}$

17 | | | | | | $h' := h \cup \{(m, f(n))\}$

18 | | | | | | $h' : s(m) \xrightarrow[surj]{inj} Y$

19 | | | | | | $s(m) \cong Y$

20 | | | | | | $\exists m \in \mathbb{N} \quad Y \cong m$

21 | | | | | | $f(n) \notin Y$

22 | | | | | | $Y = Y - \{f(n)\}$

23 | | | | | | $h : m \xrightarrow[surj]{inj} Y$

24 | | | | | | $m \cong Y$

25 | | | | | | $\exists m \in \mathbb{N} \quad Y \cong m$

26 | | | | $\exists m \in \mathbb{N} \quad Y \cong m$

27 | | $X \cong s(n) \wedge Y \subseteq X \to \exists m \in \mathbb{N} \quad Y \cong m$

28 $\forall n \in \mathbb{N}(X \cong n \wedge Y \subseteq X \to \exists m \in \mathbb{N} \quad Y \cong m)$

29 $\exists n \in \mathbb{N} \quad X \cong n \wedge Y \subseteq X \to \exists m \in \mathbb{N} \quad Y \cong m$

30 $Finite(X) \wedge Y \subseteq X \to Finite(Y)$

9.3 Countable sets

definition 9.3.1. *set "A" is* ***"Countable"*** *if "$A \cong \mathbb{N}$". set "A" is* ***"At most countable"*** *if "$A \precsim \mathbb{N}$"*
Lemma 9.3.1. *$Countable(A), Infinite(B), B \subseteq A / \therefore Countable(B)$*
Proof Is Left As An Exercise To The Reade.

Proof. Let A be countable and let $B \subseteq A$ be infinite.

Then there exists a bijective enumeration

$$f : \mathbb{N} \xrightarrow[surj]{inj} A.$$

Define recursively a sequence $(k_n)_{n\in\mathbb{N}}$ in $\mathbb{N}$:

$$k_0 := \min\{k \in \mathbb{N} : f(k) \in B\},$$
$$k_{n+1} := \min\{k \in \mathbb{N} : f(k) \in B \wedge f(k) \neq f(k_i)\ \forall i \leq n\}.$$

Define

$$b_n := f(k_n).$$

1 $b_n \in B$
2 | $x \in B$
3 | $\exists k \in \mathbb{N}\ f(k) = x$
4 | $\exists n \in \mathbb{N}\ b_n = x$
5 $B \subseteq \{b_n : n \in \mathbb{N}\}$
6 $\{b_n : n \in \mathbb{N}\} \subseteq B$
7 $B = \{b_n : n \in \mathbb{N}\}$
8 $\forall m < n\ (b_m \neq b_n)$
9 $(b_n)_{n\in\mathbb{N}}$ injective
10 $\mathbb{N} \xrightarrow[surj]{inj} B$
11 $Countable(B)$

□

Corollary 9.3.1. *a set is at most countable iff it's finite or countable.*
Theorem 9.3.1. *The range of an infinite sequence $(a_n)_{n\in\mathbb{N}}$ is at most countable.*

Proof. Let $(a_n)_{n\in\mathbb{N}}$ be a sequence in X.

We construct recursively a sequence (b_n) such that $\text{Range}(b_n) = \text{Range}(a_n)$ and (b_n) is injective.

Define

$$b_0 := a_0.$$

Having defined $b_0, \ldots, b_n$, define k_{n+1} as the least $k \in \mathbb{N}$ such that

$$a_k \neq b_i \quad \forall i \leq n.$$

If such k_{n+1} exists, set

$$b_{n+1} := a_{k_{n+1}}.$$

Otherwise stop the construction.

1 $b_n \in \text{Range}(a_n)$
2 | $x \in \text{Range}(a_n)$
3 | $\exists k \in \mathbb{N}\ (a_k = x)$
4 | $\exists i\ (b_i = x)$
5 $\text{Range}(a_n) \subseteq \{b_i\}$
6 $\{b_i\} \subseteq \text{Range}(a_n)$
7 $\text{Range}(b_n) = \text{Range}(a_n)$
8 $\forall i < j\ (b_i \neq b_j)$
9 (b_n) injective
10 $\text{Range}(a_n)$ is image of an injective sequence
11 $\text{Range}(a_n)$ is at most countable

□

Theorem 9.3.2. *The union of two countable sets is countable.*

Proof. Let $A = \{a_n : n \in \mathbb{N}\}$ and $B = \{b_n : n \in \mathbb{N}\}$.

Define a sequence $(c_n)_{n\in\mathbb{N}}$ by

$$c_{2k} := a_k, \qquad c_{2k+1} := b_k.$$

1 $x \in A \cup B$
2 | $x \in A$
3 | $\exists k \in \mathbb{N}\ (a_k = x)$
4 | $\exists k \in \mathbb{N}\ (c_{2k} = x)$
5 | $\exists n \in \mathbb{N}\ (c_n = x)$
6 | $x \in B$
7 | $\exists k \in \mathbb{N}\ (b_k = x)$
8 | $\exists k \in \mathbb{N}\ (c_{2k+1} = x)$
9 | $\exists n \in \mathbb{N}\ (c_n = x)$
10 $\forall x \in A \cup B\ \exists n \in \mathbb{N}\ (c_n = x)$
11 $\{c_n : n \in \mathbb{N}\} = A \cup B$
12 $c : \mathbb{N} \xrightarrow[surj]{} A \cup B$
13 $AtMostCountable(A \cup B)$
14 $Infinite(A)$
15 $A \precsim A \cup B$
16 $Infinite(A \cup B)$
17 $Countable(A \cup B)$

□

Corollary 9.3.2. *Finite union of countable sets is countable.*

Proof. Let $A_1, \ldots, A_n$ be countable.

1 $Countable(A_1)$
2 | $\forall k < n \quad Countable(A_k) \to Countable(\bigcup_{i \in n} A_i)$
3 | | $\forall k < s(n) \quad Countable(A_k)$
4 | | $Countable(A_n)$
5 | | $Countable(\bigcup_{i \in s(n)} A_i)$
6 $Countable(\bigcup_{i \in s(n)} A_i)$

□

Theorem 9.3.3. *If A and B are countable, then $A \times B$ is countable.*

Proof. It suffices to show $\mathbb{N} \times \mathbb{N}$ is countable.

Define

$$f : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$$

by

$$f(m,n) := \begin{cases} n^2 + m & \text{if } m \leq n, \\ m^2 + 2m - n & \text{if } n < m. \end{cases}$$

1 $(m_1, n_1), (m_2, n_2) \in \mathbb{N} \times \mathbb{N}$

2 $f(m_1, n_1) = f(m_2, n_2)$

3 $m \leq n \to n^2 = f(0, n) \leq f(m, n) \leq f(n, n) \leq f(n, m) < f(0, n + 1) = (n + 1)^2$

4 $k = max\{m, n\} \to k^2 \leq f(m, n) < (k + 1)^2$

5 $k' = max\{m', n'\} \to k'^2 \leq f(m', n') < (k' + 1)^2$

6 $f(m, n) = f(m', n')$

7 $k = k'$

8 | $m \leq n = k \wedge m' \leq n' = k'$

9 | $n = n'$

10 | $n^2 + m = n'^2 + m'$

11 | $n^2 + m = n^2 + m'$

12 | $m = m'$

13 | $n < m = k \wedge n' < m' = k'$

14 | $m = m'$

15 | $m^2 + 2m - n = m'^2 + 2m' - n'$

16 | $m^2 + 2m - n = m^2 + 2m - n'$

17 | $n = n'$

18 | $n < m = k \wedge n' = k' > m'$

19 | $m = n'$

20 | $f(m, n) > f(m, m)$

21 | $f(n', n') > f(m', n')$

22 | $f(n', n') = f(m, m)$

23 | $f(m, n) > f(m', n')$

24 | $\bot$

25 $(m_1, n_1) = (m_2, n_2)$

26 $\mathfrak{f} is injective$

27 | $(m, n) \in \mathbb{N} \times \mathbb{N}$

28 | $\exists k \in \mathbb{N}\ (f(k) = (m, n))$

29 | $\mathbb{N} \times \mathbb{N} \cong \mathbb{N}$

30 $Countable(\mathbb{N} \times \mathbb{N})$

□

Corollary 9.3.3. *Finite Cartesian product of countable sets is countable, and A^n is countable for any countable A and $n > 0$.*

Proof. Let $A_1, \ldots, A_n$ be countable.

1 $Countable(A_1)$
2 | $\forall k < n \quad Countable(A_k) \to Countable(\prod_{i\in n} A_i)$
3 | | $\forall k < s(n) \quad Countable(A_k)$
4 | | $Countable(A_n)$
5 | | $Countable(\prod_{i\in s(n)} A_i)$
6 $Countable(\prod_{i\in s(n)} A_i)$

□

Lemma 9.3.2. *If "$(A_i)_{i\in\mathbb{N}}$" is family of countable sets with a family of sequences "$(a_i)_{i\in\mathbb{N}}$" s.t. "$\forall n \in \mathbb{N} \quad a_n : \mathbb{N} \xrightarrow[surj]{inj} A_n$" then "$\bigcup_{i\in\mathbb{N}} A_i$" is countable.*
Proof Is Left As An Exercise To The Reader.😀

Proof. Define

$$h : \mathbb{N} \times \mathbb{N} \to \bigcup_{i\in\mathbb{N}} A_i \quad \text{by} \quad h(i, n) = a_i(n).$$

1 $x \in \bigcup_{i\in\mathbb{N}} A_i$
2 | $\exists i \in \mathbb{N}\ (x \in A_i)$
3 | $\exists n \in \mathbb{N}\ a_i(n) = x$
4 | $h(i, n) = x$
5 $h : \mathbb{N} \times \mathbb{N} \xrightarrow[surj]{inj} \bigcup_{i\in\mathbb{N}} A_i$
6 $Countable(\mathbb{N} \times \mathbb{N})$
7 $Countable(\bigcup_{i\in\mathbb{N}} A_i)$

□

Theorem 9.3.4. *If A is countable, then $Seq(A)$, the set of all finite sequences of elements of A, is countable.*

Proof. It suffices to prove the theorem for $A = \mathbb{N}$.

Since

$$Seq(\mathbb{N}) = \bigcup_{n\in\mathbb{N}} \mathbb{N}^n,$$

it is enough to construct, for each $n \geq 1$, an enumeration

$$a_n : \mathbb{N} \xrightarrow[surj]{inj} \mathbb{N}^n.$$

Let

$$g : \mathbb{N} \xrightarrow[surj]{inj} \mathbb{N} \times \mathbb{N}.$$

Define recursively:

$$a_1(i) := (i),$$

and

$$a_{n+1}(i) := (b_0, \ldots, b_{n-1}, i_2),$$

where

$$g(i) = (i_1, i_2)$$

and

$$a_n(i_1) = (b_0, \ldots, b_{n-1}).$$

1 $a_1 : \mathbb{N} \xrightarrow[surj]{inj} \mathbb{N}$
2 | $a_n : \mathbb{N} \xrightarrow[surj]{inj} \mathbb{N}^n$
3 | | $(b_0, \ldots, b_n) \in \mathbb{N}^{n+1}$
4 | | $\exists i_1 \in \mathbb{N} \quad a_n(i_1) = (b_0, \ldots, b_{n-1})$
5 | | $\exists i \in \mathbb{N} \quad g(i) = (i_1, b_n)$
6 | | $a_{n+1}(i) = (b_0, \ldots, b_n)$
7 | $a_{n+1} : \mathbb{N} \xrightarrow[surj]{inj} \mathbb{N}^{n+1}$
8 $\forall n \geq 1 \quad a_n : \mathbb{N} \xrightarrow[surj]{inj} \mathbb{N}^n$
9 $\forall n \in \mathbb{N} \quad \mathbb{N}^n$ is countable
10 $\bigcup_{n \in \mathbb{N}} \mathbb{N}^n$ is countable
11 $Seq(\mathbb{N})$ is countable

Since every countable set is equipotent to a subset of $\mathbb{N}$,

$$Seq(A)$$

is countable.

□

Corollary 9.3.4. *The set of all finite subsets of a countable set is countable.*

Proof. Let A be countable and define

$$\mathcal{F}(A) := \{F \subseteq A : F \text{ finite}\}.$$

For each $n \in \mathbb{N}$ define

$$\mathcal{F}_n(A) := \{F \subseteq A : |F| = n\}.$$

1 A^n is countable

2 $\{(a_1, \ldots, a_n) \in A^n : a_i \text{ distinct}\}$ is countable

3 $\mathcal{F}_n(A)$ is countable

4 | $F \in \mathcal{F}(A)$

5 | $\exists n \in \mathbb{N}\ (F \in \mathcal{F}_n(A))$

6 $\mathcal{F}(A) = \bigcup_{n\in\mathbb{N}} \mathcal{F}_n(A)$

7 $\forall n\ \mathcal{F}_n(A)$ is countable

8 $\bigcup_{n\in\mathbb{N}} \mathcal{F}_n(A)$ is countable

9 $\mathcal{F}(A)$ is countable

□

Corollary 9.3.5. *The set of all finite subsets of a countable set is countable.*

Proof. Let

$$\mathcal{F}(A) := \{X \subseteq A : X \text{ is finite}\}.$$

Define

$$F : Seq(A) \to \mathcal{F}(A)$$

by

$$F((a_0, \ldots, a_{n-1})) = \{a_0, \ldots, a_{n-1}\}.$$

1 $Seq(A)$ is countable
2 | $X \in \mathcal{F}(A)$
3 | $X = \{a_0, \ldots, a_{n-1}\}$
4 | $F((a_0, \ldots, a_{n-1})) = X$
5 $F : Seq(A) \xrightarrow[surj]{} \mathcal{F}(A)$
6 $\mathcal{F}(A)$ is at most countable
7 $\mathcal{F}(A)$ is countable

□

Theorem 9.3.5. *An equivalence relation on a countable set has at most countably many equivalence classes.*

Proof. Let $\sim$ be an equivalence relation on a countable set A.

Define

$$F : A \to A/\sim$$

by

$$F(a) := [a]_\sim$$

1 A is countable
2 | $C \in A/\sim$
3 | $\exists a \in A \quad C = [a]_E$
4 | $F(a) = C$
5 $F : A \xrightarrow[surj]{} A/\sim$
6 $A/\sim$ is at most countable

□

Corollary 9.3.6. *The set of all integers $\mathbb{Z}$ and the set of all rational numbers $\mathbb{Q}$ are countable.*

Proof. $\mathbb{Z}$ **is countable.**

By definition

$$\mathbb{Z} = \mathbb{N} \times \mathbb{N}/\sim .$$

1 $\mathbb{N} \times \mathbb{N}$ is countable

2 $\mathbb{Z} := (\mathbb{N} \times \mathbb{N})/\sim$

3 $\mathbb{Z}$ is at most countable

4 $\mathbb{Z}$ is infinite

5 $\mathbb{Z}$ is countable

$\mathbb{Q}$ is countable.

By definition

$$\mathbb{Q} = \mathbb{Z} \times (\mathbb{Z} - \{0\})/\sim .$$

1 $\mathbb{Z}$ is countable

2 $\mathbb{Z} - \{0\}$ is countable

3 $\mathbb{Z} \times (\mathbb{Z} - \{0\})$ is countable

4 $F : \mathbb{Z} \times (\mathbb{Z} - \{0\}) \to \mathbb{Z} \times (\mathbb{Z} - \{0\})/\sim$

5 $\mathbb{Q} = (\mathbb{Z} \times (\mathbb{Z} - \{0\}))/\sim$

6 $\mathbb{Q}$ is at most countable

7 $\mathbb{Q}$ is infinite

8 $\mathbb{Q}$ is countable

□

9.4 Uncountable sets

Lemma 9.4.1. $/\therefore \mathbb{N} \prec 2^{\mathbb{N}}$

Proof. **Claim 0:** $\mathbb{N} \precsim 2^{\mathbb{N}}$

Define

$$f : \mathbb{N} \to 2^{\mathbb{N}}$$

by

$$f(n)(k) = \begin{cases} 1 & k = n, \\ 0 & k \neq n. \end{cases}$$

1 | $f(m) = f(n)$
2 | | $m \neq n$
3 | | $f(m)(m) = 1 \neq 0 = f(n)(m)$
4 | | $\bot$
5 | $m = n$
6 $f : \mathbb{N} \xrightarrow{inj} 2^{\mathbb{N}}$
7 $\mathbb{N} \precsim 2^{\mathbb{N}}$

Claim 1: $\mathbb{N} \prec 2^{\mathbb{N}}$

1 | $g : \mathbb{N} \xrightarrow[surj]{} 2^{\mathbb{N}}$
2 | $d(n) := 1 - g(n)(n)$
3 | $d \in 2^{\mathbb{N}}$
4 | $\exists k \in \mathbb{N} \quad g(k) = d$
5 | $d(k) = g(k)(k)$
6 | $d(k) = 1 - g(k)(k)$
7 | $g(k)(k) = 1 - g(k)(k)$
8 | $\bot$
9 $\nexists g : \mathbb{N} \xrightarrow[surj]{} 2^{\mathbb{N}}$
10 $\mathbb{N} \ncong 2^{\mathbb{N}}$
11 $\mathbb{N} \prec 2^{\mathbb{N}}$

□

Lemma 9.4.2. $/ \therefore 2^{\mathbb{N}} \cong \mathcal{P}(\mathbb{N})$
Proof Is Left As An Exercise To The Reader.☺
Theorem 9.4.1.

$$\mathcal{P}(\mathbb{N}) \cong 2^{\mathbb{N}}$$

Proof. For each

$$S \subseteq \mathbb{N}$$

define its characteristic function

$$\chi_S : \mathbb{N} \to 2$$

by

$$\chi_S(n) := \begin{cases} 1 & n \in S, \\ 0 & n \notin S. \end{cases}$$

Define

$$\chi_{()} : \mathcal{P}(\mathbb{N}) \to 2^{\mathbb{N}}$$

1 | $\chi_S = \chi_T$
2 | $\forall n \in \mathbb{N} \quad (n \in S \leftrightarrow n \in T)$
3 | $S = T$
4 $\chi_{()}$ is injective
5 | $f \in 2^{\mathbb{N}}$
6 | $S := \{n \in \mathbb{N} : f(n) = 1\}$
7 | $\chi_{(S)} = f$
8 $\chi_{()} : \mathcal{P}(\mathbb{N}) \xrightarrow[surj]{inj} 2^{\mathbb{N}}$
9 $\mathcal{P}(\mathbb{N}) \cong 2^{\mathbb{N}}$

□

Lemma 9.4.3. $/ \therefore 2^{\mathbb{N}} \cong \mathbb{R}$
Proof Is Left As An Exercise To The Reader.😀
Theorem 9.4.2. ***Cantor theorem*** $/ \therefore \forall X \quad Type_0(X) \to X \prec 2^X$
Proof Is Left As An Exercise To The Reader.😀
Theorem 9.4.3. $/ \therefore \forall X \quad Type_0(X) \to 2^X \cong \mathcal{P}(X)$
Proof Is Left As An Exercise To The Reader.😀

Chapter 10

Ordinals

10.1 Well ordered sets

definition 10.1.1. *if "$(W, <)$" is well ordered and $a \in W$* ***"Initial segment"*** *is*

$$W[a] := \{x \in W \mid x < a\}$$

before get to next lemma Let's have a drfinition. assume "$(W_1, <_1)$" and "$(W_2, <_2)$" are two ordered structures function "f" is increasing when "$f : W_1 \to W_2$" is a (weak) homomorphism.
Lemma 10.1.1. *if "$(W, <)$" is well ordered and $f : W \to W$ is increasing then $\forall x \in W \quad x \leq f(x)$. Proof Is Left As An Exercise To The Reader.*☺
Corollary 10.1.1. .

1. *no well ordered structure is isomorph to it's initial segment of itself.*
2. *each well ordered structure has only one automorphism, the identity.*
3. *if "$(W_1, <_1)$" and "$(W_2, <_2)$" are two isomorphic well ordered structures then the isomorphism is unique.*

Proof Is Left As An Exercise To The Reader.☺
Theorem 10.1.1. *if "$(W_1, <_1)$" and "$(W_2, <_2)$" are two well ordered structures then exactly one of the following holds.*

1. $(W_1, <_1) \cong (W_2, <_2)$
2. $\exists a \in W_1 \quad (W_1[a], <_1) \cong (W_2, <_2)$
3. $\exists a \in W_2 \quad (W_1, <_1) \cong (W_2[a], <_2)$

Proof Is Left As An Exercise To The Reader.☺

Proof. Let $(W_1, <_1)$ and $(W_2, <_2)$ be well ordered sets.

Define

$$f \subseteq W_1 \times W_2 \quad \text{by} \quad (x,y) \in f \leftrightarrow W_1[x] \cong W_2[y].$$

Claim 1: f is a function.

1	$(x,y) \in f \wedge (x,y') \in f$	assumption
2	$W_2[y] \cong W_2[y']$	definition of f
3	$y = y'$	uniqueness of isomorphism of well orders

Claim 2: f is injective.

1	$(x,y) \in f \wedge (x',y) \in f$	assumption
2	$W_1[x] \cong W_1[x']$	definition of f
3	$x = x'$	uniqueness of isomorphism of well orders

Claim 3: f is increasing.

1	$x <_1 x'$	assumption
2	$W_1[x'] \cong W_2[f(x')]$	definition of f
3	$W_1[x] \cong W_2[f(x)]$	definition of f
4	$f(x) <_2 f(x')$	comparison of initial segments

Claim 4: f is isomorphism between $S = \mathrm{dom}(f)$ and $T = \mathrm{ran}(f)$.

1	$f : S \to T$ bijective	Claims 1,2
2	$x <_1 x' \Rightarrow f(x) <_2 f(x')$	Claim 3
3	f order isomorphism	definition

Claim 5: S initial segment of W_1.

1	$x \in S \wedge z <_1 x$	assumption
2	$W_1[x] \cong W_2[f(x)]$	definition of S
3	$z \in S$	restriction of initial segments

Claim 6: T initial segment of W_2 or $T = W_2$.

1	$y \in T$	definition of T
2	$T \neq W_2$	assumption of case split
3	$T = W_2[b] \ \vee \ T = W_2$	well-order trichotomy

Claim 7: exclusion of mixed case.

1	$S = W_1[a] \wedge T = W_2[b]$	assumption
2	$f : W_1[a] \cong W_2[b]$	restriction of f
3	$(a, b) \in f$	definition of domain/range
4	$a \in W_1[a]$	initial segment property
5	$a < a$	element characterization
6	$\perp$	irreflexivity

Conclusion:

$$(W_1 \cong W_2) \ \vee \ (W_1[a] \cong W_2) \ \vee \ (W_1 \cong W_2[a]).$$

□

Lemma 10.1.2. *Every element of an ordinal number is an ordinal number.*

10.2 Ordinal numbers

definition 10.2.1. ***Order in sets***

$$<_A := \{(x, y) \in A \times A \mid x \in y\}$$

definition 10.2.2. *A set "α" is a* ***Ordinal*** *if*

1. *α is transitive.*
2. *$(\alpha, <_\alpha)$ is well ordered.*

we denote it by Ord so we have the set of ordinals(Ord).
Lemma 10.2.1. *Every natural number is an ordinal.*

Proof. Let $\mathbb{N}$ be the set of natural numbers.

1	$\forall n \in \mathbb{N} \ (n \subseteq \mathbb{N})$	von Neumann construction
2	$Transitive(\mathbb{N})$	definition of transitive set
3	$<_{\mathbb{N}} = \in \cap(\mathbb{N} \times \mathbb{N})$	definition
4	$(\mathbb{N}, <_{\mathbb{N}})$ is well ordered	standard well-order of $\mathbb{N}$
5	$Ordinal(\mathbb{N})$	definition of ordinal

Hence $\mathbb{N}$ is an ordinal.

□

In ordinal numbers we show "$\mathbb{N}$" with "ω". so it's just a renaming.
Lemma 10.2.2. *if α is an ordinal then $s(\alpha)$ is an ordinal too.*
Proof Is Left As An Exercise To The Reader.😀

Proof.

$$s(\alpha) = \alpha \cup \{\alpha\}.$$

1	α transitive	ordinal assumption
2	$s(\alpha) = \alpha \cup \{\alpha\}$	definition
3	$x \in y \wedge y \in s(\alpha) \Rightarrow x \in s(\alpha)$	case split on $y \in \alpha \cup \{\alpha\}$
4	$Transitive(s(\alpha))$	definition of transitive set
5	$<_\alpha = \in \cap(\alpha \times \alpha)$	definition
6	α is greatest element of $s(\alpha)$	construction
7	$<_{s(\alpha)}$ extends $<_\alpha$	restriction
8	$(s(\alpha), <_{s(\alpha)})$ is well ordered	extension by maximum element
9	$Ordinal(s(\alpha))$	definition of ordinal

Hence $s(\alpha)$ is an ordinal number.

□

Lemma 10.2.3. *prove*

1. $Transitive(Ord)$.
2. $\forall \alpha, \beta \in Ord \quad \alpha \subset \beta \rightarrow \alpha \in \beta$.

Proof. **(1):** $Transitive(Ord)$**.**

Claim 1: x **is transitive.**

1	$u \in v \wedge v \in x$	
2	$x \in \alpha$	assumption
3	$v \in \alpha$	transitivity of α
4	$u \in \alpha$	transitivity of α
5	$u, v, x \in \alpha$	closure
6	$u <_\alpha v <_\alpha x$	definition
7	$u <_\alpha x$	linearity of $<_\alpha$

Hence $Transitive(x)$.

Claim 2: $(x, <_x)$ **is well ordered.**

1	$x \subseteq \alpha$	transitivity of α
2	$<_x = <_\alpha \restriction x$	definition
3	$<_\alpha$ well order on α	ordinal assumption
4	$<_x$ well order on x	restriction of well-order

Thus x is transitive and well ordered, hence

$$x \in Ord.$$

1	$\alpha \in Ord$
2	$x \in y \wedge y \in \alpha$
3	α transitive
4	$x \in \alpha$

Hence $Transitive(\alpha)$ for all $\alpha \in Ord$, so $Transitive(Ord)$.

(2): $\alpha \subset \beta \Rightarrow \alpha \in \beta$**.**

Let $\alpha, \beta \in Ord$ and $\alpha \subset \beta$.

1	$\alpha \subset \beta$	assumption
2	$\beta - \alpha \neq \varnothing$	set difference
3	$\exists \gamma \in \beta - \alpha \; \forall \delta \in \beta - \alpha \; (\gamma \leq \delta)$	well-order of β
4	$\gamma \in \beta \wedge \gamma \notin \alpha$	definition of set difference
5	$\forall \delta \in \alpha \; (\delta < \gamma)$	minimality of γ
6	$\alpha \subseteq \gamma$	transitivity of $<$
7	$\gamma \in \beta$	step 8
8	$\alpha = \gamma$	ordinal comparison lemma ($\alpha, \gamma \in Ord$)
9	$\alpha \in \beta$	since $\gamma \in \beta$ and $\alpha = \gamma$

Hence

$$\alpha \subset \beta \Rightarrow \alpha \in \beta.$$

□

Theorem 10.2.1. ***Order of Ordinals*** *prove*

1. *$(Ord, <)$ is linearly ordered.*
2. *$(Ord, <)$ is Well ordered.*

Proof Is Left As An Exercise To The Reader.☺

Theorem 10.2.2. ***Order of Ordinals.*** *prove*

1. $(Ord, <)$ *is linearly ordered.*
2. $(Ord, <)$ *is well ordered.*

Proof. **Claim 1:** $(Ord, <)$ **is linear.**

1	$\alpha \in \beta$	assumption
2	$\beta \in \gamma$	assumption
3	γ transitive	$\gamma \in Ord$
4	$\alpha \in \gamma$	transitivity of γ

1	$\alpha \in \alpha$	assumption
2	$(\alpha, <_\alpha)$ is linearly ordered	α is an ordinal
3	$x := \alpha$	definition
4	$x \in x$	1,3
5	$\bot$	asymmetry of $<_\alpha$

6	$\alpha \notin \alpha$	

Let $\alpha, \beta, \gamma \in Ord$.

1	$\alpha \subseteq \beta \wedge \beta \subseteq \gamma$	assumption
2	$\alpha \subseteq \gamma$	transitivity of $\subseteq$
3	$\alpha < \gamma$	Lemma 2.6(a)
4	$\alpha \not< \beta \wedge \beta \not< \alpha \Rightarrow \alpha = \beta$	Lemma 2.6(b),(c)
5	$\alpha < \beta \ \vee \ \alpha = \beta \ \vee \ \beta < \alpha$	Lemma 2.6(c)

Hence $(Ord, <)$ is linearly ordered.

Claim 2: $(Ord, <)$ **is well ordered.**

Let $X \subseteq Ord$, $X \neq \varnothing$.

1	$X \neq \varnothing$	assumption
2	$\exists \alpha \in X$	nonempty
3	$\exists \beta \in X \ \forall \gamma \in X \ (\beta \leq \gamma)$	Lemma 2.6(d)
4	β is $<$-least element of X	definition

Hence every nonempty set of ordinals has a least element.

Conclusion:

$(Ord, <)$ is linearly ordered and well ordered.

□

Theorem 10.2.3. *The natural numbers are exactly the finite ordinal numbers.*

Proof. **Claim 1: Every natural number is a finite ordinal.**

1	$n \in \mathbb{N}$	
2	$Ordinal(n)$	Theorem 2.3
3	$Finite(n)$	definition of $\mathbb{N}$

Claim 2: Every ordinal that is not a natural number is infinite.

1	$Ordinal(\alpha) \wedge \alpha \notin \mathbb{N}$	assumption
2	$\alpha \neq \omega$	$\alpha \notin \mathbb{N}$
3	$\alpha < \omega \ \vee \ \alpha = \omega \ \vee \ \omega < \alpha$	Theorem 2.6(c)
4	$\omega < \alpha$	Theorem 2.6(b),5,6
5	$\omega \in \alpha$	definition of $<$
6	$\omega \subseteq \alpha$	transitivity of α
7	$Infinite(\omega)$	definition
8	$Infinite(\alpha)$	$\omega \subseteq \alpha$

Hence every finite ordinal is a natural number.

□

Theorem 10.2.4. *Every well ordered structure is isomorphic to a unique ordinal.*

Proof. Let $(W, <)$ be a well ordered structure.

Define

$$A := \{a \in W \mid \exists \alpha \in Ord \ ((W[a], <) \cong (\alpha, <))\}.$$

For each $a \in A$, let α_a be the unique ordinal such that

$$(W[a], <) \cong (\alpha_a, <).$$

Define

$$S := \{\alpha_a \mid a \in A\}.$$

Claim 1: $Transitive(S)$**.**

1	$\alpha_a \in S$	
2	$\gamma \in \alpha_a$	
3	$h : (W[a], <) \cong (\alpha_a, <)$	definition of α_a
4	$c := h^{-1}(\gamma)$	
5	$(W[c], <) \cong (\gamma, <)$	restriction of h
6	$c \in A$	definition of A
7	$\gamma = \alpha_c$	
8	$\gamma \in S$	

Hence $Transitive(S)$.

Claim 2: S **is an ordinal.**

1	$S \subseteq Ord$	definition of S
2	$(S, \in)$ is well ordered	Order of Ordinals
3	$Transitive(S)$	Claim 1
4	$Ordinal(S)$	definition of ordinal

Let

$$S = \alpha.$$

Claim 3: $b < a \wedge a \in A \Rightarrow b \in A$**.**

1	$a \in A \wedge b < a$	
2	$h : (W[a], <) \cong (\alpha_a, <)$	definition of A
3	$h[W[b]]$ is an initial segment of α_a	
4	$\exists \beta \in \alpha_a \quad h[W[b]] = \beta$	initial segments of ordinals
5	$(W[b], <) \cong (\beta, <)$	
6	$b \in A$	definition of A
7	$\alpha_b = \beta$	
8	$\alpha_b \in \alpha_a$	

Claim 4: $A = W$ **or** $A = W[c]$ **for some** $c \in W$**.**

1	$A \subseteq W$	
2	$\forall a \in A\ \forall b < a\ (b \in A)$	Claim 3
3	A is an initial segment of W	
4	$A = W\ \vee\ \exists c \in W\ (A = W[c])$	Lemma 1.2

Define

$$f : A \to \alpha$$

by

$$f(a) := \alpha_a.$$

Claim 5: f is an isomorphism.

1	$a, b \in A$	
2	$a < b \Rightarrow \alpha_a \in \alpha_b$	Claim 3
3	$a < b \Rightarrow f(a) < f(b)$	
4	f injective	uniqueness of ordinals
5	$Range(f) = \alpha$	definition of $\alpha = S$
6	$f : (A, <) \cong (\alpha, <)$	

Claim 6: $A = W$.

1	$A = W[c]$	
2	$f : (W[c], <) \cong (\alpha, <)$	Claim 5
3	$c \in A$	definition of A
4	$c \in W[c]$	31
5	$\bot$	definition of initial segment
6	$A = W$	

Since $A = W$, Claim 5 yields

$$f : (W, <) \cong (\alpha, <).$$

The ordinal α is unique since isomorphic ordinals are equal.

□

10.3 Transfinite Induction

Theorem 10.3.1 (Second Version of Transfinite Induction)**.** *Let $P(\alpha)$ be a property. Assume*

1. $P(0)$.
2. $\forall \alpha \in Ord \quad P(\alpha) \to P(s(\alpha))$.
3. $\forall \alpha \in Ord \quad \big(Limit(\alpha) \wedge \alpha \neq 0 \wedge \forall \beta < \alpha \; P(\beta)\big) \to P(\alpha)$.

Then

$$\forall \alpha \in Ord \quad P(\alpha).$$

Proof. We prove the hypothesis of Transfinite Induction.

1	$\alpha \in Ord$	
2	$\forall \beta < \alpha \quad P(\beta)$	
3	$\quad \alpha = 0$	
4	$\quad P(\alpha)$	assumption (a)
5	$\quad \exists \beta < \alpha \quad \alpha = s(\beta)$	
6	$\quad P(\beta)$	2
7	$\quad P(\alpha)$	assumption (b)
8	$\quad Limit(\alpha) \wedge \alpha \neq 0$	
9	$\quad P(\alpha)$	assumption (c),2
10	$P(\alpha)$	trichotomy of ordinals
11	$\forall \alpha \in Ord\Big(\forall \beta < \alpha \; P(\beta) \to P(\alpha)\Big)$	
12	$\forall \alpha \in Ord \quad P(\alpha)$	Transfinite Induction

□

Theorem 10.3.2 (Transfinite Induction Principle)**.** *Let $P(\alpha)$ be a property.*

Assume

$$\forall \alpha \in Ord \quad \Big(\forall \beta < \alpha \quad P(\beta) \to P(\alpha)\Big).$$

Then

$$\forall \alpha \in Ord \quad P(\alpha).$$

Proof. Define

$$S := \{\beta \leq \gamma \mid \neg P(\beta)\}.$$

1	$\exists \gamma \in Ord \quad \neg P(\gamma)$	assumption
2	$\gamma \in S$	definition of S
3	$S \neq \varnothing$	2
4	$\exists \alpha \in S \quad \forall \delta \in S \ (\alpha \leq \delta)$	well ordering of Ord
5	$\alpha \in S$	4
6	$\neg P(\alpha)$	definition of S
7	$\forall \beta < \alpha \quad P(\beta)$	minimality of α
8	$P(\alpha)$	hypothesis
9	$\bot$	6,8
10	$\forall \alpha \in Ord \quad P(\alpha)$	contradiction

□

10.4 Transfinite Recurtion

Theorem 10.4.1 (Transfinite Recursion Theorem)**.** *Let G be an operation. Then there exists a unique operation F such that*

$$F(\alpha) = G(F \restriction \alpha) \quad \textit{for all ordinals } \alpha.$$

Proof. We say t is a **computation of length α based on G** if:

$$dom(t) = \alpha + 1 \quad \text{and} \quad \forall \beta \leq \alpha \ t(\beta) = G(t \restriction \beta).$$

Define the property:

$$P(x, y) \ : \begin{cases} x \in Ord \ \wedge \ y = t(x) \text{ for some computation } t \text{ of length } x, \\ \text{or } x \notin Ord \ \wedge \ y = \varnothing. \end{cases}$$

We show P defines a function.

Claim 1: For every ordinal α there exists a unique computation of length α.

1	$\forall \beta < \alpha \; \exists! t_\beta$ computation of length β	induction hypothesis
2	$T := \{t_\beta \mid \beta < \alpha\}$	Replacement
3	T is a compatible system of functions	uniqueness
4	$t := \bigcup T$	definition
5	$dom(t) = \alpha$	union of domains
6	t is a function	compatibility of , **??**
7	$\forall \beta < \alpha \; t(\beta) = G(t \restriction \beta)$	compatibility
8	$t(\alpha) = G(t \restriction \alpha)$	definition extension step
9	t computation of length α	definition

Claim 2: Uniqueness of computation of length α.

1	t, s computations of length α	
2	$\forall \beta < \alpha \; t(\beta) = s(\beta)$	induction on β
3	$t = s$	extensionality

Hence P defines a function F.

Define

$$F(\alpha) := t(\alpha)$$

where t is the unique computation of length α.

Claim 3: Recursion equation.

1	$\alpha \in Ord$	
2	t computation of length α	
3	$F(\alpha) = t(\alpha)$	definition
4	$t(\alpha) = G(t \restriction \alpha)$	computation rule
5	$F(\alpha) = G(F \restriction \alpha)$	definition of , **??**

Claim 4: Uniqueness of F.

1	F, S satisfy recursion equation	
2	$\forall \beta < \alpha \; F(\beta) = S(\beta)$	induction
3	$F = S$	extensionality

□

Theorem 10.4.2 (Parametric Transfinite Recursion)**.** *Let G be an operation. Then there exists a unique operation F such that*

$$F(z, \alpha) = G\big(z, F(z) \restriction \alpha\big) \quad \textit{for all sets } z \textit{ and ordinals } \alpha.$$

Proof. Fix a set z.

We say t is a **computation of length α based on G and z** if:

$$dom(t) = \alpha + 1 \quad \wedge \quad \forall \beta \leq \alpha \; t(\beta) = G\big(z, t \restriction \beta\big).$$

Define the property

$$Q(z,x,y) : \begin{cases} x \in Ord \ \wedge \ y = t(x) \text{ for some computation } t \text{ of length } x \text{ based on } G, z, \\ \text{or } x \notin Ord \ \wedge \ y = \varnothing. \end{cases}$$

Claim 1: For every ordinal α there exists a unique computation of length α based on G and z.

1	$\forall \beta < \alpha \; \exists! t_\beta$	induction hypothesis
2	$T := \{t_\beta \mid \beta < \alpha\}$	
3	T is compatible	uniqueness
4	$t := \bigcup T$	
5	$dom(t) = \alpha$	
6	t is a function	compatibility
7	$\forall \beta < \alpha \; t(\beta) = G(z, t \restriction \beta)$	
8	$t(\alpha) = G(z, t \restriction \alpha)$	
9	t computation of length α	

Claim 2: Uniqueness.

1	t, s computations of length α	
2	$\forall \beta < \alpha \; t(\beta) = s(\beta)$	
3	$t = s$	

Hence Q defines a function F.

Define

$$F(z, \alpha) := t(\alpha)$$

where t is the unique computation of length α based on G and z.

Claim 3: Recursion equation.

1	$\alpha \in Ord$	
2	t computation of length α	
3	$F(z, \alpha) = t(\alpha)$	
4	$t(\alpha) = G(z, t \restriction \alpha)$	
5	$F(z, \alpha) = G(z, F(z) \restriction \alpha)$	

Claim 4: Uniqueness of F.

1 F, S satisfy recursion

2 $\forall \beta < \alpha \; F(z, \beta) = S(z, \beta)$

3 $F = S$

□

10.5 Arithmetic of Ordinals

Theorem 10.5.1. ***Summation*** *there is a unique function "$+ : Ord \times Ord \to Ord$" s.t.*

1. $\forall \alpha \in Ord \quad 0 + \alpha = \alpha = \alpha + 0$
2. $\forall \alpha, \beta \in Ord \quad \alpha + s(\beta) = s(\alpha + \beta) = s(\alpha) + \beta$
3. $\forall \alpha, \beta \in Ord \quad limit(\alpha) \wedge limit(\beta) \to \alpha + \beta = (\bigcup_{\beta' \in \beta} \alpha + \beta') \cup (\bigcup_{\beta' \in \beta} \beta' + \alpha)$

Proof Is Left As An Exercise To The Reader.☺

Theorem 10.5.2. ***Multiplication*** *there is a unique function "$\times : Ord \times Ord \to Ord$" s.t.*

1. $\forall \alpha \in Ord \quad \alpha \times 0 = 0$
2. $\forall \alpha, \beta \in Ord \quad \alpha \times s(\beta) = \alpha \times \beta + \alpha$
3. $\forall \alpha, \beta \in Ord \quad limit(\beta) \wedge \alpha < \beta \to \alpha \times \beta = \beta \times \alpha$
4. $\forall \alpha, \beta \in Ord \quad limit(\beta) \wedge \beta \leq \alpha \to \alpha \times \beta = \bigcup_{\beta' \in trunk(\beta)} (\alpha \times \beta') + \alpha \times branch(\beta)$

Proof Is Left As An Exercise To The Reader.☺

we denote also "Multiplication" with "$\alpha \cdot \beta$" and "$\alpha\beta$" too.

10.6 Transfinite Peano arithmetic

assume a structure "$\mathfrak{N} := (N, +, \cdot, s, 0)$"(here "$+, \cdot, s, 0$" are arbitrary functions and constants and doesn't have meaning we defined before). with "$Limit(x) : \forall y \in N \quad s(y) \neq x$ " and these properties.

1. $\forall x, y \in N \quad s(x) = s(y) \to x = y$
2. $\forall x \in N \quad s(x) \neq 0$
3. $\forall x \in N \quad x + 0 = x$
4. $\forall x \in N \quad x \cdot 0 = 0$
5. $\forall x, y \in N \quad x + s(y) = s(x + y)$
6. $\forall x, y \in N \quad x \cdot s(y) = x \cdot y + x$
7. $\forall x, y \in N \quad Limit(y) \to x + y = y + x$
8. $\forall x, y \in N \quad Limit(y) \to xy = yx$

9. $\forall x, y, z \in N \quad Limit(z) \to (x + y) + z = y + (x + z)$
10. $\forall x, y, z \in N \quad Limit(z) \to (xy)z = y(xz)$
11. $\forall x, y, z \in N \quad Limit(z) \to x(y + z) = xy + xz$
12. $\forall x, y, z \in N \quad Limit(z) \to x + z = y + z \to x = y$
13. $\forall x, y \in N \quad Limit(y) \to xy = 0 \to x = 0 \vee y = 0$
14. $\Big(\forall x \in N \quad Limit(x) \to \phi(x)\Big) \wedge \Big(\forall y \in N \quad \big(\phi(y) \to \phi(s(y))\big)\Big) \to \forall y \in N \quad \phi(y)$ (this is true for arbitrary "ϕ")

we call "$\mathfrak{N}$" a model of "Hyper Peano arithmetic".

Lemma 10.6.1. *prove for "$(N, +)$"*

1. $\forall x, y, z \in N \quad (x + y) + z = x + (y + z)$ *(associative $+$)*
2. $\forall x \in N \quad x + 0 = 0 + x = x$ *(identity $+$)*
3. $\forall x, y, z \in N \quad x + z = y + z \to x = y$ *(cancellative $+$)*
4. $\forall x, y \in N \quad x + y = y + x$ *(commutative(abelian) $+$)*

Proof Is Left As An Exercise To The Reader.

Lemma 10.6.2. *prove for "$(N, +, \cdot)$"*

1. $\forall x, y, z \in N \quad (xy)z = x(yz)$ *(associative $\times$)*
2. $\forall x \in N \quad x \cdot 1 = 1 \cdot x = x$ *(identity $\times$)*
3. $\forall x, y \in N \quad xy = 0 \to x = 0 \vee y = 0$ *(zero divisor free $\times$)*
4. $\forall x, y \in N \quad xy = yx$ *(commutative(abelian) $\times$)*
5. $\forall x, y, z \in N \quad \Big(x(y + z) = xy + xz\Big) \wedge \Big((x + y)z = xz + yz\Big)$ *(distributive $\times$)*

Proof Is Left As An Exercise To The Reader.

Now define "$x < y : \exists z \neq 0 \quad x + z = y$". You can prove for natural numbers

$$\forall \alpha, \beta \in Ord \quad \alpha < \beta \leftrightarrow \exists k \neq 0 \quad \alpha + k = \beta$$

Lemma 10.6.3. *prove for "$(N, +, \times, <)$"*

1. $\forall x, y, z \in N \quad x < y \to x + z < y + z$
2. $\forall x, y, z \in N \quad x < y \wedge 0 < z \to xz < yz$

Proof Is Left As An Exercise To The Reader.

as you can see "$(Ord, +, \times, s, 0)$" is a model of hyper peano arithmetic. so properties are true for them.

This page intentionally left blank.

Chapter 11

Cardinals

This page intentionally left blank.

Chapter 12

Axiom of choice

This page intentionally left blank.

Chapter 13

fantacy numbers$(\mathbb{Z}(Ord), \mathbb{Q}(Ord), \mathbb{R}(Ord), \mathbb{C}(Ord))$

13.1 fantacy Integers"$\mathbb{Z}(Ord)$"

we learned about "Ord" before and we constructed structure "$(Ord, +, \cdot, <)$". if you assume "$Ord \times Ord$" we can define an equivalence relation "$(\alpha_1, \beta_1) \sim (\alpha_2, \beta_2) : \alpha_1 + \beta_2 = \alpha_2 + m_1$"(check it's an equivalence relation).

definition 13.1.1. ***fantacy Integers"$\mathbb{Z}(Ord)$"*** *is a set*

$$\mathbb{Z}(Ord) := Ord \times Ord/\sim$$

we denote it's members with "$z_0, z_1, \ldots$"

in this system "$0 := [(0,0)]$" and "$\forall \beta \in Ord \quad \beta := [(\beta, 0)]$" and "$\forall \beta \in Ord \quad -\beta := [(0, \beta)]$". in general"$\forall \alpha, \beta \in Ord \quad \alpha - \beta := [(\alpha, \beta)]$". so we can say "$Ord \subseteq \mathbb{Z}(Ord)$".

Theorem 13.1.1. ***Summation*** *there is a unique function "$+_{\mathbb{Z}}(Ord) : \mathbb{Z}(Ord) \times \mathbb{Z}(Ord) \to \mathbb{Z}(Ord)$" s.t.*

$$\forall \alpha_1, \alpha_2, \beta_1, \beta_2 \in Ord \quad [(\alpha_1, \beta_1)] +_{\mathbb{Z}(Ord)} [(\alpha_2, \beta_2)] = [(\alpha_1 + \alpha_2, \beta_1 + \beta_2)]$$

Proof Is Left As An Exercise To The Reader.

Theorem 13.1.2. ***Multiplication*** *there is a unique function "$\times_{\mathbb{Z}(Ord)} : \mathbb{Z}(Ord) \times \mathbb{Z}(Ord) \to \mathbb{Z}(Ord)$" s.t.*

$$\forall \alpha_1, \alpha_2, \beta_1, \beta_2 \in Ord \quad [(\alpha_1, \beta_1)] \times_{\mathbb{Z}(Ord)} [(\alpha_2, \beta_2)] = [(\alpha_1\alpha_2 + \beta_1\beta_2, \alpha_1\beta_2 + \alpha_2\beta_1)]$$

Proof Is Left As An Exercise To The Reader.

definition 13.1.2. ***Order in*** $\mathbb{Z}(Ord)$

$$[(\alpha_1, \beta_1)] <_{\mathbb{Z}(Ord)} [(\alpha_2, \beta_2)] : \alpha_1 + \beta_2 < \alpha_2 + \beta_1$$

this lemma let us use "$<$" instead of "$<_{\mathbb{Z}(Ord)}$", "$+$" instead of "$+_{\mathbb{Z}(Ord)}$" and "$\times$" instead of "$\times_{\mathbb{Z}(Ord)}$".

Lemma 13.1.1. *prove.*

1. $/\ \therefore \forall \alpha, \beta \in Ord \quad \alpha <_{\mathbb{Z}(Ord)} \beta \leftrightarrow \alpha < \beta$
2. $/\ \therefore \forall \alpha, \beta \in Ord \quad \alpha +_{\mathbb{Z}(Ord)} \beta = \alpha + \beta$
3. $/\ \therefore \forall \alpha, \beta \in Ord \quad \alpha \times_{\mathbb{Z}(Ord)} \beta = \alpha \times \beta$

Proof Is Left As An Exercise To The Reader.

we denote "Multiplication" with "$z \cdot y$" and "zy" too.

Lemma 13.1.2. *prove for "$(\mathbb{Z}(Ord), +)$"*

1. $\forall x, y, z \in \mathbb{Z}(Ord) \quad (x + y) + z = x + (y + z)$ *(associative $+$)*
2. $\forall z \in \mathbb{Z}(Ord) \quad z + 0 = 0 + z = z$ *(identity $+$)*
3. $\forall z \in \mathbb{Z}(Ord) \quad \exists y \in \mathbb{Z}(Ord) \quad z + y = y + z = 0$ *(inverse $+$)*
4. $\forall x, y \in \mathbb{Z}(Ord) \quad x + y = y + x$ *(commutative(abelian) $+$)*

Proof Is Left As An Exercise To The Reader.

Lemma 13.1.3. *prove for "$(\mathbb{Z}(Ord), +, \times)$"*

1. $\forall x, y, z \in \mathbb{Z}(Ord) \quad (xy)z = x(yz)$ *(associative $\times$)*
2. $\forall z \in \mathbb{Z}(Ord) \quad z \cdot 1 = 1 \cdot z = z$ *(identity $\times$)*
3. $\forall x, y \in \mathbb{Z}(Ord) \quad xy = 0 \rightarrow x = 0 \vee y = 0$ *(zero divisor free $\times$)*
4. $\forall x, y \in \mathbb{Z}(Ord) \quad xy = yx$ *(commutative(abelian) $\times$)*
5. $\forall x, y, z \in \mathbb{Z}(Ord) \quad \Big(x(y + z) = xy + xz\Big) \wedge \Big((x + y)z = xz + yz\Big)$ *(distributive $\times$)*

Proof Is Left As An Exercise To The Reader.

Theorem 13.1.3. ***Order of fantacy Integers($\mathbb{Z}(Ord)$)*** *prove for "$(\mathbb{Z}(Ord), +, \times, <_{\mathbb{Z}(Ord)})$"*

1. *is linearly ordered.*
2. $\Big(\forall x \in \mathbb{Z}(Ord) \quad \exists y \in \mathbb{Z}(Ord) \quad x < y\Big) \wedge \Big(\forall x \in \mathbb{Z}(Ord) \quad \exists y \in \mathbb{Z}(Ord) \quad y < x\Big)$ *(doesn't have endpoints)*
3. $\forall x, y, z \in \mathbb{Z}(Ord) \quad x < y \rightarrow x + z < y + z$
4. $\forall x, y, z \in \mathbb{Z}(Ord) \quad x < y \wedge 0 < z \rightarrow xz < y\gamma$

Proof Is Left As An Exercise To The Reader.

13.2 fantacy Rational numbers "$\mathbb{Q}(Ord)$"

we learned about "$\mathbb{Z}(Ord)$" before and we constructed structure "$(\mathbb{Z}(Ord), +, \cdot, <)$". if you assume "$\mathbb{Z}(Ord) \times (\mathbb{Z}(Ord) - \{0\})$" we can define an equivalence relation "$(y_1, x_1) \sim (y_2, x_2) : y_1 \cdot x_2 = y_2 \cdot x_1$"(check it's an equivalence relation).

definition 13.2.1. ***fantacy Rational numbers*** *"$\mathbb{Q}(Ord)$" is a set*

$$\mathbb{Q}(Ord) := \mathbb{Z}(Ord) \times (\mathbb{Z}(Ord) - \{0\})/\sim$$

we denote it's members with "$q_0, q_1, ...$"

in this system "$0 := [(0,1)]$" and "$1 := [(1,1)]$" and "$\forall x \in \mathbb{Z}(Ord) \quad x := [(x,1)]$" and "$\forall x \in \mathbb{Z}(Ord) \quad \frac{1}{x} := [(1,x)]$". in general "$\forall x, y \in \mathbb{Z}(Ord) \quad \frac{x}{y} := [(x,y)]$". so we can say "$Ord \subseteq \mathbb{Z}(Ord) \subseteq \mathbb{Q}(Ord)$".

Theorem 13.2.1. ***Summation*** *there is a unique function "$+_{\mathbb{Q}(Ord)} : \mathbb{Q}(Ord) \times \mathbb{Q}(Ord) \to \mathbb{Q}(Ord)$" s.t.*

$$\forall x, y, z, u \in \mathbb{Z}(Ord) \quad [(x,z)] +_{\mathbb{Q}(Ord)} [(y,u)] = [(xu + zy, zu)]$$

Proof Is Left As An Exercise To The Reader.

Theorem 13.2.2. ***Multiplication*** *there is a unique function "$\times_{\mathbb{Q}(Ord)} : \mathbb{Q}(Ord) \times \mathbb{Q}(Ord) \to \mathbb{Q}(Ord)$" s.t.*

$$\forall x, y, z, u \in \mathbb{Z}(Ord) \quad [(x,z)] \times_{\mathbb{Q}(Ord)} [(y,u)] = [(xy, zu)]$$

Proof Is Left As An Exercise To The Reader.

definition 13.2.2. ***Order in*** $\mathbb{Q}(Ord)$.
if $(z > 0, u > 0)$ or $(z < 0, u < 0)$

$$[(x,z)] <_{\mathbb{Q}(Ord)} [(y,u)] : x \cdot u < y \cdot z$$

if $(z > 0, u < 0)$ or $(z < 0, u > 0)$

$$[(x,z)] <_{\mathbb{Q}(Ord)} [(y,u)] : x \cdot u > y \cdot z$$

this lemma let us use "$<$" instead of "$<_{\mathbb{Q}(Ord)}$", "$+$" instead of "$+_{\mathbb{Q}(Ord)}$" and "$\times$" instead of "$\times_{\mathbb{Q}(Ord)}$".

Lemma 13.2.1. *prove.*

1. $/ \therefore \forall z, y \in \mathbb{Z}(Ord) \quad z <_{\mathbb{Q}(Ord)} y \leftrightarrow z < y$
2. $/ \therefore \forall z, y \in \mathbb{Z}(Ord) \quad z +_{\mathbb{Q}(Ord)} y = z + y$
3. $/ \therefore \forall z, y \in \mathbb{Z}(Ord) \quad z \times_{\mathbb{Q}(Ord)} y = z \times y$

Proof Is Left As An Exercise To The Reader.

we denote "Multiplication" with "$x \cdot y$" and "xy" too.

Lemma 13.2.2. *prove for "$(\mathbb{Q}(Ord), +)$"*

1. $\forall x, y, z \in \mathbb{Q}(Ord) \quad (x + y) + z = x + (y + z)$ *(associative $+$)*
2. $\forall x \in \mathbb{Q}(Ord) \quad x + 0 = 0 + x = x$ *(identity $+$)*
3. $\forall x \in \mathbb{Q}(Ord) \quad \exists y \in \mathbb{Q}(Ord) \quad x + y = y + x = 0$ *(inverse $+$)*
4. $\forall x, y \in \mathbb{Q}(Ord) \quad x + y = y + x$ *(commutative(abelian) $+$)*

Proof Is Left As An Exercise To The Reader.☺
Lemma 13.2.3. *prove for "$(\mathbb{Q}(Ord), +, \times)$"*

1. $\forall x, y, z \in \mathbb{Q}(Ord) \quad (xy)z = x(yz)$ *(associative $\times$)*
2. $\forall x \in \mathbb{Q}(Ord) \quad x \cdot 1 = 1 \cdot x = x$ *(identity $\times$)*
3. $\forall x \in \mathbb{Q}(Ord) - \{0\} \quad \exists y \in \mathbb{Q}(Ord) \quad xy = yx = 1$ *(inverse $\times$)*
4. $\forall x, y \in \mathbb{Q}(Ord) \quad xy = yx$ *(commutative(abelian) $\times$)*
5. $\forall x, y, z \in \mathbb{Q}(Ord) \quad \Big(x(y+z) = xy + xz\Big) \wedge \Big((x+y)z = xz + yz\Big)$ *(distributive $\times$)*

Proof Is Left As An Exercise To The Reader.☺
Theorem 13.2.3. ***Order of fantacy Rational numbers($\mathbb{Q}(Ord)$)*** *prove for "$(\mathbb{Q}(Ord), <_{\mathbb{Q}(Ord)})$"*

1. *is linearly ordered.*
2. $\Big(\forall x \in \mathbb{Q}(Ord) \quad \exists y \in \mathbb{Q}(Ord) \quad x < y\Big) \wedge \Big(\forall x \in \mathbb{Q}(Ord) \quad \exists y \in \mathbb{Q}(Ord) \quad y < x\Big)$ *(doesn't have endpoints)*
3. $\forall x, y \in \mathbb{Q}(Ord) \quad \exists z \in \mathbb{Q}(Ord) \quad x \neq y \to x < z < y$ *(dense)*
4. $\forall x, y, z \in \mathbb{Q}(Ord) \quad x < y \to x + z < y + z$
5. $\forall x, y, z \in \mathbb{Q}(Ord) \quad x < y \wedge 0 < z \to xz < y\gamma$

Proof Is Left As An Exercise To The Reader.☺

13.3 fantacy Real numbers "$\mathbb{R}(Ord)$"

before we we get to lesson define function "$|_| : \mathbb{Q}(Ord) \times \mathbb{Q}(Ord) \to \mathbb{Q}(Ord)$" like this.

$$|x| := \begin{cases} x & 0 \leq x \\ -x & x < 0 \end{cases}$$

if you are not familiar to this notation it means "$|_| := \{(x, y) \in \mathbb{Q}(Ord) \times \mathbb{Q}(Ord) \mid (0 \leq x \to y = x) \wedge (x < 0 \to y = -x)\}$". learn this notation because we want to use later.

we learned about "$\mathbb{Q}(Ord)$" before and we constructed structure "$(\mathbb{Q}(Ord), +, \cdot, <)$". now assume $\mathbb{Q}(Ord)^{Ord}$(we know what does it mean by the definition. it means all the sequences of $\mathbb{Q}(Ord)$). we don't want it all we just want a subset.
definition 13.3.1. ***Cauchy sequence*** *is a sequence like "$(a_i)_{i \in Ord} \in \mathbb{Q}(Ord)^{Ord}$" s.t. elements of it get infinitely close to each other i.e.*

$$Cauchy(a_i)_{i \in Ord} : \forall \varepsilon > 0 \quad \exists N \quad \forall n, m \quad N < n, m < roof(N) \to |a_n - a_m| < \varepsilon$$

we usually give "N" natural numbers but it's not necessary.

we define an equivalence relation "$(a_i)_{i \in Ord} \sim (b_i)_{i \in Ord} : \forall \varepsilon > 0 \quad \exists N \quad \forall n \quad N < n < roof(N) \to |a_n - b_n| < \varepsilon$"(check it's an equivalence relation). we denote every $[(a_i)_{i \in Ord}]$ with $[a_i]_{i \in Ord}$.

definition 13.3.2. ***fantacy Real numbers*** *"$\mathbb{R}(Ord)$" is a set*

$$\mathbb{R}(Ord) := Cauchy/\sim$$

we denote it's members with "$r_0, r_1, ...$"

in this system "$0 := [0]_{i\in Ord}$" and "$1 := [1]_{i\in Ord}$" and "$\forall q \in \mathbb{Q}(Ord) \quad q := [q]_{i\in Ord}$". so we can say "$Ord \subseteq \mathbb{Z}(Ord) \subseteq \mathbb{Q}(Ord) \subseteq \mathbb{R}(Ord)$".

Theorem 13.3.1. ***Summation*** *there is a unique function "$+_{\mathbb{R}(Ord)} : \mathbb{R}(Ord) \times \mathbb{R}(Ord) \to \mathbb{R}(Ord)$" s.t.*

$$[a_i]_{i\in Ord} +_{\mathbb{R}(Ord)} [b_i]_{i\in Ord} = [a_i + b_i]_{i\in Ord}$$

Proof Is Left As An Exercise To The Reader.☺

Theorem 13.3.2. ***Multiplication*** *there is a unique function "$\times_{\mathbb{R}(Ord)} : \mathbb{R}(Ord) \times \mathbb{R}(Ord) \to \mathbb{R}(Ord)$" s.t.*

$$[a_i]_{i\in Ord} \times_{\mathbb{R}(Ord)} [b_i]_{i\in Ord} = [a_i b_i]_{i\in Ord}$$

Proof Is Left As An Exercise To The Reader.☺

definition 13.3.3. ***Order in*** $\mathbb{R}(Ord)$.

$$[a_i]_{i\in Ord} <_{\mathbb{R}(Ord)} [b_i]_{i\in Ord} : \exists \varepsilon > 0 \quad \exists N \quad \forall n \quad N < n < roof(N) \to b_n - a_n > \varepsilon$$

this lemma let us use "$<$" instead of "$<_{\mathbb{R}(Ord)}$", "$+$" instead of "$+_{\mathbb{R}(Ord)}$" and "$\times$" instead of "$\times_{\mathbb{R}(Ord)}$".

Lemma 13.3.1. *prove.*

1. $/ \therefore \forall x, y \in \mathbb{Q}(Ord) \quad x <_{\mathbb{R}(Ord)} y \leftrightarrow x < y$
2. $/ \therefore \forall x, y \in \mathbb{Q}(Ord) \quad x +_{\mathbb{R}(Ord)} y = x + y$
3. $/ \therefore \forall x, y \in \mathbb{Q}(Ord) \quad x \times_{\mathbb{R}(Ord)} y = x \times y$

Proof Is Left As An Exercise To The Reader.☺

we denote "Multiplication" with "$x \cdot y$" and "xy" too.

Lemma 13.3.2. *prove.*

1. $\forall x, y, z \in \mathbb{R}(Ord) \quad (x + y) + z = x + (y + z)$ *(associative +)*
2. $\forall x \in \mathbb{R}(Ord) \quad x + 0 = 0 + x = x$ *(identity +)*
3. $\forall x \in \mathbb{R}(Ord) \quad \exists y \in \mathbb{R}(Ord) \quad x + y = y + x = 0$ *(inverse +)*
4. $\forall x, y \in \mathbb{R}(Ord) \quad x + y = y + x$ *(commutative(abelian) +)*

Proof Is Left As An Exercise To The Reader.☺

Lemma 13.3.3. *prove.*

1. $\forall x, y, z \in \mathbb{R}(Ord) \quad (xy)z = x(yz)$ *(associative ×)*
2. $\forall x \in \mathbb{R}(Ord) \quad x \cdot 1 = 1 \cdot x = x$ *(identity ×)*

3. $\forall x \in \mathbb{R}(Ord) - \{0\} \quad \exists y \in \mathbb{R}(Ord) \quad xy = yx = 1$ *(inverse $\times$)*
4. $\forall x, y \in \mathbb{R}(Ord) \quad xy = yx$ *(commutative(abelian) $\times$)*
5. $\forall x, y, z \in \mathbb{R}(Ord) \quad \Big(x(y+z) = xy + xz\Big) \wedge \Big((x+y)z = xz + yz\Big)$ *(distributive $\times$)*

Proof Is Left As An Exercise To The Reader.

Theorem 13.3.3. ***Linear order of fantacy Real numbers($\mathbb{R}(Ord)$)*** *prove "$(\mathbb{R}(Ord), <_{\mathbb{R}(Ord)})$"*

1. *is linearly ordered.*
2. $\Big(\forall x \in \mathbb{R}(Ord) \quad \exists y \in \mathbb{R}(Ord) \quad x < y\Big) \wedge \Big(\forall x \in \mathbb{R}(Ord) \quad \exists y \in \mathbb{R}(Ord) \quad y < x\Big)$ *(doesn't have endpoints)*
3. $\forall x, y \in \mathbb{R}(Ord) \quad \exists z \in \mathbb{R}(Ord) \quad x \neq y \to x < z < y$ *(dense)*
4. $\forall x, y, z \in \mathbb{R}(Ord) \quad x < y \to x + z < y + z$
5. $\forall x, y, z \in \mathbb{R}(Ord) \quad x < y \wedge 0 < z \to xz < yk$

Proof Is Left As An Exercise To The Reader.

13.4 fantacy Complex numbers"$\mathbb{C}(Ord)$"

we learned about "$\mathbb{R}(Ord)$" before and we constructed structure "$(\mathbb{R}(Ord), +, \cdot, <)$".

definition 13.4.1. ***fantacy Complex numbers"$\mathbb{C}(Ord)$"*** *is a set*

$$\mathbb{C}(Ord) := \mathbb{R}(Ord) \times \mathbb{R}(Ord)$$

we denote it's members with "$z_0, x, \dots$"

in this system "$0 := (0,0)$" and "$\forall n \in \mathbb{R}(Ord) \quad r := (r, 0)$" . in general "$\forall x, y \in \mathbb{R}(Ord) \quad x + iy := (x, y)$". so we can say "$Ord \subseteq \mathbb{Z}(Ord) \subseteq \mathbb{Q}(Ord) \subseteq \mathbb{R}(Ord) \subseteq \mathbb{C}(Ord)$".

Theorem 13.4.1. ***Summation*** *there is a unique function "$+_{\mathbb{C}(Ord)} : \mathbb{C}(Ord) \times \mathbb{C}(Ord) \to \mathbb{C}(Ord)$" s.t.*

$$\forall x, y, z, u \in \mathbb{R}(Ord) \quad (x, z) +_{\mathbb{C}(Ord)} (y, u) = (x + y, z + u)$$

Proof Is Left As An Exercise To The Reader.

Theorem 13.4.2. ***Multiplication*** *there is a unique function "$\times_{\mathbb{C}(Ord)} : \mathbb{C}(Ord) \times \mathbb{C}(Ord) \to \mathbb{C}(Ord)$" s.t.*

$$\forall x, y, z, u \in \mathbb{R}(Ord) \quad (x, z) \times_{\mathbb{C}(Ord)} (y, u) = (xy - zu, xu + yz)$$

Proof Is Left As An Exercise To The Reader.

this lemma let us use "$+$" instead of "$+_{\mathbb{C}(Ord)}$" and "$\times$" instead of "$\times_{\mathbb{C}(Ord)}$".

Lemma 13.4.1. *prove.*

1. $/ \therefore \forall x, y \in \mathbb{R}(Ord) \quad x +_{\mathbb{C}(Ord)} y = x + y$

2. $/\therefore \forall x, y \in \mathbb{R}(Ord) \quad x \times_{\mathbb{C}(Ord)} y = x \times y$

Proof Is Left As An Exercise To The Reader.☺

we denote "Multiplication" with "$z \cdot y$" and "zy" too.

Lemma 13.4.2. *prove for "$(\mathbb{C}(Ord), +)$"*

1. $\forall x, y, z \in \mathbb{C}(Ord) \quad (x + y) + z = x + (y + z)$ *(associative +)*
2. $\forall z \in \mathbb{C}(Ord) \quad z + 0 = 0 + z = z$ *(identity +)*
3. $\forall z \in \mathbb{C}(Ord) \quad \exists y \in \mathbb{C}(Ord) \quad z + y = y + z = 0$ *(inverse +)*
4. $\forall x, y \in \mathbb{C}(Ord) \quad x + y = y + x$ *(commutative(abelian) +)*

Proof Is Left As An Exercise To The Reader.☺

Lemma 13.4.3. *prove for "$(\mathbb{C}(Ord), +, \times)$"*

1. $\forall x, y, z \in \mathbb{C}(Ord) \quad (xy)z = x(yz)$ *(associative ×)*
2. $\forall z \in \mathbb{C}(Ord) \quad z \cdot 1 = 1 \cdot z = z$ *(identity ×)*
3. $\forall x, y \in \mathbb{C}(Ord) \quad xy = yx$ *(commutative(abelian) ×)*
4. $\forall x, y, z \in \mathbb{C}(Ord) \quad \Big(x(y + z) = xy + xz\Big) \wedge \Big((x + y)z = xz + yz\Big)$ *(distributive ×)*

Proof Is Left As An Exercise To The Reader.☺

Theorem 13.4.3. ***Exponents*** *there is a unique function "$(_)^{(_)} : \mathbb{C}(Ord) \times Ord \to \mathbb{C}(Ord)$" s.t.*

1. $\forall x \in \mathbb{C}(Ord) \quad x^0 = 1$
2. $\forall x \in \mathbb{C}(Ord) \quad \forall n \in Ord \quad x^{s(n)} = x^n \cdot x$

Proof Is Left As An Exercise To The Reader.☺

Lemma 13.4.4. *prove for "$(\mathbb{C}(Ord), +, \cdot, (_)^{(_)})$"*

1. $\forall x \in \mathbb{Z}(Ord) \quad \forall m, n \in Ord \quad x^m \cdot x^n = x^{m+n}$
2. $\forall x \in \mathbb{Z}(Ord) \quad \forall m, n \in Ord \quad (x^m)^n = x^{mn}$

Proof Is Left As An Exercise To The Reader.☺

Theorem 13.4.4. ***Exponents*** *there is a unique function "$(_)^{(_)} : \mathbb{C}(Ord) \times \mathbb{Z}(Ord) \to \mathbb{C}(Ord)$" s.t.*

1. $\forall x \in \mathbb{C}(Ord) \quad x^0 = 1$
2. $\forall x \in \mathbb{C}(Ord) \quad \forall m, n \in Ord \quad x^{m-n} = \frac{x^m}{x^n}$

Proof Is Left As An Exercise To The Reader.☺

Lemma 13.4.5. *prove for "$(\mathbb{C}(Ord), +, \cdot)$"*

1. $\forall x \in \mathbb{C}(Ord) \quad \forall y, z \in \mathbb{Z}(Ord) \quad x^y \cdot x^z = x^{y+z}$
2. $\forall x \in \mathbb{C}(Ord) \quad \forall y, z \in \mathbb{Z}(Ord) \quad (x^y)^z = x^{yz}$

Proof Is Left As An Exercise To The Reader.☺

Theorem 13.4.5. ***Big Sum*** *if "$a : Ord \to \mathbb{C}(Ord)$" there is a unique function "$\sum_{i=0}^{(_)}(_) : \mathbb{C}(Ord) \times Ord \to \mathbb{C}(Ord)$" s.t.*

1. $\forall a \quad \sum_{i=0}^{0}(a_0) = a_0$
2. $\forall a \quad \forall n \in Ord \quad \sum_{i=0}^{s(n)} a_{s(n)} = \sum_{i=0}^{n} a_n + a_{s(n)}$

Proof Is Left As An Exercise To The Reader.☺

Lemma 13.4.6. *prove for "$(\mathbb{C}(Ord), +, \cdot)$"*

1. $\forall n \in Ord \quad \forall a, b \quad \sum_{i=0}^{n} a_n + \sum_{i=0}^{n} b_n = \sum_{i=0}^{n}(a_n + b_n)$
2. $\forall n \in Ord \quad \forall a \quad k\sum_{i=0}^{n} a_n = \sum_{i=0}^{n} ka_n$

Proof Is Left As An Exercise To The Reader.☺

Theorem 13.4.6. ***Big Product*** *if "$a : Ord \to \mathbb{C}(Ord)$" there is a unique function "$\prod_{i=0}^{(_)}(_) : \mathbb{C}(Ord) \times Ord \to \mathbb{C}(Ord)$" s.t.*

1. $\forall a \quad \prod_{i=0}^{0}(a_0) = a_0$
2. $\forall a \quad \forall n \in Ord \quad \prod_{i=0}^{s(n)} a_{s(n)} = \prod_{i=0}^{n} a_n \cdot a_{s(n)}$

Proof Is Left As An Exercise To The Reader.☺

Lemma 13.4.7. *prove for "$(\mathbb{C}(Ord), +, \cdot)$"*

1. $\forall n \in Ord \quad \forall a, b \quad \prod_{i=0}^{n} a_n \prod_{i=0}^{n} b_n = \prod_{i=0}^{n}(a_n b_n)$
2. $\forall n \in Ord \quad \forall a \quad (\prod_{i=0}^{n} a_n)^k = \prod_{i=0}^{n}(a_n)^k$

Proof Is Left As An Exercise To The Reader.☺

Chapter 14

hyperreal numbers($^*\mathbb{N}, ^*\mathbb{Z}, ^*\mathbb{Q}, ^*\mathbb{R}$)

14.1 Filters

Filters are mathematical objects used to formalize the idea of "large subsets" of a set.
definition 14.1.1 (Filter)**.** *Let X be a nonempty set. A* filter $\mathcal{F}$ *on X is a collection of subsets of X such that:*

1. $\varnothing \notin \mathcal{F}$.
2. $X \in \mathcal{F}$.
3. *If $A, B \in \mathcal{F}$, then $A \cap B \in \mathcal{F}$.*
4. *If $A \in \mathcal{F}$ and $A \subseteq B \subseteq X$, then $B \in \mathcal{F}$.*

Intuitively, the sets in $\mathcal{F}$ are considered "large". The intersection of two large sets is still large, and every superset of a large set is also large.
Example 14.1.1 (Trivial Filter)**.** *For any set X, $\mathcal{F} = \{X\}$ is a filter on X.*
Example 14.1.2 (Principal Filter)**.** *Fix $A \subseteq X$ with $A \neq \varnothing$. Define*

$$\mathcal{F}_A := \{B \subseteq X : A \subseteq B\}.$$

Then $\mathcal{F}_A$ is a filter on X. This is called the principal filter generated by A.
Example 14.1.3 (Fréchet Filter)**.** *Let $X = \mathbb{N}$. Define*

$$\mathcal{F} = \{A \subseteq \mathbb{N} : \mathbb{N} - A \text{ is finite}\}.$$

Then $\mathcal{F}$ is a filter. Its elements are the cofinite subsets of $\mathbb{N}$. This is called the Fréchet filter.

14.2 Ultrafilters

Ultrafilters are maximal filters.

definition 14.2.1 (Ultrafilter)**.** *A filter $\mathcal{U}$ on X is called an* ultrafilter *if it is maximal among filters; that is, whenever $\mathcal{F}$ is a filter on X and $\mathcal{U} \subseteq \mathcal{F}$, then $\mathcal{F} = \mathcal{U}$.*

There is a very important characterization of ultrafilters.
Theorem 14.2.1 (Ultrafilter Characterization)**.** *Let $\mathcal{U}$ be a filter on X. Then $\mathcal{U}$ is an ultrafilter if and only if for every subset $A \subseteq X$, exactly one of the following holds:*

$$A \in \mathcal{U}, \quad \text{or} \quad X - A \in \mathcal{U}.$$

Proof. Suppose first that $\mathcal{U}$ is an ultrafilter. Let $A \subseteq X$. We cannot have both $A \in \mathcal{U}$ and $X - A \in \mathcal{U}$, since then $\varnothing = A \cap (X - A) \in \mathcal{U}$, which is impossible. Suppose that neither A nor $X - A$ belongs to $\mathcal{U}$. Then the filter generated by $\mathcal{U} \cup \{A\}$ is strictly larger than $\mathcal{U}$, contradicting the maximality of $\mathcal{U}$. Hence exactly one of them belongs to $\mathcal{U}$.

Conversely, suppose that for every $A \subseteq X$, exactly one of A and $X - A$ belongs to $\mathcal{U}$. Let $\mathcal{F}$ be a filter on X such that $\mathcal{U} \subseteq \mathcal{F}$. If $\mathcal{F} \neq \mathcal{U}$, choose $A \in \mathcal{F} - \mathcal{U}$. By assumption, $X - A \in \mathcal{U} \subseteq \mathcal{F}$. Therefore

$$A \cap (X - A) = \varnothing \in \mathcal{F},$$

a contradiction. Hence $\mathcal{F} = \mathcal{U}$, so $\mathcal{U}$ is an ultrafilter.

□

Thus an ultrafilter makes a complete decision about every subset of X: exactly one of A and $X - A$ is "large".
Example 14.2.1. *Fix $x \in X$. Define*

$$\mathcal{U}_x := \{A \subseteq X : x \in A\}.$$

Then $\mathcal{U}_x$ is an ultrafilter. It is called the principal ultrafilter at x.

Proof. First, $\varnothing \notin \mathcal{U}_x$ and $X \in \mathcal{U}_x$. If $A, B \in \mathcal{U}_x$, then $x \in A \cap B$, so $A \cap B \in \mathcal{U}_x$. If $A \in \mathcal{U}_x$ and $A \subseteq B$, then $x \in B$, so $B \in \mathcal{U}_x$. Thus $\mathcal{U}_x$ is a filter.

Moreover, for every $A \subseteq X$, either $x \in A$ or $x \notin A$. Hence either

$$A \in \mathcal{U}_x, \qquad X - A \in \mathcal{U}_x.$$

By the Ultrafilter Characterization, $\mathcal{U}_x$ is an ultrafilter.

□

14.3 Nonprincipal Ultrafilters

An ultrafilter on $\mathbb{N}$ is called *nonprincipal* if it contains no finite set.

Equivalently, it extends the Fréchet filter.

Nonprincipal ultrafilters cannot be explicitly constructed in ordinary mathematics; their existence usually requires the Axiom of Choice.

definition 14.3.1 (Extension of a Filter)**.** *Let $\mathcal{F}$ and $\mathcal{G}$ be filters on a set X.*

We say that $\mathcal{G}$ extends $\mathcal{F}$ if $\mathcal{F} \subseteq \mathcal{G}$.

This means every set considered "large" by $\mathcal{F}$ is also considered "large" by $\mathcal{G}$.

Theorem 14.3.1 (Ultrafilter Lemma)**.** *Every filter on a set X can be extended to an ultrafilter.*

In other words, if $\mathcal{F}$ is a filter on X, then there exists an ultrafilter $\mathcal{U}$ such that $\mathcal{F} \subseteq \mathcal{U}$.

So every filter can be enlarged until it becomes maximal.

Example 14.3.1. *Let*

$$\mathcal{F} = \{A \subseteq \mathbb{N} : \mathbb{N} - A \text{ is finite}\}.$$

This is the Fréchet filter. It is not an ultrafilter, because for the set of even numbers $E = \{0, 2, 4, 6, \dots\}$, neither $E \in \mathcal{F}$ nor $\mathbb{N} - E \in \mathcal{F}$. By the Ultrafilter Lemma, there exists an ultrafilter $\mathcal{U}$ such that $\mathcal{F} \subseteq \mathcal{U}$. Since $\mathcal{U}$ is an ultrafilter, exactly one of

$$E \in \mathcal{U}, \quad \text{or} \quad \mathbb{N} - E \in \mathcal{U}$$

holds.

The Ultrafilter Lemma is usually proved using Zorn's Lemma.

Theorem 14.3.2 (Ultrafilter Lemma)**.** *Let $\mathcal{F}$ be a filter on a set X. Then there exists an ultrafilter $\mathcal{U}$ on X such that $\mathcal{F} \subseteq \mathcal{U}$.*

Proof. Let $\mathcal{P} = \{\mathcal{G} : \mathcal{G} \text{ is a filter on } X \text{ and } \mathcal{F} \subseteq \mathcal{G}\}$. We partially order $\mathcal{P}$ by inclusion. Let $\mathcal{C} \subseteq \mathcal{P}$ be a chain and define

$$\mathcal{H} := \bigcup_{\mathcal{G} \in \mathcal{C}} \mathcal{G}.$$

Clearly $\varnothing \notin \mathcal{H}$ and $X \in \mathcal{H}$. If $A, B \in \mathcal{H}$, then $A \in \mathcal{G}_1$ and $B \in \mathcal{G}_2$ for some $\mathcal{G}_1, \mathcal{G}_2 \in \mathcal{C}$. Since $\mathcal{C}$ is a chain, one of $\mathcal{G}_1 \subseteq \mathcal{G}_2$ or $\mathcal{G}_2 \subseteq \mathcal{G}_1$ holds, so A, B belong to the larger filter and hence $A \cap B \in \mathcal{H}$. If $A \in \mathcal{H}$ and $A \subseteq B \subseteq X$, then $A \in \mathcal{G}$ for some $\mathcal{G} \in \mathcal{C}$, and since $\mathcal{G}$ is a filter, $B \in \mathcal{G} \subseteq \mathcal{H}$. Thus $\mathcal{H}$ is a filter. Moreover, $\mathcal{F} \subseteq \mathcal{H}$, so $\mathcal{H} \in \mathcal{P}$. Therefore every chain in $\mathcal{P}$ has an upper bound, and by Zorn's Lemma, $\mathcal{P}$ has a maximal element, say $\mathcal{U}$.

We claim that $\mathcal{U}$ is an ultrafilter. Suppose $\mathcal{U} \subseteq \mathcal{V}$, where $\mathcal{V}$ is a filter on X. Then $\mathcal{F} \subseteq \mathcal{U} \subseteq \mathcal{V}$, so $\mathcal{V} \in \mathcal{P}$. By the maximality of $\mathcal{U}$, $\mathcal{V} = \mathcal{U}$. Therefore $\mathcal{U}$ is an ultrafilter extending $\mathcal{F}$. □

Theorem 14.3.3. *Let $\mathcal{U}$ be a nonprincipal ultrafilter on $\mathbb{N}$. Then $\mathcal{F}_{\mathrm{Fr}} \subseteq \mathcal{U}$. That is, every cofinite subset of $\mathbb{N}$ belongs to $\mathcal{U}$.*

Proof. Let $A \in \mathcal{F}_{\mathrm{Fr}}$. Then $\mathbb{N} - A$ is finite. Suppose, toward a contradiction, that $A \notin \mathcal{U}$. Since $\mathcal{U}$ is an ultrafilter, exactly one of

$$A \in \mathcal{U}, \quad \text{or} \quad \mathbb{N} - A \in \mathcal{U}$$

holds. Hence $\mathbb{N} - A \in \mathcal{U}$. Thus $\mathcal{U}$ contains a finite set, say $F = \{n_1, \ldots, n_k\} \in \mathcal{U}$.

Suppose that $\{n_i\} \notin \mathcal{U}$ for every i. Then, since $\mathcal{U}$ is an ultrafilter, $\mathbb{N} - \{n_i\} \in \mathcal{U}$ for every i. By closure under finite intersections,

$$\mathbb{N} - F = \bigcap_{i=1}^{k} (\mathbb{N} - \{n_i\}) \in \mathcal{U}.$$

Since also $F \in \mathcal{U}$, we obtain $F \cap (\mathbb{N} - F) = \varnothing \in \mathcal{U}$, a contradiction. Therefore $\{n_i\} \in \mathcal{U}$ for some i, so $\mathcal{U}$ is principal, contradicting our assumption.

Hence $A \in \mathcal{U}$. Since A was arbitrary, $\mathcal{F}_{\mathrm{Fr}} \subseteq \mathcal{U}$. □

14.4 Definition of ${}^*\mathbb{N}$, ${}^*\mathbb{Z}$, ${}^*\mathbb{Q}$, ${}^*\mathbb{R}$, and ${}^*\mathbb{R}_{Ord}$

definition 14.4.1 (Ultrapower Construction)**.** *Let X be a set and let $\mathcal{U}$ be a nonprincipal ultrafilter on ω. Consider the set X^ω of all functions $a : \omega \to X$. We usually write such a function as $a = (a_i)_{i \in \omega}$, where $a_i \in X$ for every $i \in \omega$.*
definition 14.4.2. *Define a relation $\sim$ on X^ω by $a \sim b$ if $\{i \in \omega : a_i = b_i\} \in \mathcal{U}$. In words, two sequences are equivalent if they agree on a set of indices belonging to the ultrafilter.*
Theorem 14.4.1. *The relation $\sim$ is an equivalence relation on X^ω.*

Proof. **Reflexivity.** For every $a \in X^\omega$,

$$\{i \in \omega : a_i = a_i\} = \omega \in \mathcal{U},$$

so $a \sim a$.

Symmetry. If $a \sim b$, then

$$\{i \in \omega : a_i = b_i\} = \{i \in \omega : b_i = a_i\},$$

so $b \sim a$.

Transitivity. Suppose $a \sim b$ and $b \sim c$. Then

$$A = \{i \in \omega : a_i = b_i\} \in \mathcal{U}, \qquad B = \{i \in \omega : b_i = c_i\} \in \mathcal{U}.$$

Since $\mathcal{U}$ is a filter, $A \cap B \in \mathcal{U}$. If $i \in A \cap B$, then $a_i = b_i = c_i$, so

$$A \cap B \subseteq \{i \in \omega : a_i = c_i\}.$$

Since $\mathcal{U}$ is closed upward,

$$\{i \in \omega : a_i = c_i\} \in \mathcal{U},$$

and therefore $a \sim c$. □

definition 14.4.3. *The quotient $X^\omega / \sim$ is called the* ultrapower *of X modulo $\mathcal{U}$. The equivalence class of $a \in X^\omega$ is denoted by $[a]$.*

We now apply this construction to the standard number systems.

definition 14.4.4. *The set of hypernatural numbers is*

$$ {}^*\mathbb{N} = \mathbb{N}^\omega / \sim . $$

definition 14.4.5. *The set of hyperintegers is*

$$ {}^*\mathbb{Z} = \mathbb{Z}^\omega / \sim . $$

definition 14.4.6. *The set of hyperrational numbers is*

$$ {}^*\mathbb{Q} = \mathbb{Q}^\omega / \sim . $$

definition 14.4.7. *The set of hyperreal numbers is*

$$ {}^*\mathbb{R} = \mathbb{R}^\omega / \sim . $$

definition 14.4.8 (The Structure ${}^*\mathbb{R}_{Ord}$)**.** *For each ordinal α, let ${}^*\mathbb{R}_\alpha$ denote the corresponding stage of hyperreal extension. Define*

$$ {}^*\mathbb{R}_{Ord} := \bigcup_{\alpha \in \mathrm{Ord}} {}^*\mathbb{R}_\alpha. $$

Since $\mathrm{Ord} \subseteq {}^*\mathbb{R}_{Ord}$*, every ordinal β is itself an element of ${}^*\mathbb{R}_{Ord}$. In particular, given $N \in {}^*\mathbb{R}_{Ord}$ and the index set $\{0, \dots, N-1\} \subseteq {}^*\mathbb{R}_{Ord}$, we say $i \in \{0, \dots, N-1\}$* ***is an ordinal*** *if $i \in \mathrm{Ord}$ (as opposed to i being a non-ordinal element of some ${}^*\mathbb{R}_\alpha$).*

The algebraic operations are defined componentwise. For example, if $[a], [b] \in {}^*\mathbb{R}$, then

$$ [a] + [b] = [(a_i + b_i)_{i \in \omega}], \qquad [a] \cdot [b] = [(a_i b_i)_{i \in \omega}]. $$

Similarly, the order relation is defined by $[a] < [b]$ if $\{i \in \omega : a_i < b_i\} \in \mathcal{U}$.

Theorem 14.4.2. *The standard structures $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, and $\mathbb{R}$ embed into ${}^*\mathbb{N}$, ${}^*\mathbb{Z}$, ${}^*\mathbb{Q}$, and ${}^*\mathbb{R}$, respectively.*

Proof. We prove the statement for $\mathbb{R}$. Define

$$ \iota : \mathbb{R} \to {}^*\mathbb{R}, \qquad \iota(r) = [(r, r, r, \dots)]. $$

Suppose $\iota(r) = \iota(s)$. Then

$$ \{i \in \omega : r = s\} \in \mathcal{U}. $$

If $r \neq s$, then $\{i \in \omega : r = s\} = \varnothing$, which cannot belong to a filter. Hence $r = s$, so ι is injective. □

Example 14.4.1. *Define $a_i = i$ for every $i \in \omega$. Then $[a] \in {}^*\mathbb{N}$ is larger than every standard natural number. Indeed, for every $n \in \mathbb{N}$,*

$$ \{i \in \omega : n < a_i\} = \{i \in \omega : n < i\} $$

is cofinite, hence belongs to $\mathcal{U}$. Therefore $n < [a]$.

14.5 The Ultrapower Construction of the Hyperreals

Let $r = (r_n)_{n\in\mathbb{N}}$ and $s = (s_n)_{n\in\mathbb{N}}$ be two sequences of real numbers. We denote the set $\{n \in \mathbb{N} : r_n = s_n\}$ by $[\![r = s]\!]$. Thus

$$[\![r = s]\!] = \{n \in \mathbb{N} : r_n = s_n\}.$$

The set $[\![r = s]\!]$ may be thought of as the interpretation, or value, of the statement "$r = s$". Instead of assigning a truth value to the statement, we assign the set of indices at which it is true.

Let $\mathcal{F}$ be a free ultrafilter on $\mathbb{N}$. We define $r \sim s$ if and only if $[\![r = s]\!] \in \mathcal{F}$.

In this case we say that $r = s$ holds **almost everywhere modulo** $\mathcal{F}$.

The same idea applies to other assertions. We define

$$[\![r < s]\!] = \{n \in \mathbb{N} : r_n < s_n\}, \qquad [\![r > s]\!] = \{n \in \mathbb{N} : r_n > s_n\},$$

and

$$[\![r \leq s]\!] = \{n \in \mathbb{N} : r_n \leq s_n\}.$$

Thus, for example, $[\![r < s]\!] \in \mathcal{F}$ means that $r_n < s_n$ for $\mathcal{F}$-almost all n. We call these sets the **Statement Values** of the corresponding statements.

Theorem 14.5.1. *The relation $\sim$ is an equivalence relation on $\mathbb{R}^{\mathbb{N}}$.*

Proof. For every $r \in \mathbb{R}^{\mathbb{N}}$, $[\![r = r]\!] = \mathbb{N} \in \mathcal{F}$, so $r \sim r$.

If $r \sim s$, then $[\![r = s]\!] = [\![s = r]\!] \in \mathcal{F}$, so $s \sim r$.

Finally, suppose $r \sim s$ and $s \sim t$. Then $[\![r = s]\!], [\![s = t]\!] \in \mathcal{F}$, so

$$[\![r = s]\!] \cap [\![s = t]\!] \in \mathcal{F}.$$

Since

$$[\![r = s]\!] \cap [\![s = t]\!] \subseteq [\![r = t]\!],$$

and $\mathcal{F}$ is closed upward, $[\![r = t]\!] \in \mathcal{F}$. Hence $r \sim t$.

□

definition 14.5.1. *The equivalence class of $r \in \mathbb{R}^{\mathbb{N}}$ under $\sim$ is denoted by*

$$[r] = \{s \in \mathbb{R}^{\mathbb{N}} : r \sim s\}.$$

definition 14.5.2. *For $r, s \in \mathbb{R}^{\mathbb{N}}$, define addition and multiplication on ${}^*\mathbb{R}$ by*

$$[r] + [s] = [r + s], \qquad [r] \cdot [s] = [r \cdot s],$$

where $(r + s)_n = r_n + s_n$ and $(r \cdot s)_n = r_n s_n$.

We define the order relation by

$$[r] < [s] \leftrightarrow [\![r < s]\!] \in \mathcal{F}.$$

These definitions are well-defined. If $r \sim r'$ and $s \sim s'$, then $r + s \sim r' + s'$ and $r \cdot s \sim r' \cdot s'$. Similarly,

$$[r] < [s] \leftrightarrow [r'] < [s'].$$

Theorem 14.5.2 (Almost-All Criterion)**.** *For sequences $r = (r_n)$, $s = (s_n)$, and $t = (t_n)$, we have*

$$[r] = [s] \leftrightarrow r_n = s_n \quad \textit{for } \mathcal{F}\textit{-almost all } n,$$

$$[r] < [s] \leftrightarrow r_n < s_n \quad \textit{for } \mathcal{F}\textit{-almost all } n,$$

$$[r] + [s] = [t] \leftrightarrow r_n + s_n = t_n \quad \textit{for } \mathcal{F}\textit{-almost all } n,$$

and

$$[r] \cdot [s] = [t] \leftrightarrow r_n s_n = t_n \quad \textit{for } \mathcal{F}\textit{-almost all } n.$$

The same principle holds for many other mathematical statements and is the basis of the ***Transfer Principle****.*

Theorem 14.5.3. *The structure $({}^*\mathbb{R}, +, \cdot, <)$ is an ordered field with zero $[0]$ and unity $[1]$.*

Proof. The ring properties follow directly from the corresponding properties of $\mathbb{R}$. In particular, the additive inverse of $[r]$ is $-[r] = [-r]$.

It remains to show that every nonzero element has a multiplicative inverse. Suppose $[r] \neq [0]$. Then $[\![r = 0]\!] \notin \mathcal{F}$. Since $\mathcal{F}$ is an ultrafilter,

$$J = \{n \in \mathbb{N} : r_n \neq 0\} \in \mathcal{F}.$$

Define

$$s_n = \begin{cases} \dfrac{1}{r_n}, & n \in J, \\ 0, & n \notin J. \end{cases}$$

Then $[\![r \cdot s = 1]\!] = J \in \mathcal{F}$, so $[r] \cdot [s] = [1]$. Thus $[s]$ is the multiplicative inverse of $[r]$.

Finally, since $\mathcal{F}$ is an ultrafilter, for any r, s, exactly one of

$$[\![r < s]\!], \qquad [\![r = s]\!], \qquad [\![s < r]\!]$$

belongs to $\mathcal{F}$. Hence the order on ${}^*\mathbb{R}$ is linear. Therefore $({}^*\mathbb{R}, +, \cdot, <)$

is an ordered field.

□

definition 14.5.3. *We identify every real number $r \in \mathbb{R}$ with the equivalence class of the constant sequence $(r, r, r, \ldots)$. Thus $r = [(r, r, r, \ldots)]$. Under this identification, $\mathbb{R} \subseteq {}^*\mathbb{R}$.*

For example, consider

$$\epsilon = \left[\left(0, 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots\right)\right].$$

For every $n \in \mathbb{N}$, $0 < \varepsilon < 1/n$, and therefore ε is a positive infinitesimal.

Similarly,

$$\omega = [(1, 2, 3, 4, \ldots)]$$

satisfies $\omega > n \qquad \forall n \in \mathbb{N}$.

Therefore ω is a positive infinite hyperreal number.

definition 14.5.4. *Let $f : A \to \mathbb{R}$, where $A \subseteq \mathbb{R}$. The **Nonstandard Extension** of f is the function ${}^*f : {}^*A \to {}^*\mathbb{R}$ defined by*

$${}^*f([x_n]) = [f(x_n)],$$

where ${}^*A = \{[x_n] : x_n \in A\}$.

The definition is well-defined. If $[x_n] = [y_n]$, then $\{n \in \mathbb{N} : x_n = y_n\} \in \mathcal{F}$. Therefore

$$\{n \in \mathbb{N} : f(x_n) = f(y_n)\} \in \mathcal{F},$$

so $[f(x_n)] = [f(y_n)]$. Thus the value of ${}^*f([x_n])$ does not depend on the representative chosen for $[x_n]$.

Moreover, if $x \in A$ is standard, then ${}^*f(x) = f(x)$. Indeed,

$${}^*f([(x, x, x, \ldots)]) = [(f(x), f(x), f(x), \ldots)] = f(x).$$

Therefore ${}^*f$ is genuinely an extension of f.

The Almost-All Criterion is the fundamental reason why this construction allows ordinary statements about real numbers to be extended to nonstandard numbers. For example, if $f(x) = x^2$ and $\omega = [(1, 2, 3, \ldots)]$, then ${}^*f(\omega) = [(1, 4, 9, \ldots)]$. Thus expressions such as ${}^*f(a + \varepsilon)$

are well-defined even when $a + \varepsilon \notin \mathbb{R}$.

This is the construction that will allow us to formulate limits, continuity, derivatives, and integrals using infinitesimals.

14.6 (Un)limited, Infinitesimal, and Appreciable Numbers

A hyperreal number $b \in {}^*\mathbb{R}$ is called:

- **limited** if $r < b < s$ for some $r, s \in \mathbb{R}$;
- **positive unlimited** if $r < b$ for every $r \in \mathbb{R}$;
- **negative unlimited** if $b < r$ for every $r \in \mathbb{R}$;
- **unlimited** if it is positive unlimited or negative unlimited;
- **positive infinitesimal** if $0 < b < r$ for every $r \in \mathbb{R}_{>0}$;
- **negative infinitesimal** if $r < b < 0$ for every $r \in \mathbb{R}_{<0}$;
- **infinitesimal** if it is positive infinitesimal, negative infinitesimal, or 0;
- **appreciable** if it is limited but not infinitesimal.

Thus all real numbers and all infinitesimals are limited. The only infinitesimal real number is 0, while every other real number is appreciable. An appreciable number is therefore a hyperreal number that is neither infinitely small nor infinitely large.

The above definitions can also be expressed using the natural numbers. For $b \in {}^*\mathbb{R}$,

$$b \text{ is limited} \leftrightarrow |b| < n \quad \text{for some } n \in \mathbb{N},$$

$$b \text{ is unlimited} \leftrightarrow |b| > n \quad \forall n \in \mathbb{N},$$

$$b \text{ is infinitesimal} \leftrightarrow |b| < \frac{1}{n} \quad \forall n \in \mathbb{N},$$

and

$$b \text{ is appreciable} \leftrightarrow \frac{1}{n} < |b| < m \quad \text{for some } n, m \in \mathbb{N}.$$

In particular, if $\varepsilon \neq 0$ is an infinitesimal, then

$$|\varepsilon| < \frac{1}{n} \qquad \forall n \in \mathbb{N}.$$

We denote the set of unlimited hypernaturals by ${}^*\mathbb{N}_\infty = {}^*\mathbb{N} - \mathbb{N}$. Similarly, we denote the sets of positive and negative unlimited hyperreals by

$$ {}^*\mathbb{R}_\infty^+ = \{x \in {}^*\mathbb{R} : x \text{ is positive unlimited}\} $$

and

$$ {}^*\mathbb{R}_\infty^- = \{x \in {}^*\mathbb{R} : x \text{ is negative unlimited}\}. $$

This notation may be adapted to an arbitrary subset $X \subseteq {}^*\mathbb{R}$. For example,

$$ X_\infty = \{x \in X : x \text{ is unlimited}\}, $$

$$ X^+ = \{x \in X : x > 0\}, \qquad X^- = \{x \in X : x < 0\}. $$

We will also use

$$ \mathcal{L} = \{x \in {}^*\mathbb{R} : x \text{ is limited}\} $$

for the set of all limited hyperreal numbers, and

$$ \mathcal{I} = \{x \in {}^*\mathbb{R} : x \text{ is infinitesimal}\} $$

for the set of all infinitesimals.

14.7 Arithmetic of Hyperreals

Let ε, δ be infinitesimals, let b, c be appreciable, and let H, K be unlimited hyperreal numbers. Then the following properties hold.

Sums: $\varepsilon + \delta$ is infinitesimal, $b + \varepsilon$ is appreciable, $b + c$ is limited (possibly infinitesimal), and $H + \varepsilon$ and $H + b$ are unlimited.

Opposites:

$$ -\varepsilon \text{ is infinitesimal}, \qquad -b \text{ is appreciable}, \qquad -H \text{ is unlimited}. $$

Products: $\varepsilon\delta$ and εb are infinitesimal, bc is appreciable, and bH and HK are unlimited.

Reciprocals: If $\varepsilon \neq 0$, then $1/\varepsilon$ is unlimited. If $b \neq 0$, then $1/b$ is appreciable. If $H \neq 0$, then $1/H$ is infinitesimal.

Quotients: If ε, δ are nonzero infinitesimals, then ε/δ may be infinitesimal, appreciable, or unlimited. If $b \neq 0$ is appreciable and ε is infinitesimal, then ε/b is infinitesimal. If $b, c \neq 0$ are appreciable, then b/c is appreciable. If H is unlimited and $b \neq 0$ is appreciable, then H/b is

unlimited. If $H \neq 0$ is unlimited and b is appreciable, then b/H is infinitesimal. If $H, K \neq 0$ are unlimited, then H/K may be infinitesimal, appreciable, or unlimited.

Roots: If $\varepsilon > 0$ is infinitesimal, then $\sqrt{\varepsilon}$ is infinitesimal. If $b > 0$ is appreciable, then $\sqrt{b}$ is appreciable. If $H > 0$ is unlimited, then $\sqrt{H}$ is unlimited.

Indeterminate Forms: The expressions

$$\frac{\varepsilon}{\delta}, \qquad \frac{H}{K}, \qquad \varepsilon H, \qquad H + K$$

do not have a determined classification from the information given alone.

For example, if $\varepsilon = 1/\omega$, then

$$\varepsilon\omega = 1,$$

which is appreciable. On the other hand, if $\varepsilon = 1/\sqrt{\omega}$, then

$$\varepsilon\omega = \sqrt{\omega},$$

which is unlimited. Thus arithmetic involving unlimited and infinitesimal numbers must be considered carefully.

In ${}^*\mathbb{R}$, the terminology is interpreted as follows:

- **Finite** means limited (bounded by a real number).
- **Infinite** means unlimited (larger in magnitude than every real number).

This distinction is relative to $\mathbb{R}$ inside ${}^*\mathbb{R}$: a number may be finite in this sense while still being nonstandard.

14.8 Halos, Galaxies, and Real Comparisons

Let $x, y \in {}^*\mathbb{R}$.
definition 14.8.1 (Infinitesimal Closeness)**.** *We say that x is* infinitely close *to y, and write*

$$x \approx y,$$

if $x - y$ is infinitesimal.
definition 14.8.2 (Halo)**.** *The* halo *of $x \in {}^*\mathbb{R}$ is the set*

$$\mathrm{hal}(x) = \{y \in {}^*\mathbb{R} : y \approx x\}.$$

definition 14.8.3 (Galaxy)**.** *The* galaxy *of $x \in {}^*\mathbb{R}$ is the set*

$$\mathrm{gal}(x) = \{y \in {}^*\mathbb{R} : x - y \text{ is limited}\}.$$

Lemma 14.8.1. *Every halo is contained in a galaxy.*

Proof. If $y \in \text{hal}(x)$, then $x - y$ is infinitesimal, hence limited. Therefore $y \in \text{gal}(x)$.

□

Lemma 14.8.2. *For standard $r, s \in \mathbb{R}$,*

$$r \approx s \quad \leftrightarrow \quad r = s.$$

Proof. If $r \approx s$, then $r - s$ is a real infinitesimal, hence $r - s = 0$. Therefore $r = s$.

□

14.9 Standard Parts

Theorem 14.9.1. *Every limited hyperreal b is infinitely close to exactly one real number. This real number is called the* *****standard part***** *of b and is denoted by $\text{st}(b)$. Thus*

$$b \approx \text{st}(b).$$

Proof. Let

$$A = \{r \in \mathbb{R} : r < b\}.$$

Since b is limited, there exist real numbers r, s such that $r < b < s$. Therefore A is nonempty and bounded above by s. By the completeness of $\mathbb{R}$, A has a least upper bound $c = \sup A \in \mathbb{R}$.

We show that $b \approx c$. Let $\epsilon > 0$ be any positive real number. Since c is an upper bound of A, we cannot have $c + \epsilon \in A$. Therefore $b \leq c + \epsilon$. On the other hand, $c - \epsilon$ cannot be an upper bound of A, since c is the least upper bound. Hence there exists some $a \in A$ such that $c - \epsilon < a < b$. Therefore $c - \epsilon < b$. Thus $c - \epsilon < b \leq c + \epsilon$, and consequently $|b - c| \leq \epsilon$. Since this holds for every positive real ϵ, we have $b \approx c$.

It remains to prove uniqueness. Suppose $b \approx c'$ for some $c' \in \mathbb{R}$. Since also $b \approx c$, we obtain $c \approx c'$. But $c, c' \in \mathbb{R}$, and the only real infinitesimal is 0. Hence $c = c'$. Therefore there exists exactly one real number infinitely close to b. We define this number to be $\text{st}(b) = c$.

□

Theorem 14.9.2. *Let b, c be limited hyperreal numbers and let $n \in \mathbb{N}$. Then*

$$\text{st}(b \pm c) = \text{st}(b) \pm \text{st}(c), \text{st}(bc) = \text{st}(b)\,\text{st}(c),$$

and

$$\text{st}\left(\frac{b}{c}\right) = \frac{\text{st}(b)}{\text{st}(c)} \qquad \textit{if } \text{st}(c) \neq 0.$$

Also,

$$\text{st}(b^n) = \text{st}(b)^n, \text{st}(|b|) = |\,\text{st}(b)|,$$

and, if $b \geq 0$,

$$\text{st}(\sqrt{b}) = \sqrt{\text{st}(b)}.$$

Furthermore,

$$b \leq c \quad \rightarrow \quad \mathrm{st}(b) \leq \mathrm{st}(c).$$

Proof. Exercise.

□

These properties show that the standard-part map

$$\mathrm{st} : \mathcal{L} \to \mathbb{R},$$

where $\mathcal{L}$ denotes the set of all limited hyperreal numbers, is an order-preserving ring homomorphism onto $\mathbb{R}$. Its kernel is precisely the set of infinitesimals:

$$\ker(\mathrm{st}) = \{b \in \mathcal{L} : \mathrm{st}(b) = 0\} = \mathcal{I}.$$

Thus two limited hyperreal numbers have the same standard part exactly when they are infinitely close:

$$\mathrm{st}(b) = \mathrm{st}(c) \leftrightarrow b \approx c.$$

Equivalently, the equivalence classes of the relation $\approx$ on the limited hyperreals are precisely the sets

$$\mathrm{hal}(b) = \{c \in \mathcal{L} : c \approx b\},$$

called the ****halos**** of b.

Therefore the quotient of the ring of limited hyperreals by the ideal of infinitesimals is isomorphic to the real number field.

Theorem 14.9.3. *The quotient ring $\mathcal{L}/\mathcal{I}$ is isomorphic to $\mathbb{R}$ by the correspondence* $\mathrm{hal}(b) \longmapsto \mathrm{st}(b)$. *In particular, $\mathcal{I}$ is a maximal ideal of the ring $\mathcal{L}$.*

This page intentionally left blank.

Chapter 15

New beginning for calculus

15.1 Limits

First we have to fix a "Main variable". It is almost all the time x.

(sometimes it is "t").
definition 15.1.1. *Let $f : A \to \mathbb{R}$ where $A \subseteq \mathbb{R}$, and let a be a limit point of A. We say that $L \in \mathbb{R}$ is the **Limit** of f as x approaches a if*

$$\forall \varepsilon > 0 \; \exists \delta > 0 \; \forall x \in A \, (0 < |x - a| < \delta \to |f(x) - L| < \varepsilon) \, .$$

We write

$$\lim_{x \to a} f(x) = L.$$

Theorem 15.1.1 (Infinitesimal Characterization of Limits)**.** *Let$f : A \to \mathbb{R}$, $A \subseteq \mathbb{R}$,and let a be a limit point of A.Then$\lim_{x\to a} f(x) = L$ if and only if, for every nonzero infinitesimal$\varepsilon \in \mathbb{R}$(Ord) such that $a + \varepsilon \in A$, we have ${}^*f(a + \varepsilon) \approx L$. Equivalently,*

$$\lim_{x \to a} f(x) = L \leftrightarrow \operatorname{st}({}^*f(a + \varepsilon)) = L.$$

Proof. Suppose first that $\lim_{x\to a} f(x) = L$. Let ε be a nonzero infinitesimal such that $a+\varepsilon \in A$. We prove that

$${}^*f(a + \varepsilon) \approx L.$$

Let $\varepsilon > 0$. By the definition of the limit, there exists $\delta > 0$ such that $0 < |x - a| < \delta$ implies $|f(x) - L| < \varepsilon$. Since ε is infinitesimal, $|\varepsilon| < \delta$. Therefore $0 < |(a+\varepsilon) - a| < \delta$.Hence$|{}^*f(a+\varepsilon) - L| < \varepsilon$.Since this holds for every $\varepsilon > 0$,${}^*f(a+\varepsilon) \approx L$. Conversely, suppose that for every nonzero infinitesimal ε such that $a + \varepsilon \in {}^*A$, we have ${}^*f(a + \varepsilon) \approx L$. Suppose, toward a contradiction, that $\lim_{x\to a} f(x) \neq L$. Then there exists $\varepsilon > 0$ such that for every $\delta > 0$ there exists $x \in A$ such that $0 < |x - a| < \delta$ and $|f(x) - L| \geq \varepsilon$.

In particular, for every $n \in \mathbb{N}$ there exists $x_n \in A$ such that$0 < |x_n - a| < \frac{1}{n}$ and $|f(x_n) - L| \geq \varepsilon$.By the nonstandard extension, taking an infinite index N, we obtain$0 < |x_N - a| < \frac{1}{N}$.Since N is infinite, $1/N$ is infinitesimal. Therefore $\varepsilon := x_N - a$ is a nonzero infinitesimal. But

$$|{}^*f(a+\varepsilon) - L| = |f(x_N) - L| \geq \varepsilon,$$

so ${}^*f(a+\varepsilon) \not\approx L$, contradicting the assumption. Therefore $\lim_{x\to a} f(x) = L$. Finally,${}^*f(a+\varepsilon) \approx L$is equivalent to $\operatorname{st}\big({}^*f(a+\varepsilon)\big) = L$.

□

Theorem 15.1.2 (Uniqueness of Limit)**.** *If* $\lim_{x\to a} f(x) = L$ *and* $\lim_{x\to a} f(x) = M$*, then* $L = M$*.*

Proof. Let ε be a nonzero infinitesimal such that $a+\varepsilon \in {}^*A$. By the Infinitesimal Characterization of Limits,$\operatorname{st}\big({}^*f(a+\varepsilon)\big) = Land\operatorname{st}\big({}^*f(a+\varepsilon)\big) = M$.Hence$L = M$.

□

Theorem 15.1.3 (Limit Laws)**.** *Suppose* $\lim_{x\to a} f(x) = L$ *and* $\lim_{x\to a} g(x) = M$*. Then* $\lim_{x\to a}(f(x) + g(x)) = L + M$*,* $\lim_{x\to a}(f(x) - g(x)) = L - M$*, and* $\lim_{x\to a} f(x)g(x) = LM$*. If* $M \neq 0$*, then* $\lim_{x\to a} \frac{f(x)}{g(x)} = \frac{L}{M}$*. Moreover, for every* $c \in \mathbb{R}$*,* $\lim_{x\to a} cf(x) = cL$*.*

Proof. Let ε be a nonzero infinitesimal such that $a+\varepsilon \in A$. By the Infinitesimal Characterization of Limits,

$${}^*f(a+\varepsilon) \approx L \qquad \text{and} \qquad {}^*g(a+\varepsilon) \approx M.$$

Therefore, by the Standard Part Theorem,

$$\operatorname{st}\big({}^*f(a+\varepsilon) + {}^*g(a+\varepsilon)\big) = L + M.$$

Similarly,

$$\operatorname{st}\big({}^*f(a+\varepsilon) - {}^*g(a+\varepsilon)\big) = L - M.$$

For the product,

$$\operatorname{st}\big({}^*f(a+\varepsilon){}^*g(a+\varepsilon)\big) = \operatorname{st}({}^*f(a+\varepsilon))\operatorname{st}({}^*g(a+\varepsilon)) = LM.$$

If $M \neq 0$, then $\operatorname{st}({}^*g(a+\varepsilon)) = M \neq 0$, and hence ${}^*g(a+\varepsilon) \neq 0$. Therefore,

$$\operatorname{st}\left(\frac{{}^*f(a+\varepsilon)}{{}^*g(a+\varepsilon)}\right) = \frac{\operatorname{st}({}^*f(a+\varepsilon))}{\operatorname{st}({}^*g(a+\varepsilon))} = \frac{L}{M}.$$

Finally, for every $c \in \mathbb{R}$,

$$\operatorname{st}\big(c^*f(a+\varepsilon)\big) = c\operatorname{st}({}^*f(a+\varepsilon)) = cL.$$

Thus all the stated limit laws hold.

□

Theorem 15.1.4 (Squeeze Theorem)**.** *Suppose that for all x sufficiently close to a, $f(x) \le g(x) \le h(x)$. If $\lim_{x\to a} f(x) = L$ and $\lim_{x\to a} h(x) = L$, then $\lim_{x\to a} g(x) = L$.*

Proof. Let ε be a nonzero infinitesimal such that $a+\varepsilon \in {}^*A$. Since the inequality holds for all x sufficiently close to a, by the Transfer Principle,${}^*f(a+\varepsilon) \le {}^*g(a+\varepsilon) \le {}^*h(a+\varepsilon)$.By the Infinitesimal Characterization of Limits,${}^*f(a+\varepsilon) \approx L$ and ${}^*h(a+\varepsilon) \approx L$.Hence, by the Squeeze Principle for infinitesimals,${}^*g(a+\varepsilon) \approx L$.Therefore, by the Infinitesimal Characterization of Limits,$\lim_{x\to a} g(x) = L$.

□

Theorem 15.1.5 (Absolute Value)**.** *If $\lim_{x\to a} f(x) = L$, then $\lim_{x\to a} |f(x)| = |L|$.*

Proof. Since ${}^*f(a+\varepsilon) \approx L$,

$$|\,|{}^*f(a+\varepsilon)| - |L|\,| \le |{}^*f(a+\varepsilon) - L|.$$

Therefore $|{}^*f(a+\varepsilon)| \approx |L|$. Hence, by the Infinitesimal Characterization of Limits,

$$\lim_{x\to a} |f(x)| = |L|.$$

□

Theorem 15.1.6 (Composition of Limits)**.** *Let $\lim_{x\to a} g(x) = b$ and $\lim_{y\to b} f(y) = L$. Suppose that $g(x) \ne b$ whenever x is sufficiently close to a and $x \ne a$. Then $\lim_{x\to a} f(g(x)) = L$.*

Proof. Let ε be a nonzero infinitesimal such that $a+\varepsilon \in {}^*A$. Since $\lim_{x\to a} g(x) = b$,

$${}^*g(a+\varepsilon) \approx b.$$

By the Transfer Principle and the assumption on g, ${}^*g(a+\varepsilon) \ne b$. Therefore, by the Infinitesimal Characterization of $\lim_{y\to b} f(y) = L$, ${}^*f({}^*g(a+\varepsilon)) \approx L$. Since ${}^*f({}^*g(a+\varepsilon)) = {}^*(f\circ g)(a+\varepsilon)$, we have

$${}^*(f\circ g)(a+\varepsilon) \approx L.$$

Hence, by the Infinitesimal Characterization of Limits,

$$\lim_{x\to a} f(g(x)) = L.$$

□

Theorem 15.1.7 (Polynomial Limit)**.** *Let $p(x) = a_0 + a_1x + \cdots + a_nx^n$. Then $\lim_{x\to a} p(x) = p(a)$.*

Proof. By the Limit Laws, $\lim_{x\to a} x^k = a^k$ for every $k \in \mathbb{N}$. Therefore,

$$\lim_{x\to a} p(x) = a_0 + a_1 a + \cdots + a_n a^n = p(a).$$

□

Theorem 15.1.8 (Rational Function Limit)**.** *Let $r(x) = \frac{p(x)}{q(x)}$ where p and q are polynomials. If $q(a) \neq 0$, then*

$$\lim_{x\to a} r(x) = \frac{p(a)}{q(a)}.$$

Proof. Let ε be a nonzero infinitesimal such that $a+\varepsilon \in {}^*A$. By the Polynomial Limit Theorem,

$$\operatorname{st}\big({}^*p(a+\varepsilon)\big) = p(a), \qquad \operatorname{st}\big({}^*q(a+\varepsilon)\big) = q(a) \neq 0.$$

Hence ${}^*q(a+\varepsilon) \neq 0$, and by the Standard Part Theorem,

$$\operatorname{st}\left(\frac{{}^*p(a+\varepsilon)}{{}^*q(a+\varepsilon)}\right) = \frac{p(a)}{q(a)}.$$

Therefore, by the Infinitesimal Characterization of Limits,

$$\lim_{x\to a} r(x) = \frac{p(a)}{q(a)}.$$

□

definition 15.1.2. *Let $f : A \to \mathbb{R}$ and let a be a limit point of A. We say that f has the* *****Right Limit***** *L at a if*

$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\,(a < x < a + \delta \to |f(x) - L| < \varepsilon)\,.$$

We write $\lim_{x\to a^+} f(x) = L$. Similarly, we say that f has the *****Left Limit***** *L at a if*

$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\,(a - \delta < x < a \to |f(x) - L| < \varepsilon)\,.$$

We write $\lim_{x\to a^-} f(x) = L$.

Theorem 15.1.9 (Infinitesimal Characterization of One-Sided Limits)**.** *We have $\lim_{x\to a^+} f(x) = L$ if and only if ${}^*f(a+\varepsilon) \approx L$ for every positive infinitesimal ε such that $a+\varepsilon \in {}^*A$. Similarly, $\lim_{x\to a^-} f(x) = L$ if and only if ${}^*f(a+\varepsilon) \approx L$ for every negative infinitesimal ε such that $a+\varepsilon \in {}^*A$.*

Theorem 15.1.10 (One-Sided Limits)**.** *The following are equivalent:*

$$\lim_{x\to a} f(x) = L \quad \leftrightarrow \quad \lim_{x\to a^-} f(x) = \lim_{x\to a^+} f(x) = L.$$

Proof. If $\lim_{x\to a} f(x) = L$, then every nonzero infinitesimal is either positive or negative. Therefore ${}^*f(a+\varepsilon) \approx L$ for both positive and negative infinitesimal ε. Hence $\lim_{x\to a^-} f(x) = L$ and $\lim_{x\to a^+} f(x) = L$.

Conversely, suppose $\lim_{x\to a^-} f(x) = L$ and $\lim_{x\to a^+} f(x) = L$. Every nonzero infinitesimal ε is either positive or negative, so ${}^*f(a+\varepsilon) \approx L$. Hence $\lim_{x\to a} f(x) = L$.

□

definition 15.1.3. *We say* $\lim_{x\to a} f(x) = +\infty$ *if*

$$\forall M > 0\ \exists \delta > 0\ \forall x \in A\,(0 < |x-a| < \delta \to f(x) > M)$$

. *Similarly,* $\lim_{x\to a} f(x) = -\infty$ *if*

$$\forall M > 0\ \exists \delta > 0\ \forall x \in A\,(0 < |x-a| < \delta \to f(x) < -M)$$

Theorem 15.1.11 (Infinitesimal Characterization of Infinite Limits)**.** *We have* $\lim_{x\to a} f(x) = +\infty$ *if and only if* ${}^*f(a+\varepsilon)$ *is positive infinite for every nonzero infinitesimal* ε *such that* $a + \varepsilon \in {}^*A$. *Similarly,* $\lim_{x\to a} f(x) = -\infty$ *if and only if* ${}^*f(a+\varepsilon)$ *is negative infinite for every such* ε.

definition 15.1.4. *Let* $f : A \to \mathbb{R}$ *where* $A \subseteq \mathbb{R}$ *is unbounded above. We say that* $\lim_{x\to\infty} f(x) = L$ *if* $\forall \epsilon > 0\ \exists M > 0\ \forall x \in A\,(x > M \to |f(x) - L| < \epsilon)$.

Theorem 15.1.12 (Infinitesimal Characterization of Limits at Infinity)**.** *We have* $\lim_{x\to\infty} f(x) = L$ *if and only if* ${}^*f(H) \approx L$ *for every positive infinite* $H \in {}^*\mathbb{R}$ *such that* $H \in {}^*A$. *Equivalently,*

$$\lim_{x\to\infty} f(x) = L \leftrightarrow \mathrm{st}({}^*f(H)) = L.$$

Proof. Suppose $\lim_{x\to\infty} f(x) = L$. Let H be positive infinite. If ${}^*f(H) \not\approx L$, then for some $\epsilon > 0$, $|{}^*f(H) - L| \geq \epsilon$. By the definition of the limit, there exists $M > 0$ such that $x > M$ implies $|f(x) - L| < \epsilon$. By Transfer, $H > M$ implies $|{}^*f(H) - L| < \epsilon$, a contradiction. Hence ${}^*f(H) \approx L$.

Conversely, suppose that ${}^*f(H) \approx L$ for every positive infinite $H \in {}^*A$. If $\lim_{x\to\infty} f(x) \neq L$, then for some $\epsilon > 0$, for every $M > 0$ there exists $x \in A$ such that $x > M$ and $|f(x) - L| \geq \epsilon$. Thus, for every $n \in \mathbb{N}$, choose $x_n \in A$ such that $x_n > n$ and $|f(x_n) - L| \geq \epsilon$. Let $H = [(x_n)]$. Then H is positive infinite, while by Transfer $|{}^*f(H) - L| \geq \epsilon$, so ${}^*f(H) \not\approx L$, a contradiction. Therefore $\lim_{x\to\infty} f(x) = L$.

□

definition 15.1.5. *We say that* $\lim_{x\to\infty} f(x) = +\infty$ *if*

$$\forall M > 0\ \exists N > 0\ \forall x \in A\,(x > N \to f(x) > M)$$

.

Similarly,$\lim_{x\to\infty} f(x) = -\infty$ *if*

$$\forall M > 0\ \exists N > 0\ \forall x \in A\,(x > N \to f(x) < -M)$$

.

Theorem 15.1.13 (Infinitesimal Characterization of Infinite Limits at Infinity)**.** *We have* $\lim_{x\to\infty} f(x) = +\infty$ *if and only if* ${}^*f(H)$ *is positive infinite for every positive infinite* $H \in {}^*A$. *Similarly,* $\lim_{x\to\infty} f(x) = -\infty$ *if and only if* ${}^*f(H)$ *is negative infinite for every positive infinite* $H \in {}^*A$.

Theorem 15.1.14 (Sequential Characterization of Limits)**.** *Let $f : A \to \mathbb{R}$ and let a be a limit point of A. Then $\lim_{x\to a} f(x) = L$ if and only if, for every sequence $(x_n) * n \in \mathbb{N}$ in A such that $x_n \neq a$ and $\lim * n \to \infty x_n = a$, we have $\lim_{n\to\infty} f(x_n) = L$.*

Theorem 15.1.15 (Limit of a Monotone Function)**.** *Suppose that f is monotone on an interval. Then the one-sided limits of f exist at every interior point of the interval, possibly taking the values $+\infty$ or $-\infty$.*

Theorem 15.1.16 (Limit of Powers)**.** *If $\lim_{x\to a} f(x) = L$, then for every $n \in \mathbb{N}$,*

$$\lim_{x\to a} f(x)^n = L^n.$$

Proof. This follows by repeated application of the Limit Laws.

□

The most important conceptual result of this section is the chain

$$\lim_{x\to a} f(x) = L \leftrightarrow {}^*f(a+\varepsilon) \approx L \leftrightarrow \mathrm{st}({}^*f(a+\varepsilon)) = L.$$

which will be used directly in the definitions of continuity, derivatives, and the Riemann integral.

Theorem 15.1.17 (Infinitesimal Characterization of Limits)**.** *Let $f : A \to \mathbb{R}$ and let a be a limit point of A. Then $\lim_{x\to a} f(x) = L$ if and only if, for every nonzero infinitesimal $\varepsilon \in {}^*\mathbb{R}$ such that $a + \varepsilon \in {}^*A$, we have ${}^*f(a+\varepsilon) \approx L$. Equivalently,*

$$\lim_{x\to a} f(x) = L \leftrightarrow \mathrm{st}\big({}^*f(a+\varepsilon)\big) = L.$$

Proof. Suppose first that $\lim_{x\to a} f(x) = L$. Let ε be a nonzero infinitesimal such that $a+\varepsilon \in {}^*A$. If ${}^*f(a+\varepsilon) \not\approx L$, then for some $\epsilon > 0$ we have $|{}^*f(a+\varepsilon) - L| \geq \epsilon$. By the definition of the limit, there exists $\delta > 0$ such that $0 < |x-a| < \delta$ implies $|f(x) - L| < \epsilon$. By Transfer, $0 < |\varepsilon| < \delta$ implies $|{}^*f(a+\varepsilon) - L| < \epsilon$, a contradiction. Hence ${}^*f(a+\varepsilon) \approx L$.

Conversely, suppose that ${}^*f(a+\varepsilon) \approx L$ for every nonzero infinitesimal ε such that $a+\varepsilon \in {}^*A$. Suppose that $\lim_{x\to a} f(x) \neq L$. Then for some $\epsilon > 0$, for every $n \in \mathbb{N}$ there exists $x_n \in A$ such that $0 < |x_n - a| < 1/n$ and $|f(x_n) - L| \geq \epsilon$. Let $H = [(x_n)]$. Then $H - a$ is a nonzero infinitesimal, but $|{}^*f(H) - L| \geq \epsilon$, so ${}^*f(H) \not\approx L$, a contradiction. Therefore $\lim_{x\to a} f(x) = L$.

Finally, ${}^*f(a+\varepsilon) \approx L$ if and only if $\mathrm{st}({}^*f(a+\varepsilon)) = L$.

□

definition 15.1.6. *Let $f : A \to \mathbb{R}$ where $A \subseteq \mathbb{R}$ is unbounded above. We say that $\lim_{x\to\infty} f(x) = L$ if*

$$\forall \epsilon > 0 \; \exists M > 0 \; \forall x \in A \, (x > M \to |f(x) - L| < \epsilon)$$

.

definition 15.1.7. *Let $x \in {}^*\mathbb{R}$. We say that x is ****Infinite**** if $|x| > n$ for every $n \in \mathbb{N}$. If $x > n$ for every $n \in \mathbb{N}$, we say that x is a ****Positive Infinite**** number.*

Theorem 15.1.18 (Infinitesimal Characterization of the Limit at Infinity)**.** *Let $f : A \to \mathbb{R}$ where $A \subseteq \mathbb{R}$ is unbounded above. Then $\lim_{x\to\infty} f(x) = L$ if and only if, for every positive infinite $H \in {}^*A$, we have ${}^*f(H) \approx L$. Equivalently,*

$$\lim_{x\to\infty} f(x) = L \leftrightarrow \mathrm{st}({}^*f(H)) = L.$$

Proof. Suppose first that $\lim_{x\to\infty} f(x) = L$. Let $H \in {}^*A$ be positive infinite. If ${}^*f(H) \not\approx L$, then for some $\epsilon > 0$, $|{}^*f(H) - L| \geq \epsilon$. By the definition of the limit, there exists $M > 0$ such that $x > M$ implies $|f(x) - L| < \epsilon$. By Transfer, $H > M$ implies $|{}^*f(H) - L| < \epsilon$, a contradiction. Hence ${}^*f(H) \approx L$.

Conversely, suppose that ${}^*f(H) \approx L$ for every positive infinite $H \in {}^*A$. If $\lim_{x\to\infty} f(x) \neq L$, then for some $\epsilon > 0$, for every $n \in \mathbb{N}$ there exists $x_n \in A$ such that $x_n > n$ and $|f(x_n) - L| \geq \epsilon$. Let $H = [(x_n)]$. Then H is positive infinite, while $|{}^*f(H) - L| \geq \epsilon$, so ${}^*f(H) \not\approx L$, a contradiction. Therefore $\lim_{x\to\infty} f(x) = L$.

□

15.2 Continuity

First we fix a "Main Point". It is almost all the time a.
definition 15.2.1. *Let $f : A \to \mathbb{R}$, where $A \subseteq \mathbb{R}$, and let $a \in A$. We say that f is* *****Continuous***** *at a if*

$$\forall \epsilon > 0\ \exists \delta > 0\ \forall x \in A\,(|x - a| < \delta \to |f(x) - f(a)| < \epsilon)$$

.
Theorem 15.2.1 (Continuity and Limits)**.** *Let $f : A \to \mathbb{R}$ and let $a \in A$ be a limit point of A. Then f is continuous at a if and only if $\lim_{x\to a} f(x) = f(a)$.*

Proof. Suppose first that f is continuous at a. Then, by definition, for every $\epsilon > 0$ there exists $\delta > 0$ such that $|x - a| < \delta$ implies $|f(x) - f(a)| < \epsilon$. In particular, this holds whenever $0 < |x - a| < \delta$. Hence $\lim_{x\to a} f(x) = f(a)$.

Conversely, suppose that $\lim_{x\to a} f(x) = f(a)$. Then for every $\epsilon > 0$ there exists $\delta > 0$ such that $0 < |x - a| < \delta$ implies $|f(x) - f(a)| < \epsilon$. If $x = a$, then $|f(x) - f(a)| = 0 < \epsilon$. Therefore $|x - a| < \delta$ implies $|f(x) - f(a)| < \epsilon$, and hence f is continuous at a.

□

Theorem 15.2.2 (Infinitesimal Characterization of Continuity)**.** *Let $f : A \to \mathbb{R}$ and let $a \in A$. Then f is continuous at a if and only if, for every infinitesimal $\varepsilon \in \mathbb{R}(\mathrm{Ord})$ such that $a + \varepsilon \in A$, ${}^*f(a + \varepsilon) \approx f(a)$.*

Equivalently,

$$f \text{ is continuous at } a \leftrightarrow \mathrm{st}\big({}^*f(a + \varepsilon)\big) = f(a)$$

for every infinitesimal ε such that $a + \varepsilon \in A$.

Proof. Suppose first that f is continuous at a. Let ε be an infinitesimal such that $a+\varepsilon \in A$. If ${}^*f(a+\varepsilon) \not\approx f(a)$, then there exists a positive real ϵ such that $|{}^*f(a+\varepsilon)-f(a)| \geq \epsilon$. By continuity, there exists $\delta > 0$ such that $|x-a| < \delta$ implies $|f(x)-f(a)| < \epsilon$. Since ε is infinitesimal, $|\varepsilon| < \delta$. By the Transfer Principle,

$|{}^*f(a+\varepsilon) - f(a)| < \epsilon$, a contradiction. Hence ${}^*f(a+\varepsilon) \approx f(a)$.

Conversely, suppose that ${}^*f(a+\varepsilon) \approx f(a)$ for every infinitesimal ε such that $a+\varepsilon \in A$. Suppose that f is not continuous at a. Then there exists $\epsilon > 0$ such that for every $\delta > 0$ there is $x \in A$ with $|x-a| < \delta$ and $|f(x)-f(a)| \geq \epsilon$.

For every $n \in \mathbb{N}$, choose $x_n \in A$ such that $|x_n - a| < 1/n$ and $|f(x_n)-f(a)| \geq \epsilon$. Taking an infinite index N gives $|x_N - a| < 1/N$, so $\varepsilon := x_N - a$ is infinitesimal. But $|{}^*f(a+\varepsilon)-f(a)| \geq \epsilon$, contradicting ${}^*f(a+\varepsilon) \approx f(a)$. Therefore f is continuous at a.

Finally, ${}^*f(a+\varepsilon) \approx f(a)$ is equivalent to $\mathrm{st}({}^*f(a+\varepsilon)) = f(a)$ by the Standard Part Theorem.

□

Theorem 15.2.3 (Algebra of Continuous Functions)**.** *Let f and g be continuous at a. Then $f+g$, $f-g$, and fg are continuous at a. If $g(a) \neq 0$, then f/g is continuous at a.*

Proof. For every infinitesimal ε, continuity gives ${}^*f(a+\varepsilon) \approx f(a)$ and ${}^*g(a+\varepsilon) \approx g(a)$. By the arithmetic of hyperreals,

${}^*(f+g)(a+\varepsilon) \approx f(a)+g(a)$, ${}^*(f-g)(a+\varepsilon) \approx f(a)-g(a)$, and ${}^*(fg)(a+\varepsilon) \approx f(a)g(a)$. If $g(a) \neq 0$, then ${}^*g(a+\varepsilon) \approx g(a) \neq 0$, and therefore ${}^*(f/g)(a+\varepsilon) \approx f(a)/g(a)$. Hence the stated functions are continuous at a.

□

Theorem 15.2.4 (Composition of Continuous Functions)**.** *Let $g : A \to \mathbb{R}$ be continuous at a, and let $f : B \to \mathbb{R}$ be continuous at $g(a)$, where $g(A) \subseteq B$. Then $f \circ g$ is continuous at a.*

Proof. Let ε be infinitesimal such that $a+\varepsilon \in A$. Since g is continuous at a, ${}^*g(a+\varepsilon) \approx g(a)$. Since f is continuous at $g(a)$, ${}^*f({}^*g(a+\varepsilon)) \approx f(g(a))$. By the composition property of nonstandard extensions,

${}^*(f \circ g)(a+\varepsilon) = {}^*f({}^*g(a+\varepsilon)) \approx f(g(a)) = (f \circ g)(a)$.

Therefore $f \circ g$ is continuous at a.

□

Theorem 15.2.5 (Continuous Functions Have Limits)**.** *Let $f : A \to \mathbb{R}$ be continuous at a, where a is a limit point of A. Then*

$$\lim_{x \to a} f(x) = f(a).$$

Proof. This follows immediately from the Continuity and Limits Theorem.

□

15.3 Derivatives

first we have to fix a "Main variable" it's almost all the time "x". (sometimes it's "t").
definition 15.3.1. *assume* $f : {}^*\mathbb{R}_{Ord} \to {}^*\mathbb{R}_{Ord}$ *we define* ***Difrential*** $(d_\varepsilon(f))$

$$d_\varepsilon(f) := f(x+\varepsilon) - f(x)$$

Main reason we choose "x" in definition because it's "Main variable". you can prove $\forall x \in {}^*\mathbb{R}_{Ord} \quad d_\varepsilon(x) = \varepsilon$.
definition 15.3.2. *we define* ***Derivative***

Let $f : \mathbb{R} \to \mathbb{R}$ *and let* $\varepsilon \in {}^*\mathbb{R}_{Ord}$ *be a nonzero infinitesimal:*

$$\varepsilon \neq 0, \qquad |\varepsilon| < \frac{1}{n} \quad \forall n \in \mathbb{N}.$$

Define

$$\frac{df}{dx} = f'(x) = \operatorname{st}\left(\frac{d_\varepsilon(f)}{d_\varepsilon(x)}\right)$$

Equivalently,

$$\frac{df}{dx} = \operatorname{st}\left(\frac{f(x+\varepsilon) - f(x)}{\varepsilon}.\right)$$

More generally,

$$\frac{d}{dx} := \{(f, st(\frac{d_\varepsilon(f)}{d_\varepsilon(x)})) \mid f : {}^*\mathbb{R}_{Ord} \to {}^*\mathbb{R}_{Ord}\}$$

there is a biggest set like A s.t. $\frac{d}{dx}A$ *is a function (you can prove it as an exercise). we call that set "Derivatible funsions" and that function "Derivative".*

15.4 Partial Derivatives

first we have to fix a "Main variable" it's almost all the time "x". (sometimes it's "t").
definition 15.4.1. *assume* $f : {}^*\mathbb{R}^n_{Ord} \to {}^*\mathbb{R}_{Ord}$ *we define* ***Partial Difrential*** $(d_\varepsilon(f))$

$$\partial_{\varepsilon,x_i}(f) := f(x_1, \ldots, x_i + \varepsilon, \ldots, x_n) - f(x_1, \ldots, x_i, \ldots, x_n)$$

Main reason we choose "x" in definition because it's "Main variable". you can prove $\forall x \in {}^*\mathbb{R}_{Ord} \quad \partial_{\varepsilon,x}(x) = \varepsilon$.
definition 15.4.2. *we define* ***Partial Derivative***

Letf $: \mathbb{R}^n \to \mathbb{R}, \qquad f = f(x_1, \ldots, x_n)$. *and let* $\varepsilon \in {}^*\mathbb{R}_{Ord}$ *be a nonzero infinitesimal:*

$$\varepsilon \neq 0, \qquad |\varepsilon| < \frac{1}{n} \quad \forall n \in \mathbb{N}.$$

define

$$\frac{\partial f}{\partial x_i} = \text{st}\left(\frac{\partial_{\varepsilon,x_i}(f)}{\partial_{\varepsilon,x_i}(x_i)}\right)$$

Equivalently,

$$\frac{\partial f}{\partial x_i} = \text{st}\left(\frac{f(x_1,\ldots,x_i+\varepsilon,\ldots,x_n)-f(x_1,\ldots,x_i,\ldots,x_n)}{\varepsilon}\right)$$

More generally,

$$\frac{\partial}{\partial x_i} := \{(f, st(\frac{\partial_{\varepsilon,x_i}(f)}{\partial_{\varepsilon,x_i}(x_i)})) \mid f : {}^*\mathbb{R}_{Ord} \to {}^*\mathbb{R}_{Ord}\}$$

15.5 Riemann Integral

definition 15.5.1. *Let $f : [a,b] \to \mathbb{R}$. A **Partition** of $[a,b]$ is a finite ordered set $P = \{x_0, x_1, \ldots, x_n\}$ such that*

$$a = x_0 < x_1 < \cdots < x_n = b.$$

*We define the **Mesh** of P by*

$$\|P\| := \max_{1\le i\le n}(x_i - x_{i-1}).$$

*For every partition P, define $M_i := \sup\{f(x) : x \in [x_{i-1}, x_i]\}$ and $m_i := \inf\{f(x) : x \in [x_{i-1}, x_i]\}$. The **Upper Darboux Sum** of f with respect to P is*

$$U(f,P) := \sum_{i=1}^{n} M_i(x_i - x_{i-1}),$$

*and the **Lower Darboux Sum** of f with respect to P is*

$$L(f,P) := \sum_{i=1}^{n} m_i(x_i - x_{i-1}).$$

*We say that f is **Riemann Integrable** on $[a,b]$ if f is bounded and*

$$\inf_P U(f,P) = \sup_P L(f,P).$$

In this case, we define

$$\int_a^b f(x)\,dx := \inf_P U(f,P) = \sup_P L(f,P).$$

Theorem 15.5.1 (Convergence of Riemann Sums)**.** *Suppose that $f : [a, b] \to \mathbb{R}$ is Riemann integrable. For each $n \in \mathbb{N}$, let*

$$x_i = a + \frac{i(b-a)}{n}, \qquad 0 \le i \le n.$$

Then

$$\lim_{n\to\infty} \sum_{i=0}^{n-1} f(x_i)\frac{b-a}{n} = \int_a^b f(x)\,dx.$$

Proof. Let $P_n = \{x_0, x_1, \ldots, x_n\}$. Since f is Riemann integrable,

$$\lim_{n\to\infty} U(f, P_n) = \lim_{n\to\infty} L(f, P_n) = \int_a^b f(x)\,dx,$$

because $\|P_n\| = (b-a)/n \to 0$.

Since $f(x_i)$ lies between the infimum and supremum of f on $[x_i, x_{i+1}]$,

$$L(f, P_n) \le \sum_{i=0}^{n-1} f(x_i)\frac{b-a}{n} \le U(f, P_n).$$

Therefore, by the Squeeze Theorem,

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{i=0}^{n-1} f(x_i)\frac{b-a}{n}$$

□

Since the Riemann sum converges to $\int_a^b f(x)\,dx$, by the Standard Part Theorem, for an infinite N and $\varepsilon = \frac{b-a}{N}$,

$$\int_a^b f(x)\,dx. = \operatorname{st}\left(\sum_{i=0}^{N-1} f(x_i)\varepsilon\right)$$

let $\varepsilon = \frac{b-a}{N}$ and $x_i = a + i\varepsilon$. we can prove for every choice of sample points $\xi_n(i) \in [x_i, x_{i+1}]$,

$$\int_a^b f(x)\,dx. = \operatorname{st}\left(\sum_{i=0}^{N-1} f(\xi_n(i))\varepsilon\right)$$

More generally,

$$\int_a^b := \left\{ \left(f, \operatorname{st}\left(\sum_{i=0}^{N-1} f(\xi_i)\varepsilon\right)\right) \middle| f \text{ is Riemann integrable on } [a, b] \right\}.$$

There is a biggest set A such that $\int_a^b A$ is a function (you can prove it as an exercise). We call that set "Riemann Integrable Functions" and that function "Riemann Integral".

definition 15.5.2 (More Generalisation for ξ)**.** *Let $\varepsilon = \frac{b-a}{N}$ and*

$$x_i = a + i\varepsilon, \qquad i \in \{0, \ldots, N-1\}.$$

Let K be an index set. For each $k \in K$, let $\xi_N^k : \{0, \ldots, N-1\} \to [a, b]$ be a choice of sample points, such that there is a permutation $\sigma : \{0, \ldots, N-1\} \to \{0, \ldots, N-1\}$ such that

$$\xi_N^k(i) \in [x_{\sigma(i)}, x_{\sigma(i)+1}].$$

then for every Riemann integrable function $f : [a, b] \to \mathbb{R}$,

$$\int_a^b f(x)\,dx = \operatorname{st}\left(\sum_{i=0}^{N-1} f(\xi_N^k(i))\varepsilon\right), \qquad k \in K.$$

Lemma 15.5.1 (Riemann Integrability and Sample Points)**.** *Let $f : [a, b] \to \mathbb{R}$ be bounded. Then f is Riemann integrable if and only if for every $k, l \in K$,*

$$\operatorname{st}\left(\sum_{i=0}^{N-1} f(\xi_N^k(i))\varepsilon\right) = \operatorname{st}\left(\sum_{i=0}^{N-1} f(\xi_N^l(i))\varepsilon\right).$$

Proof. Suppose first that f is Riemann integrable and let

$$I = \int_a^b f(x)\,dx.$$

For every $k \in K$, since σ is a permutation, the points $\xi_N^k(i)$ give one sample point in each interval of the partition $P_N = x_0, \ldots, x_N$. Hence, if

$$m_i = \inf_{x \in [x_i, x_{i+1}]} f(x), \qquad M_i = \sup_{x \in [x_i, x_{i+1}]} f(x),$$

then

$$\sum_{i=0}^{N-1} m_i\varepsilon \le \sum_{i=0}^{N-1} f(\xi_N^k(i))\varepsilon \le \sum_{i=0}^{N-1} M_i\varepsilon.$$

Taking standard parts and using the Riemann integrability of f, both the lower and upper sums have standard part I. Therefore

$$\operatorname{st}\left(\sum_{i=0}^{N-1} f(\xi_N^k(i))\varepsilon\right) = I$$

for every $k \in K$, and hence the required equality holds for every $k, l \in K$.

Conversely, suppose that

$$\operatorname{st}\left(\sum_{i=0}^{N-1} f(\xi_N^k(i))\varepsilon\right) = \operatorname{st}\left(\sum_{i=0}^{N-1} f(\xi_N^l(i))\varepsilon\right)$$

for every $k, l \in K$. Then the standard part of the Riemann sum is independent of the choice of sample points. Hence the upper and lower sums have the same standard part, so

$$\operatorname{st}(U(f, P_N)) = \operatorname{st}(L(f, P_N)).$$

Thus

$$\operatorname{st}\big(U(f, P_N) - L(f, P_N)\big) = 0,$$

and therefore f is Riemann integrable. □

This page intentionally left blank.

Chapter 16

Well-Ordered Integral(more powerful than Lebesgue Integral)

definition 16.0.1 (Admissible Sample Points)**.** *Let* $\varepsilon = \dfrac{b-a}{N}$ *and* $x_i = a + i\varepsilon$ *for* $i \in \{0, \dots, N-1\} \subseteq {}^*\mathbb{R}_{Ord}$*. For each* $k \in K$ *let*

$$\xi_N^k : \{0, \dots, N-1\} \to [a, b]$$

be a choice of sample points. We say ξ_N^k *is* ***admissible*** *if:*

1. *For every* $i \in \{0, \dots, N-1\}$ *that is an ordinal,* $\xi_N^k(i) \in \mathbb{R}$ *is a standard real number.*
2. *For every* $i, j \in \{0, \dots, N-1\}$ *such that* $\xi_N^k(i)$ *and* $\xi_N^k(j)$ *lie in the same halo, we have*

$$\xi_N^k(j) = \xi_N^k(i) \pm \varepsilon_N$$

   for some infinitesimal ε_N*.*

3. *For every* $i \in \{0, \dots, N-1\}$ *that is not an ordinal, there exists an ordinal* $j < i$*,* $j \in \{0, \dots, N-1\}$*, such that*

$$\xi_N^k(i) \approx \xi_N^k(j),$$

   where $\approx$ *denotes infinite closeness.*

Let $K_{\text{adm}} \subseteq K$ *denote the set of indices* k *for which* ξ_N^k *is admissible.*

definition 16.0.2 (Well-Ordered Integral)**.** *Let* $f : [a, b] \to \mathbb{R}$ *be bounded. We say* f *is* ***well-ordered integrable*** *on* $[a, b]$ *if for every* $k, l \in K_{\text{adm}}$*,*

$$\operatorname{st}\left(\sum_{i=0}^{N-1} f(\xi_N^k(i))\,\varepsilon\right) = \operatorname{st}\left(\sum_{i=0}^{N-1} f(\xi_N^l(i))\,\varepsilon\right).$$

In this case we define

$${}^{\circledast}\int_a^b f(x)\,dx := \operatorname{st}\left(\sum_{i=0}^{N-1} f(\xi_N^k(i))\,\varepsilon\right), \qquad k \in K_{\text{adm}}.$$

Remark 16.0.1. *Since $K_{\text{adm}} \subseteq K$, agreement over K_{adm} alone is a weaker requirement than agreement over all of K. Hence every Riemann integrable function is well-ordered integrable, and*

$$\circledast \int_a^b f(x)\,dx = \int_a^b f(x)\,dx$$

whenever f is Riemann integrable.

Chapter 17

New beginning for Probability and Measeure theory

17.1 Finite Approach

We begin with the simplest form of probability, where the sample space is a finite set. In this case, probability can be understood directly in terms of counting.

definition 17.1.1 (Finite Probability Space)**.** *Let S be a finite nonempty set. A* ***probability function*** *on S is a function*

$$P_S : \mathcal{P}(S) \to [0, 1]$$

such that $P_S(S) = 1$ and, for every pair of disjoint subsets $A, B \subseteq S$,

$$P_S(A \cup B) = P_S(A) + P_S(B).$$

The pair (S, P_S) is called a ***finite probability space****. The elements of S are called* ***outcomes****, and the subsets of S are called* ***events****.*

The simplest probability function is obtained when all outcomes are equally likely.

definition 17.1.2 (Uniform Probability)**.** *Let S be a finite nonempty set. The* ***uniform probability*** *on S is defined by*

$$P_S(A) = \frac{|A|}{|S|}$$

for every $A \subseteq S$.

Thus, the probability of an event is the ratio between the number of favorable outcomes and the total number of outcomes.

Example 17.1.1. *Consider tossing a fair coin. Let $S = \{H, T\}$. Then $P_S(H) = P_S(T) = \frac{1}{2}$. If $A = H$ is the event of obtaining heads, then*

$$P_S(A) = \frac{|A|}{|S|} = \frac{1}{2}.$$

The uniform probability satisfies the axioms of a probability function.

Theorem 17.1.1. *Let S be a finite nonempty set and define $P_S(A) = |A|/|S|$ for every $A \subseteq S$. Then P_S is a probability function.*

Proof. Clearly, $P_S(S) = |S|/|S| = 1$. Let $A, B \subseteq S$ be disjoint. Since A and B are disjoint finite sets, $|A \cup B| = |A| + |B|$. Therefore

$$P_S(A \cup B) = \frac{|A \cup B|}{|S|} = \frac{|A| + |B|}{|S|} = P_S(A) + P_S(B).$$

Hence P_S is a probability function. □

The finite case also gives the probability of a single outcome. If S has n elements and the probability is uniform, then every outcome has probability $1/n$. Consequently, for every event $A \subseteq S$,

$$P_S(A) = \sum_{s \in A} P_S(\{s\}) = \frac{|A|}{|S|}.$$

This suggests that probability can be viewed as a generalization of counting. Counting assigns to a finite set A its cardinality $|A|$, while probability assigns to an event A a normalized value between 0 and 1.

definition 17.1.3 (Weighted Finite Probability Space)**.** *Let S be a finite nonempty set. A function $p : S \to [0, 1]$ is called a* ***probability mass function*** *if*

$$\sum_{s \in S} p(s) = 1.$$

For every $A \subseteq S$, define

$$P_S(A) = \sum_{s \in A} p(s).$$

Then (S, P_S) is called a ***finite probability space****.*

The uniform probability space is the special case in which $p(s) = 1/|S|$ for every $s \in S$.

Example 17.1.2. *Suppose $S = \{1, 2, 3, 4, 5, 6\}$ is the set of possible outcomes of a die. For a fair die, $p(s) = 1/6$ for every $s \in S$. Hence, if $A = \{2, 4, 6\}$, then*

$$P_S(A) = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{1}{2}.$$

Thus the probability of obtaining an even number is $1/2$.

17.1.1 Expected Value in the Finite Approach

Probability allows us not only to assign probabilities to events, but also to describe the average value of a random outcome. This leads to the notion of expected value.

definition 17.1.4 (Random Variable)**.** *Let* (S, P_S) *be a finite probability space. A* *****random variable***** *is a function*

$$X : S \to \mathbb{R}.$$

definition 17.1.5 (Expected Value)**.** *Let* (S, P_S) *be a finite probability space and let* $X : S \to \mathbb{R}$ *be a random variable. The* *****expected value***** *of* X *is*

$$\mathbb{E}_{P_S}[X] = \sum_{s \in S} X(s) P_S(\{s\}).$$

Thus, the expected value is a weighted average of the possible values of X, where the weight of each outcome is its probability.

For a uniform finite probability space, we have

$$P_S(\{s\}) = \frac{1}{|S|},$$

and therefore

$$\mathbb{E}_{P_S}[X] = \frac{1}{|S|} \sum_{s \in S} X(s).$$

Example 17.1.3. *Consider a fair six-sided die. Let*

$$S = \{1, 2, 3, 4, 5, 6\}$$

and define $X(s) = s$*. Since every outcome has probability* $1/6$*,*

$$\mathbb{E}_{P_S}[X] = \frac{1}{6}(1 + 2 + 3 + 4 + 5 + 6) = \frac{21}{6} = \frac{7}{2}.$$

Thus the expected value of the result of a fair die is $7/2$.

Example 17.1.4. *Suppose a game has three possible outcomes with payoffs*

$$X(s_1) = 0, \qquad X(s_2) = 10, \qquad X(s_3) = 20,$$

and suppose

$$P_S(\{s_1\}) = \frac{1}{2}, \qquad P_S(\{s_2\}) = \frac{1}{3}, \qquad P_S(\{s_3\}) = \frac{1}{6}.$$

Then

$$\mathbb{E}_{P_S}[X] = 0 \cdot \frac{1}{2} + 10 \cdot \frac{1}{3} + 20 \cdot \frac{1}{6} = \frac{20}{3}.$$

Thus the expected payoff is $20/3$.

17.2 Continuous Approach

The finite approach is based on counting the elements of a finite set. For continuous spaces, however, counting is no longer an appropriate notion of size. Instead, we use a measure.

definition 17.2.1 (Measure)**.** *Let S be a set and let $\mathcal{F} \subseteq \mathcal{P}(S)$ be a σ-algebra. A* *****measure***** *on $(S, \mathcal{F})$ is a function*

$$\mu : \mathcal{F} \to [0, \infty]$$

such that $\mu(\varnothing) = 0$ and, for every sequence $(A_n)_{n\in\mathbb{N}}$ of pairwise disjoint sets in $\mathcal{F}$,

$$\mu\left(\bigcup_{n=0}^{\infty} A_n\right) = \sum_{n=0}^{\infty} \mu(A_n).$$

The most important measure on $\mathbb{R}$ for probability theory is the Lebesgue measure, denoted by λ. For an interval,

$$\lambda([a, b]) = b - a.$$

We can normalize the Lebesgue measure on a finite interval to obtain a probability measure.

definition 17.2.2 (Probability Measure)**.** *Let $(S, \mathcal{F})$ be a measurable space. A* *****probability measure***** *on $(S, \mathcal{F})$ is a measure P satisfying*

$$P(S) = 1.$$

The triple $(S, \mathcal{F}, P)$ is called a *****probability space*****.

In the finite case, the uniform probability of an event A is obtained by dividing its cardinality by the cardinality of the whole space. In a continuous space, cardinality is no longer an appropriate notion of size. Instead, we use a measure.

For example, consider the interval $[a, b]$. Its natural size is its length,

$$\mu([a, b]) = b - a.$$

Normalizing this measure gives the uniform probability on $[a, b]$.

definition 17.2.3 (Uniform Probability on an Interval)**.** *Let $a < b$ and let $\mathcal{F}$ be the collection of Lebesgue measurable subsets of $[a, b]$. Define*

$$P_S(A) = \frac{\lambda(A)}{b-a}$$

for every $A \in \mathcal{F}$, where $S = [a, b]$. We call P_S the *****uniform probability***** *on $[a, b]$.*

We first verify that this is indeed a probability measure.
Theorem 17.2.1. *Let $S = [a, b]$ with $a < b$. The function*

$$P_S(A) = \frac{\lambda(A)}{b-a}$$

is a probability measure on the measurable subsets of S.

Proof. Since $\lambda(\varnothing) = 0$,

$$P_S(\varnothing) = \frac{\lambda(\varnothing)}{b-a} = 0.$$

Also,

$$P_S(S) = \frac{\lambda([a,b])}{b-a} = \frac{b-a}{b-a} = 1.$$

Now let $(A_n)_{n\in\mathbb{N}}$ be a sequence of pairwise disjoint measurable subsets of S. Since Lebesgue measure is countably additive,

$$\lambda\left(\bigcup_{n=0}^{\infty} A_n\right) = \sum_{n=0}^{\infty} \lambda(A_n).$$

Therefore,

$$P_S\left(\bigcup_{n=0}^{\infty} A_n\right) = \frac{1}{b-a}\sum_{n=0}^{\infty} \lambda(A_n) = \sum_{n=0}^{\infty} \frac{\lambda(A_n)}{b-a} = \sum_{n=0}^{\infty} P_S(A_n).$$

Hence P_S is countably additive and satisfies $P_S(S) = 1$. Thus it is a probability measure. □

The formula

$$P_S(A) = \frac{\lambda(A)}{b-a}$$

is therefore the continuous analogue of the finite formula

$$P_S(A) = \frac{|A|}{|S|}.$$

In the finite case, cardinality measures the size of a set; in the continuous case, Lebesgue measure plays the same role.

Example 17.2.1. *Choose a real number uniformly from* $[0,1]$. *What is the probability that the chosen number belongs to* $[0, \frac{1}{4}]$*?*

Here $S = [0,1]$ *and* $A = [0, \frac{1}{4}]$. *Therefore*

$$P_S(A) = \frac{\lambda([0, \frac{1}{4}])}{\lambda([0,1])} = \frac{\frac{1}{4}}{1} = \frac{1}{4}.$$

Example 17.2.2. *Choose a real number uniformly from* $[0,10]$. *What is the probability that the chosen number belongs to* $[3,7]$*?*

Since

$$\lambda([3,7]) = 7 - 3 = 4$$

and

$$\lambda([0,10]) = 10,$$

we obtain

$$P_S([3,7]) = \frac{4}{10} = \frac{2}{5}.$$

Example 17.2.3. *Choose a real number uniformly from* $[0,1]$. *What is the probability that the chosen number is greater than* $\frac{1}{2}$*?*

Let

$$A = \left(\frac{1}{2}, 1\right].$$

Then

$$P_S(A) = \frac{\lambda((\frac{1}{2}, 1])}{\lambda([0,1])} = \frac{\frac{1}{2}}{1} = \frac{1}{2}.$$

Notice that whether the endpoint $\frac{1}{2}$ *is included does not affect the probability, since*

$$P_S(\{\tfrac{1}{2}\}) = 0.$$

Example 17.2.4. *Choose a real number uniformly from* $[0,1]$. *What is the probability that the chosen number is rational?*

Let

$$A = \mathbb{Q} \cap [0,1].$$

The set A *is countable and therefore has Lebesgue measure zero. Hence*

$$P_S(A) = \frac{\lambda(\mathbb{Q} \cap [0,1])}{\lambda([0,1])} = 0.$$

Thus a randomly chosen real number from $[0,1]$ *is rational with probability* 0.
Example 17.2.5. *Choose a real number uniformly from* $[0,1]$. *What is the probability that the chosen number belongs to*

$$A = [0, \tfrac{1}{3}] \cup [\tfrac{2}{3}, 1]?$$

The two intervals are disjoint except possibly at endpoints, which have measure zero. Therefore

$$\lambda(A) = \frac{1}{3} + \frac{1}{3} = \frac{2}{3}.$$

Consequently,

$$P_S(A) = \frac{2}{3}.$$

17.2.1 Expected Value in the Continuous Approach

The same idea extends to continuous probability spaces. Instead of summing over individual outcomes, we integrate with respect to the probability measure.
definition 17.2.4 (Expected Value in a Continuous Probability Space)**.** *Let* $(S, \mathcal{F}, P)$ *be a probability space and let* $X : S \to \mathbb{R}$ *be an integrable random variable. The* *****expected value***** *of* X *is*

$$\mathbb{E}_{P_S}[X] = \int_S X \, dP.$$

Theorem 17.2.2. *Let* $X : S \to \mathbb{R}$ *be integrable with respect to Lebesgue measure. Then*

$$\mathbb{E}_{P_S}[X] = \frac{1}{b-a} \int_a^b X(x) \, dx.$$

Proof. We first prove the result for nonnegative simple functions. Let

$$\phi = \sum_{i=1}^{n} c_i \chi_{A_i},$$

where the A_i are pairwise disjoint measurable subsets of S and $c_i \geq 0$. By the definition of integration of a simple function,

$$\int_S \phi \, dP_S = \sum_{i=1}^{n} c_i P_S(A_i).$$

Since

$$P_S(A_i) = \frac{\lambda(A_i)}{b-a},$$

we obtain

$$\int_S \phi \, dP_S = \frac{1}{b-a} \sum_{i=1}^{n} c_i \lambda(A_i) = \frac{1}{b-a} \int_S \phi \, d\lambda.$$

Now let $X : S \to [0, \infty]$ be measurable. There exists a sequence of nonnegative simple functions (ϕ_n) such that $\phi_n \uparrow X$. By the Monotone Convergence Theorem,

$$\int_S X \, dP_S = \lim_{n\to\infty} \int_S \phi_n \, dP_S.$$

Using the result for simple functions,

$$\begin{aligned} \int_S X \, dP_S &= \lim_{n\to\infty} \frac{1}{b-a} \int_S \phi_n \, d\lambda \\ &= \frac{1}{b-a} \int_S X \, d\lambda, \end{aligned}$$

where the last equality follows again from the Monotone Convergence Theorem.

Finally, let $X : S \to \mathbb{R}$ be integrable. Write

$$X = X^+ - X^-,$$

where

$$X^+ = \max\{X, 0\}, \qquad X^- = \max\{-X, 0\}.$$

Since X is integrable, both X^+ and X^- are integrable. Applying the result above to each of them gives

$$\begin{aligned}\int_S X\,dP_S &= \int_S X^+\,dP_S - \int_S X^-\,dP_S \\ &= \frac{1}{b-a}\left(\int_S X^+\,d\lambda - \int_S X^-\,d\lambda\right) \\ &= \frac{1}{b-a}\int_S X\,d\lambda.\end{aligned}$$

Since $S = [a, b]$,

$$\int_S X\,d\lambda = \int_a^b X(x)\,dx.$$

Therefore,

$$\boxed{\mathbb{E}_{P_S}[X] = \int_S X\,dP_S = \frac{1}{b-a}\int_a^b X(x)\,dx}.$$

□

Example 17.2.6. *Let $S = [0, 1]$ and define the random variable*

$$X : S \to \mathbb{R}, \qquad X(x) = x.$$

Then, by the previous proposition,

$$\mathbb{E}_{P_S}[X] = \int_0^1 x\,dx = \frac{1}{2}.$$

Thus the expected value of the coordinate function $X(x) = x$ on $[0, 1]$ is $1/2$.
Example 17.2.7. *Let $S = [0, 10]$ and define*

$$X : S \to \mathbb{R}, \qquad X(x) = x.$$

By the previous proposition,

$$\mathbb{E}_{P_S}[X] = \frac{1}{10}\int_0^{10} x\,dx = \frac{1}{10}\cdot\frac{10^2}{2} = 5.$$

Thus the expected value of the coordinate function on $[0, 10]$ is the midpoint of the interval.
Example 17.2.8. *Let $S = [0, 1]$ and define*

$$X : S \to \mathbb{R}, \qquad X(x) = x^2.$$

Then

$$\mathbb{E}_{P_S}[X] = \int_0^1 x^2\,dx = \frac{1}{3}.$$

On the other hand, if $Y : S \to \mathbb{R}$ is defined by $Y(x) = x$, then

$$\mathbb{E}_{P_S}[Y] = \frac{1}{2}.$$

Consequently,

$$\mathbb{E}_{P_S}[X] = \frac{1}{3} \neq \frac{1}{4} = \left(\mathbb{E}_{P_S}[Y]\right)^2.$$

This illustrates that, in general,

$$\mathbb{E}[X^2] \neq (\mathbb{E}[X])^2.$$

Example 17.2.9. *Let $S = [0, 1]$ and define*

$$X : S \to \mathbb{R}, \qquad X(x) = x(1 - x).$$

Then

$$\begin{aligned}\mathbb{E}_{P_S}[X] &= \int_0^1 x(1-x)\,dx \\ &= \int_0^1 (x - x^2)\,dx \\ &= \frac{1}{2} - \frac{1}{3} = \frac{1}{6}.\end{aligned}$$

Example 17.2.10. *Let $S = [0, 1]$ and let*

$$A = \left[0, \frac{1}{2}\right].$$

Define the indicator random variable

$$X : S \to \mathbb{R}, \qquad X(x) = \chi_A(x).$$

Thus,

$$X(x) = \begin{cases} 1, & 0 \leq x \leq \frac{1}{2}, \\ 0, & \frac{1}{2} < x \leq 1. \end{cases}$$

By the previous proposition,

$$\mathbb{E}_{P_S}[X] = \int_0^1 \chi_A(x)\,dx = \lambda(A) = \frac{1}{2}.$$

This agrees with the general identity

$$\mathbb{E}_{P_S}[\chi_A] = P_S(A).$$

Example 17.2.11. *Let $S = [0,1]$ and define the Dirichlet function*

$$X : S \to \mathbb{R}, \qquad X(x) = \begin{cases} 1, & x \in \mathbb{Q}, \\ 0, & x \notin \mathbb{Q}. \end{cases}$$

The function X is measurable because $\mathbb{Q} \cap [0,1]$ is countable and therefore Lebesgue measurable. Moreover,

$$\lambda(\mathbb{Q} \cap [0,1]) = 0.$$

Hence, using the previous proposition,

$$\mathbb{E}_{P_S}[X] = \int_0^1 X(x)\,dx = \lambda(\mathbb{Q} \cap [0,1]) = 0.$$

Notice that $X(x) = 1$ for every rational x, but the expected value is still 0, because the rationals have Lebesgue measure zero.

Example 17.2.12. *Let $S = [0,1]$ and define*

$$X : S \to \mathbb{R}, \qquad X(x) = \begin{cases} x, & x \in \mathbb{Q}, \\ x^2, & x \notin \mathbb{Q}. \end{cases}$$

Since $\mathbb{Q} \cap [0,1]$ has Lebesgue measure zero, the functions X and x^2 are equal almost everywhere. Therefore,

$$\mathbb{E}_{P_S}[X] = \int_0^1 X(x)\,dx = \int_0^1 x^2\,dx = \frac{1}{3}.$$

Thus changing a function on a set of probability zero does not change its expectation.

Example 17.2.13. *Let $S = [0,1]$ and define*

$$X : S \to \mathbb{R}, \qquad X(x) = \begin{cases} 1, & x \in [0, \frac{1}{4}], \\ 2, & x \in (\frac{1}{4}, \frac{3}{4}], \\ 4, & x \in (\frac{3}{4}, 1]. \end{cases}$$

Then

$$\begin{aligned}\mathbb{E}_{P_S}[X] &= \int_0^1 X(x)\,dx \\ &= 1 \cdot \frac{1}{4} + 2 \cdot \frac{1}{2} + 4 \cdot \frac{1}{4} \\ &= \frac{1}{4} + 1 + 1 = \frac{9}{4}.\end{aligned}$$

This gives the continuous analogue of the usual weighted-average formula for a discrete random variable.

Example 17.2.14. *Let $S = [0, 1]$ and define*

$$X : S \to \mathbb{R}, \qquad X(x) = x^2.$$

Then X is integrable on $[0, 1]$. By the previous proposition,

$$\mathbb{E}_{P_S}[X] = \int_0^1 x^2\,dx = \frac{1}{3}.$$

Example 17.2.15. *Let $S = [-1, 1]$ and define*

$$X : S \to \mathbb{R}, \qquad X(x) = x^3.$$

Then

$$\mathbb{E}_{P_S}[X] = \frac{1}{2}\int_{-1}^1 x^3\,dx = 0,$$

because x^3 is an odd function.

Thus a random variable can take both positive and negative values while having expected value 0.

Example 17.2.16. *Let $S = [0, 1]$ and define*

$$X : S \to \mathbb{R}, \qquad X(x) = x(1 - x).$$

Then

$$\begin{aligned}\mathbb{E}_{P_S}[X] &= \int_0^1 x(1-x)\,dx \\ &= \int_0^1 (x - x^2)\,dx \\ &= \frac{1}{2} - \frac{1}{3} = \frac{1}{6}.\end{aligned}$$

Example 17.2.17. *Let $S = [0, 2]$ and define*

$$X : S \to \mathbb{R}, \qquad X(x) = \begin{cases} x, & 0 \le x \le 1, \\ 2 - x, & 1 < x \le 2. \end{cases}$$

Then

$$\begin{aligned} \mathbb{E}_{P_S}[X] &= \frac{1}{2}\left(\int_0^1 x\,dx + \int_1^2 (2-x)\,dx\right) \\ &= \frac{1}{2}\left(\frac{1}{2} + \frac{1}{2}\right) \\ &= \frac{1}{2}. \end{aligned}$$

Example 17.2.18. *Let $S = [0, 1]$ and let*

$$A = \left[\frac{1}{4}, \frac{3}{4}\right].$$

Define

$$X : S \to \mathbb{R}, \qquad X(x) = \chi_A(x).$$

Thus,

$$X(x) = \begin{cases} 1, & \frac{1}{4} \le x \le \frac{3}{4}, \\ 0, & \textit{otherwise.} \end{cases}$$

Then

$$\begin{aligned} \mathbb{E}_{P_S}[X] &= \int_0^1 \chi_A(x)\,dx \\ &= \int_{1/4}^{3/4} 1\,dx \\ &= \frac{1}{2}. \end{aligned}$$

In particular,

$$\mathbb{E}_{P_S}[\chi_A] = P_S(A).$$

Example 17.2.19. *Let $S = [0, 1]$ and define*

$$X : S \to \mathbb{R}, \qquad X(x) = \begin{cases} 1, & 0 \le x < \frac{1}{2}, \\ 3, & \frac{1}{2} \le x \le 1. \end{cases}$$

Then

$$\begin{aligned}\mathbb{E}_{P_S}[X] &= \int_0^{1/2} 1\,dx + \int_{1/2}^1 3\,dx \\ &= \frac{1}{2} + \frac{3}{2} \\ &= 2.\end{aligned}$$

Thus the expected value is the average of the two values 1 *and* 3*, weighted according to the lengths of the regions on which they occur.*

Example 17.2.20. *Let* $S = [0,1]$ *and define*

$$X : S \to \mathbb{R}, \qquad X(x) = \begin{cases} x, & 0 \le x \le \frac{1}{2}, \\ x^2, & \frac{1}{2} < x \le 1. \end{cases}$$

Then

$$\begin{aligned}\mathbb{E}_{P_S}[X] &= \int_0^{1/2} x\,dx + \int_{1/2}^1 x^2\,dx \\ &= \frac{1}{8} + \left[\frac{x^3}{3}\right]_{1/2}^1 \\ &= \frac{1}{8} + \frac{1}{3} - \frac{1}{24} \\ &= \frac{5}{12}.\end{aligned}$$

Example 17.2.21. *Let* $S = [0,1]$ *and define*

$$X : S \to \mathbb{R}, \qquad X(x) = \frac{1}{1+x}.$$

Since X *is continuous on* $[0,1]$*, it is integrable. By the previous proposition,*

$$\begin{aligned}\mathbb{E}_{P_S}[X] &= \int_0^1 \frac{1}{1+x}\,dx \\ &= [\ln(1+x)]_0^1 \\ &= \ln 2.\end{aligned}$$

Example 17.2.22. *Let* $S = [0,1]$ *and define*

$$X : S \to \mathbb{R}, \qquad X(x) = e^x.$$

Then

$$\begin{aligned}\mathbb{E}_{P_S}[X] &= \int_0^1 e^x \, dx \\ &= [e^x]_0^1 \\ &= e - 1.\end{aligned}$$

We can therefore see the common structure between the finite and continuous cases. In the finite case, expected value is obtained by a weighted sum,

$$\mathbb{E}_{P_S}[X] = \sum_{s \in S} X(s) P_S(\{s\}),$$

while in the continuous case it is obtained by integration,

$$\mathbb{E}_{P_S}[X] = \int_S X \, dP.$$

For a uniform finite space, this is an ordinary average,

$$\mathbb{E}_{P_S}[X] = \frac{1}{|S|} \sum_{s \in S} X(s),$$

whereas for a uniform interval it is an average integral,

$$\mathbb{E}_{P_S}[X] = \frac{1}{b-a} \int_a^b X(x) \, dx.$$

This gives another important connection:

$$\text{finite average} \longrightarrow \text{integral average}.$$

Later, the nonstandard approach will allow these averages to be represented using hyperfinite sums, providing a direct connection between finite, continuous, and nonstandard probability.

17.3 Nonstandard Approach

The finite probability space provides a natural starting point for a nonstandard construction of probability. Recall that if S is finite and the probability is uniform, then for every event $A \subseteq S$ we have $P_S(A) = |A|/|S|$. The idea is to keep this same definition when the size of the sample space is an infinite hyperinteger.

Let ${}^*\mathbb{N}$ denote the hypernatural numbers. Since ${}^*\mathbb{N}$ contains infinite elements, we may choose

$$H \in {}^*\mathbb{N}_\infty$$

We call such an H an **infinite hypernatural**.

Consider the hyperfinite set

$$^*\mathbb{N}_H = \{0, 1, 2, \ldots, H-1\}.$$

Although $^*\mathbb{N}_H$ is infinite from the standard point of view, it behaves internally like a finite set with H elements. In particular, every internal subset $A \subseteq {}^*\mathbb{N}_H$ has an internal cardinality $|A| \in {}^*\mathbb{N}$.

This allows us to extend the finite formula for uniform probability without changing its form.
definition 17.3.1 (Hyperfinite Probability)**.** *Let $H \in {}^*\mathbb{N}_\infty$ and let $^*\mathbb{N}_H = 1, \ldots, H$. For every internal set $A \subseteq {}^*\mathbb{N}_H$, define its* ***hyperfinite probability*** *by*

$$P_H(A) = \frac{|A_H|}{H}.$$

This is exactly the same formula as in the finite case. The only difference is that H and $|A|$ may now be infinite hyperintegers. Consequently, $P_H(A)$ is generally a hyperreal number rather than a standard real number.

For example, consider the set of even elements of $^*\mathbb{N}_H$,

$$A = \{0, 2, 4, \ldots, H-2\},$$

assuming H is even. Then $|A| = H/2$, and therefore $P_H(A) = 1/2$. More generally, if A contains approximately half of the elements of $^*\mathbb{N}_H$, its hyperfinite probability is approximately $1/2$.

The finite additivity of probability follows immediately from the corresponding property of finite cardinality.
Theorem 17.3.1. *Let $A, B \subseteq {}^*\mathbb{N}_H$ be internal and disjoint. Then*

$$P_H(A \cup B) = P_H(A) + P_H(B).$$

Moreover, $P_H(^*\mathbb{N}_H) = 1$.

Proof. Since A and B are internal and disjoint, the transfer principle gives $|A \cup B| = |A| + |B|$. Hence

$$P_H(A \cup B) = \frac{|A \cup B|}{H} = \frac{|A| + |B|}{H} = P_H(A) + P_H(B).$$

Also, $|^*\mathbb{N}_H| = H$, so $P_H(^*\mathbb{N}_H) = H/H = 1$. □

There is, however, an important difference between the finite and hyperfinite cases. In the finite case, $P_S(A)$ is always a standard real number. In the hyperfinite case, $P_H(A)$ may be infinitesimally close to a real number without being equal to it. For example, if $A \subseteq {}^*\mathbb{N}_H$ satisfies $|A| = H - 1$, then

$$P_H(A) = 1 - \frac{1}{H}.$$

Since H is infinite, $1/H$ is infinitesimal. Thus $P_H(A) \approx 1$.

This suggests that the standard probability associated with a hyperfinite event should be obtained by taking its standard part.

definition 17.3.2 (Standard Part Probability)**.** *Let $A \subseteq {}^*\mathbb{N}_H$ be internal. If $P_H(A)$ is finite, define*

$$P(A) = \mathrm{st}\big(P_H({}^*A_H)\big).$$

For the previous example, $P_H(A) = 1 - 1/H \approx 1$, and therefore $P(A) = 1$.

The important point is that the nonstandard approach does not replace counting by a completely different construction. Instead, it extends finite counting to a hyperfinite set and then uses the standard-part map to return to ordinary real probabilities.

Thus the basic construction is

$$\text{finite counting} \longrightarrow \text{hyperfinite counting} \longrightarrow \text{standard probability}.$$

17.4 Countable Probability

We now want to extend the finite construction to a countable sample space. The first example is the set of natural numbers. Let $S = \mathbb{N}$. One might try to define the probability of an event $A \subseteq \mathbb{N}$ by the same formula as in the finite case, namely $P_S(A) = |A|/|S|$. However, this expression is not well-defined in the usual theory, since $\mathbb{N}$ has no finite cardinality and there is no uniform probability on $\mathbb{N}$ assigning the same positive probability to every element.

A natural way to approximate $\mathbb{N}$ by finite sets is to consider its initial segments. For every $n \in \mathbb{N}$, let $n = \{0, 1, \ldots, n-1\}$. For an event $A \subseteq \mathbb{N}$, define $A_n = A \cap n = \{a \in A : a < n\}$. Thus, A_n contains the elements of A that occur among the first n natural numbers. Since n is finite, its uniform probability of A_n is

$$P_n(A_n) = \frac{|A_n|}{n}.$$

This gives a natural finite approximation to the probability of A. If the limit exists, we define the probability of A by

$$P_{\mathbb{N}}(A) = \lim_{n\to\infty} \frac{|A_n|}{n}.$$

This quantity is usually called the **natural density** of A.

For example, let A be the set of even natural numbers. Then A_n contains approximately half of the elements of n, and therefore

$$\lim_{n\to\infty} \frac{|A_n|}{n} = \frac{1}{2}.$$

Thus the natural density of the even numbers is 1/2. Similarly, the set of odd numbers has probability 1/2.

Now consider the nonstandard construction. Let $H \in {}^*\mathbb{N} - \mathbb{N}$ be an infinite hypernatural and let

$${}^*\mathbb{N}_H = \{0, 1, \ldots, H-1\}.$$

For an internal set $A \subseteq {}^*\mathbb{N}_H$, its hyperfinite probability is

$$P_H(A) = \frac{|{}^*A_H|}{H}.$$

The finite approximations $A_n = a \in A : a < n$ suggest that the standard value of this hyperfinite probability should correspond to the limiting density of the finite approximations. In fact, whenever the limit exists,

$$\lim_{n\to\infty} \frac{|A_n|}{n} = \mathrm{st}\left(\frac{|{}^*A_H|}{H}\right).$$

Thus the nonstandard formulation replaces the limiting process by a single hyperfinite computation. Instead of letting n tend to infinity, we choose one infinite H and count the elements of A inside the hyperfinite set S_H. Taking the standard part then gives the same value as the classical limit.

This also explains why the nonstandard approach is useful: the expression $|A|/H$ is defined before taking the standard part, while the standard expression $|A|/|\mathbb{N}|$ has no meaning as a uniform probability.

We now extend this idea from $\mathbb{N}$ to an arbitrary countable set.
definition 17.4.1 (Countable Probability)**.** *Let S be a countably infinite set equipped with a fixed well-ordering*

$$S = \{s_0, s_1, s_2, \ldots\}.$$

For $A \subseteq S$, define

$$A_n = \{s_k \in A : k < n\}.$$

If the limit exists, the ***countable density*** *of A is*

$$P_S(A) = \lim_{n\to\infty} \frac{|A_n|}{n}.$$

The definition depends, in general, on the chosen well-ordering of S. Therefore, a countable set by itself does not determine a unique uniform probability. The ordering specifies how the finite approximations n approach the whole set.

In the nonstandard formulation, the same ordering gives the hyperfinite sample space

$$\{{}^*s_i\}_{i<H}, \qquad H \in {}^*\mathbb{N} - \mathbb{N}.$$

For an internal event ${}^*A_H \subseteq \{{}^*s_i\}_{i<H}$, we define

$$P_H({}^*A_H) = \frac{|{}^*A_H|}{H}.$$

Whenever the corresponding standard density exists, we again have

$$P_S(A) = \operatorname{st}\left(P_H({}^*A_H)\right) = \operatorname{st}\left(\frac{|{}^*A_H|}{H}\right).$$

Thus, after fixing a well-ordering, the passage from a finite probability space to a countable one can be summarized as

$$\text{finite counting} \longrightarrow \lim_{n\to\infty} \frac{|A_n|}{n} \longleftrightarrow \operatorname{st}\left(\frac{|{}^*A_H|}{H}\right).$$

17.5 Decision Theory

Decision theory is the mathematical study of the choice of an action when the consequences of that action depend on an uncertain state of the world.

A decision problem can be formally described by a tuple

$$\mathcal{D} = (S, A, u, P),$$

where S, A are sets, u is a function, and P is a probability measure. We describe each of these objects formally.

17.5.1 The State Space

Let S be a nonempty set. The elements of S are called states of nature. A state $s \in S$ represents a possible condition of the world relevant to the decision.

For example, suppose that we must decide whether to take an umbrella. We may have

$$S = \{\text{Rain}, \text{NoRain}\}.$$

Thus the actual state of the world is an element of S, although the decision-maker may not know which element it is.

17.5.2 The Action Space

Let A be a nonempty set. The elements of A are called actions or decisions. An element $a \in A$ represents an action available to the decision-maker.

For example,

$$A = \{\text{Take}, \text{Leave}\},$$

where Take means taking an umbrella and Leave means leaving it at home.

17.5.3 The Utility Function

Let

$$u : A \times S \longrightarrow \mathbb{R}.$$

Thus u is a set-theoretic function whose domain is the Cartesian product $A \times S = \{(a, s) \mid a \in A,\ s \in S\}$ and whose codomain is $\mathbb{R}$.

For every pair $(a, s) \in A \times S$, there is exactly one real number $u(a, s) \in \mathbb{R}$. The number $u(a, s)$ represents the utility of choosing action a when the state of nature is s.

For example, let $S = \{\text{Rain}, \text{NoRain}\}$ and $A = \{\text{Take}, \text{Leave}\}$.

Suppose that the utility function is given by

$u(a, s)$	Rain	NoRain
Take	10	8
Leave	0	10

This means

$u(\text{Take}, \text{Rain}) = 10,$

$u(\text{Take}, \text{NoRain}) = 8,$

$u(\text{Leave}, \text{Rain}) = 0,$

and

$u(\text{Leave}, \text{NoRain}) = 10.$

The function u therefore completely describes the utility associated with every possible combination of an action and a state.

17.5.4 Probability on the State Space

Suppose that uncertainty about the state of nature is represented by a probability measure $P : \mathcal{F} \longrightarrow [0, 1]$, where $\mathcal{F} \subseteq \mathcal{P}(S)$ is a σ-algebra on S, and $P(S) = 1$. The pair $(S, \mathcal{F}, P)$

is called a probability space.

For a finite state space, we may simply assign a probability $P(\{s\})$ to every $s \in S$, satisfying

$$\sum_{s \in S} P(\{s\}) = 1.$$

In the umbrella example, suppose $P(\{\text{Rain}\}) = 0.3$ and $P(\{\text{NoRain}\}) = 0.7$

17.5.5 Expected Utility

For every action $a \in A$, its expected utility is defined by $EU(a) = \mathbb{E}_{P_S}[u_a]$.

Here S in $u(a, S)$ denotes the random state of nature. More formally, let $u_a : S \longrightarrow \mathbb{R}$ be the function defined by $u_a(s) = u(a, s)$. Then $EU(a) = \mathbb{E}_{P_S}[u_a]$. Equivalently,

$$\boxed{EU(a) = \mathbb{E}_{P_S}[u_a]}$$

where the expectation is taken with respect to the probability measure P.

The expected utility of an action $a \in A$ is then defined by

$$EU(a) = \mathbb{E}_P[u_a] = \int_S u(a, s)\, dP(s),$$

provided that u_a is integrable.

Equivalently, $EU : A \longrightarrow \mathbb{R}$ is the function defined by

$$EU(a) = \int_S u(a, s)\, dP(s).$$

When S is finite, the expectation becomes

$$EU(a) = \sum_{s \in S} P(\{s\})u(a, s).$$

For the action Take, we have

$$\begin{aligned} EU(\text{Take}) &= P(\{\text{Rain}\})u(\text{Take}, \text{Rain}) + P(\{\text{NoRain}\})u(\text{Take}, \text{NoRain}) \\ &= 0.3(10) + 0.7(8) \\ &= 8.6 \end{aligned}$$

Similarly,

$$\begin{aligned} EU(\text{Leave}) &= P(\{\text{Rain}\})u(\text{Leave}, \text{Rain}) + P(\{\text{NoRain}\})u(\text{Leave}, \text{NoRain}) \\ &= 0.3(0) + 0.7(10) \\ &= 7 \end{aligned}$$

Thus $EU(\text{Take}) > EU(\text{Leave})$.

17.5.6 The Maximizing Action

The set of actions with maximal expected utility is denoted by $\arg\max_{a \in A} EU(a)$. If the maximum exists, this can equivalently be written as

$$\arg\max_{a \in A} EU(a) = \left\{ a \in A \mid EU(a) = \max_{b \in A} EU(b) \right\}.$$

Notice the distinction between $\max_{a \in A} EU(a)$ and $\arg\max_{a \in A} EU(a)$. The first is a real number, namely the largest expected utility. The second is a subset of A, namely the set of actions that attain that largest value.

In the umbrella example, $EU(\text{Take}) = 8.6$ and $EU(\text{Leave}) = 7$. Therefore, $\max_{a \in A} EU(a) = 8.6$ while $\arg\max_{a \in A} EU(a) = \{\text{Take}\}$. Consequently, $a^* \in \arg\max_{a \in A} EU(a)$ means that a^* is an action whose expected utility is maximal. In this example, $a^* = \text{Take}$. The notation $a^* \in \arg\max_{a \in A} EU(a)$ does not necessarily imply that there is a unique optimal action. For example, if $EU(a_1) = EU(a_2) = 10$ and every other action has expected utility less than 10, then $\arg\max_{a \in A} EU(a) = \{a_1, a_2\}$.

Thus the expected-utility rule may determine a set of optimal actions rather than a unique action.

17.5.7 A Decision Problem with Infinitesimal Probabilities

The previous construction shows that a countably infinite set does not admit a uniform countably additive probability. Nevertheless, the finite approximations suggest a natural way to interpret the statement that a natural number is chosen "uniformly at random". This interpretation becomes particularly interesting when we compare standard limits with hyperfinite probabilities.

Consider the following problem. A natural number N is chosen randomly, and one of the following three rules is used:

1. Every natural number makes the player lose.
2. Only 0 makes the player win.
3. Every number from 0 through 999 makes the player win.

The first rule has probability 0 of winning, while the other two rules raise the question of what "randomly choosing a natural number" should mean.

There is no uniform countably additive probability on $\mathbb{N}$. Indeed, suppose that every $n \in \mathbb{N}$ has the same probability p. If $p > 0$, then

$$P_{\mathbb{N}}(\mathbb{N}) = \sum_{n=0}^{\infty} p = \infty,$$

while if $p = 0$, then

$$P_{\mathbb{N}}(\mathbb{N}) = \sum_{n=0}^{\infty} 0 = 0.$$

In neither case is $P_{\mathbb{N}}(\mathbb{N}) = 1$.

The previous example can be formulated precisely as a decision problem. The purpose is to show how infinitesimal probabilities can distinguish events which all have standard probability zero.

We now consider the following decision problem. Let $S = \mathbb{N}$ and let $\mathcal{A} = \{a_1, a_2, a_3\}$. Define the payoff function $u : \mathbb{N} \times \mathcal{A} \longrightarrow \mathbb{R}$ by $u(s, a_1) = 0$, and $u(s, a_2) = \begin{cases} 1, & s = 0, \\ 0, & s \neq 0, \end{cases}$ and $u(s, a_3) = \begin{cases} 1, & 0 \leq s \leq 999, \\ 0, & s > 999. \end{cases}$ Thus a_1 never gives a positive payoff, a_2 gives payoff 1 precisely when $s = 0$, and a_3 gives payoff 1 precisely when $s \in \{0, 1, \ldots, 999\}$. Define the corresponding winning events by $W_1 = \varnothing, \qquad W_2 = \{0\}, \qquad W_3 = \{0, 1, \ldots, 999\}$. Then $u(s, a_i) = \chi_{W_i}(s)$, where χ_{W_i} denotes the indicator function of W_i.

Suppose first that we attempt to formulate this problem using an ordinary countably additive probability measure on $\mathbb{N}$.

There is no countably additive probability measure P on $\mathbb{N}$ which assigns the same probability to every singleton.

Proof. Suppose that such a probability measure exists. Since all singletons have the same probability, there is some $p \in [0, 1]$ such that $P(\{n\}) = p$

for every $n \in \mathbb{N}$.

If $p > 0$, then for every $N \in \mathbb{N}$, $P(\{0, 1, \ldots, N\}) = \sum_{n=0}^{N} P(\{n\}) = (N + 1)p$.

For sufficiently large N, we have $(N + 1)p > 1$, contradicting $P(\mathbb{N}) = 1$.

Therefore $p = 0$.

But countable additivity gives $P(\mathbb{N}) = P\left(\bigcup_{n\in\mathbb{N}}\{n\}\right) = \sum_{n\in\mathbb{N}} P(\{n\}) = \sum_{n\in\mathbb{N}} 0 = 0$,

contradicting $P(\mathbb{N}) = 1$.

Hence no such probability measure exists. □

In fact, for any countably additive probability measure P on $\mathbb{N}$, every singleton may have probability zero only if the total probability is zero. Therefore the ordinary standard model cannot provide a uniform countably additive probability on $\mathbb{N}$.

We therefore replace $\mathbb{N}$ by a hyperfinite initial segment.

Let $H \in {}^*\mathbb{N} - \mathbb{N}$ be an infinite hypernatural number and define $S_H = \{0, 1, \ldots, H-1\}$. The set S_H is hyperfinite and has internal cardinality $|S_H| = H$. We equip S_H with its internal uniform probability. For every internal subset $B \subseteq S_H$, define $P_{S_H}(B) = \frac{|B|}{H}$. In particular, for every $s \in S_H$, $P_{S_H}(\{s\}) = \frac{1}{H}$. Because H is an infinite hypernatural, $H > n$ for every standard $n \in \mathbb{N}$. Consequently, $0 < \frac{1}{H} < \frac{1}{n}$ for every standard positive integer n. Hence $\frac{1}{H}$

is a positive infinitesimal.

We now extend the payoff function to $S_H \times \mathcal{A}$. Since 999 is standard and $H > 999$, we have $\{0, 1, \ldots, 999\} \subseteq S_H$. The three payoff functions are therefore $u(\cdot, a_1) = 0,$, $u(s, a_2) = \begin{cases} 1, & s = 0, \\ 0, & s \neq 0, \end{cases}$ and $u(s, a_3) = \begin{cases} 1, & 0 \leq s \leq 999, \\ 0, & 999 < s < H. \end{cases}$ For an action $a \in \mathcal{A}$, its hyperfinite expected payoff is $EU_H(a) = \mathbb{E}_{P_{S_H}}[u(\cdot, a)]$. Since S_H is hyperfinite, the expectation is given by the hyperfinite sum

$$EU_H(a) = \sum_{s=0}^{H-1} u(s, a) P_{S_H}(\{s\}) = \frac{1}{H} \sum_{s=0}^{H-1} u(s, a).$$

We now calculate the three expected payoffs separately.

For a_1, we have $u(s, a_1) = 0$ for every $s \in S_H$. Therefore

$$EU_H(a_1) = \frac{1}{H} \sum_{s=0}^{H-1} 0 = 0.$$

For a_2, the only state at which the payoff is 1 is $s = 0$. Hence $\sum_{s=0}^{H-1} u(s, a_2) = 1$. Therefore

$$EU_H(a_2) = \frac{1}{H}.$$

Equivalently, since $W_2 = \{0\}$, we have

$$EU_H(a_2) = P_{S_H}(W_2) = P_{S_H}(\{0\}) = \frac{1}{H}.$$

For a_3, the payoff is 1 at exactly the 1000 states $0, 1, \ldots, 999$. Thus $\sum_{s=0}^{H-1} u(s, a_3) = 1000$, and therefore

$$EU_H(a_3) = \frac{1000}{H}.$$

Equivalently, $W_3 = \{0, 1, \ldots, 999\}$, so $EU_H(a_3) = P_{S_H}(W_3) = \frac{|W_3|}{H} = \frac{1000}{H}$. Consequently, $EU_H(a_1) = 0, \quad EU_H(a_2) = \frac{1}{H}, \quad EU_H(a_3) = \frac{1000}{H}$. Since $H > 0$, we have $0 < 1 < 1000$. Multiplication by the positive hyperreal number $1/H$ gives $0 < \frac{1}{H} < \frac{1000}{H}$. Hence

$$\boxed{EU_H(a_1) < EU_H(a_2) < EU_H(a_3).}$$

Therefore

$$\arg\max_{a \in \mathcal{A}} EU_H(a) = \{a_3\}.$$

In particular, a_3 is the unique expected-payoff maximizing action.

Chapter 18

Measure

18.1 Construction of Lebesgue Measure

1. Interval length and outer measure. For every bounded interval (a, b), let

$$\ell((a, b)) = b - a.$$

thus

$$\ell([a, b]) = b - a.$$

For every $A \subseteq \mathbb{R}$, define

$$\boxed{\ell^*(A) = \inf\left\{\sum_{n=1}^{\infty} \ell(I_n) : A \subseteq \bigcup_{n=1}^{\infty} I_n,\ I_n \text{ open intervals}\right\}.}$$

Here, the infimum is taken over all countable coverings of A by open intervals. We allow intervals of infinite length, and we set $\ell^*(\varnothing) = 0$.

We first show that ℓ^* is an outer measure.

Clearly, $\ell^*(A) \geq 0$ for every $A \subseteq \mathbb{R}$, and $\ell^*(\varnothing) = 0$. If $A \subseteq B$, every open-interval cover of B also covers A, so

$$\ell^*(A) \leq \ell^*(B).$$

Now let $(A_n)_{n\geq 1}$ be a sequence of subsets of $\mathbb{R}$. If $\sum_{n=1}^{\infty} \ell^*(A_n) = \infty$, countable subadditivity is immediate. Otherwise, for every $\varepsilon > 0$, choose open intervals $(I_{n,k})_{k\geq 1}$ covering A_n such that

$$\sum_{k=1}^{\infty} \ell(I_{n,k}) < \ell^*(A_n) + \frac{\varepsilon}{2^n}.$$

The collection of all these intervals covers $\bigcup_{n=1}^{\infty} A_n$. Hence

$$\ell^*\left(\bigcup_{n=1}^{\infty} A_n\right) \leq \sum_{n=1}^{\infty}\sum_{k=1}^{\infty} \ell(I_{n,k}) \leq \sum_{n=1}^{\infty} \ell^*(A_n) + \varepsilon.$$

Letting $\varepsilon \to 0$, we obtain

$$\ell^*\left(\bigcup_{n=1}^{\infty} A_n\right) \leq \sum_{n=1}^{\infty} \ell^*(A_n).$$

Thus ℓ^* is an outer measure.

Lemma 18.1.1 (The outer measure of an interval)**.** *We prove that, for $a < b$,*

$$\boxed{\ell^*([a,b]) = b - a.}$$

thus

$$\ell^*((a,b)) = b - a.$$

Proof. For every $\varepsilon > 0$, the open interval

$$I = \left(a - \frac{\varepsilon}{2}, b + \frac{\varepsilon}{2}\right)$$

covers $[a,b]$. Therefore

$$\ell^*([a,b]) \leq b - a + \varepsilon.$$

Since ε is arbitrary,

$$\ell^*([a,b]) \leq b - a.$$

For the reverse inequality, let $(I_n)_{n\geq 1}$ be any countable cover of $[a,b]$ by open intervals. We must prove

$$\sum_{n=1}^{\infty} \ell(I_n) \geq b - a.$$

If this sum is infinite, the inequality is immediate. Otherwise, since $[a,b]$ is compact, there is a finite subcover

$$[a,b] \subseteq I_1 \cup \cdots \cup I_N.$$

We use the elementary fact that if finitely many open intervals cover $[a,b]$, then the sum of their lengths is at least $b - a$. Consequently,

$$b - a \leq \sum_{n=1}^{N} \ell(I_n) \leq \sum_{n=1}^{\infty} \ell(I_n).$$

Since the cover was arbitrary, taking the infimum gives

$$\ell^*([a,b]) \geq b - a.$$

Combining both inequalities yields

$$\boxed{\ell^*([a,b]) = b - a.}$$

□

The same value holds for (a,b), $(a,b]$, and $[a,b)$, since each can be covered by open intervals of length arbitrarily close to $b-a$, and each contains closed intervals whose lengths approach $b-a$.

5. Carathéodory measurability. A set $E \subseteq \mathbb{R}$ is called ℓ^*-measurable if, for every $X \subseteq \mathbb{R}$,

$$\boxed{\ell^*(X) = \ell^*(X \cap E) + \ell^*(X - E).}$$

Let $\mathcal{L}$ denote the collection of all ℓ^*-measurable sets.

Theorem 18.1.1 (Carathéodory's Theorem)**.** *If ℓ^* is an outer measure on X, then the collection $\mathcal{L}$ of ℓ^*-measurable sets is a σ-algebra, and the restriction $\ell^*|_{\mathcal{L}}$ is a complete measure.*

Proof. First, $\varnothing \in \mathcal{L}$, since for every $E \subseteq X$,

$$\ell^*(E) = \ell^*(E \cap \varnothing) + \ell^*(E \cap \varnothing^c) = 0 + \ell^*(E).$$

Moreover, $\mathcal{L}$ is closed under complements, since the definition of measurability is symmetric in A and A^c.

Next, let $A, B \in \mathcal{L}$. For every $E \subseteq X$, applying the measurability of A and B gives

$$\ell^*(E) = \ell^*(E \cap A \cap B) + \ell^*(E \cap A \cap B^c) + \ell^*(E \cap A^c \cap B) + \ell^*(E \cap A^c \cap B^c).$$

By subadditivity,

$$\ell^*(E) \geq \ell^*(E \cap (A \cup B)) + \ell^*(E \cap (A \cup B)^c)$$

Since

$$A \cup B = (A \cap B) \cup (A \cap B^c) \cup (A^c \cap B),$$

by subadditivity,

$$\ell^*(E \cap A \cap B) + \ell^*(E \cap A \cap B^c) + \ell^*(E \cap A^c \cap B) \geq \ell^*(E \cap (A \cup B))$$

Consequently,

$$\ell^*(E\cap A\cap B)+\ell^*(E\cap A\cap B^c)+\ell^*(E\cap A^c\cap B)+\ell^*(E\cap A^c\cap B^c) \geq \ell^*(E\cap(A\cup B))+\ell^*(E\cap(A\cup B)^c)$$

The reverse inequality follows from subadditivity. Hence

$$\ell^*(E) = \ell^*(E \cap (A \cup B)) + \ell^*(E \cap (A \cup B)^c),$$

so $A \cup B \in \mathcal{L}$.

The reverse inequality follows from subadditivity. Hence $A \cup B \in \mathcal{L}$, so $\mathcal{L}$ is an algebra.

Moreover, if $A, B \in \mathcal{L}$ and $A \cap B = \varnothing$, then

$$\ell^*(A \cup B) = \ell^*((A \cup B) \cap A) + \ell^*((A \cup B) \cap A^c) = \ell^*(A) + \ell^*(B)$$

Thus ℓ^* is finitely additive on $\mathcal{L}$.

To show that $\mathcal{L}$ is a σ-algebra, it suffices to show that it is closed under countable disjoint unions. Let $(A_n)_{n\geq 1}$ be a sequence of pairwise disjoint sets in $\mathcal{L}$. For every $E \subseteq X$, the measurability of A_n gives

$$\ell^*\left(E \cap \bigcup_{j=1}^{n} A_j\right) \quad = \ell^*(E \cap A_n) + \ell^*\left(E \cap \bigcup_{j=1}^{n-1} A_j\right).$$

Therefore, by induction,

$$\ell^*\left(E \cap \bigcup_{j=1}^{n} A_j\right) = \sum_{j=1}^{n} \ell^*(E \cap A_j).$$

It follows that

$$\ell^*(E) = \sum_{j=1}^{n} \ell^*(E \cap A_j) + \ell^*\left(E \cap \left(\bigcup_{j=1}^{n} A_j\right)^c\right) \geq \sum_{j=1}^{n} \ell^*(E \cap A_j) + \ell^*\left(E \cap \left(\bigcup_{j=1}^{\infty} A_j\right)^c\right).$$

Letting $n \to \infty$ and using countable subadditivity, we obtain

$$\ell^*(E) \geq \sum_{j=1}^{\infty} \ell^*(E \cap A_j) + \ell^*\left(E \cap \left(\bigcup_{j=1}^{\infty} A_j\right)^c\right) \geq \ell^*\left(E \cap \bigcup_{j=1}^{\infty} A_j\right) + \ell^*\left(E \cap \left(\bigcup_{j=1}^{\infty} A_j\right)^c\right) \geq \ell^*(E).$$

Hence all inequalities are equalities, so

$$\ell^*(E) = \ell^*\left(E \cap \bigcup_{j=1}^{\infty} A_j\right) + \ell^*\left(E \cap \left(\bigcup_{j=1}^{\infty} A_j\right)^c\right).$$

Thus $\bigcup_{j=1}^{\infty} A_j \in \mathcal{L}$, and $\mathcal{L}$ is a σ-algebra.

Taking $E = \bigcup_{j=1}^{\infty} A_j$ in the preceding equality gives

$$\ell^*\left(\bigcup_{j=1}^{\infty} A_j\right) = \sum_{j=1}^{\infty} \ell^*(A_j).$$

Therefore $\ell^*|_{\mathcal{L}}$ is a measure on $\mathcal{L}$.

Finally, suppose $A \in \mathcal{L}$ and $\ell^*(A) = 0$. If $N \subseteq A$, then $\ell^*(N) = 0$. Thus, for every $E \subseteq X$,

$$\ell^*(E \cap N) = 0,$$

and consequently

$$\ell^*(E) = \ell^*(E \cap N) + \ell^*(E \cap N^c).$$

Hence $N \in \mathcal{L}$ and $\lambda(N) = 0$. Therefore λ is complete.

□

4. Lebesgue measure. The measure

$$\boxed{\lambda = \ell^*|_{\mathcal{L}}}$$

is called Lebesgue measure on $\mathbb{R}$. Since intervals are Lebesgue measurable and their outer measure equals their length, we have

$$\boxed{\lambda([a,b]) = b - a.}$$

Moreover, λ is a complete measure, and every Borel subset of $\mathbb{R}$ is Lebesgue measurable.